Causal Clifford Theory (CL10): a quantum cellular automaton on $\mathrm{Cl}(10,0)$ — Standard Model kinematics, exactly unitary causal dynamics, and a lattice gravitational sector

Mohammed Haouach *Independent researcher; correspondence: haouach.mohammed@gmail.com*

Abstract

A quantum cellular automaton is defined on a Planck lattice whose 2×2×2 cells carry the 36-level cell module of $\mathrm{Cl}(10,0)$ — the 32-dimensional Dirac module and two neutral singlet sectors, conjugate pairs of levels — on eight sub-sites (on-site space $\mathbb{C}^{288}$), updated by one six-stage product of Spin(10) plane rotations and swap-rotations, the singlet pairs carrying the ladder masses as on-site phases. From this root follow, with no added field: a (1+3) Minkowski subalgebra; the 16-component Weyl module whose charges are exactly one Standard Model generation, with all eight gauge and mixed anomalies cancelling; the **45** adjoint containing the twelve gauge bosons; and the 5/3 hypercharge normalisation. The step is exactly unitary ($\det U = 1$, ±-paired spectrum, exact matter/antimatter spectral mirror) and strictly causal, with front speed exactly one Planck cell per stage. The closure axiom — the update must be an element of the root — quantizes the rotation angle to $g \in \{\pi/2, \pi/4\}$: the fundamental step has order 8, the internal spectrum is the finite algebraic set $\{\pm\pi/4\,(10),\ \pm 3\pi/4\,(6)\}$, and the on-site update keeps no free continuous angle. One loop factor then organises four scales over thirty orders of magnitude: the dark-matter mass 118.57 GeV and the seesaw scale 9.6×10^{13} GeV are on-site levels of the cell; the neutrino mass follows from their ratio; the electroweak hierarchy carries one named residual, +30%, whose measured address is the clock at which the loop functional is read.

What does **not** follow is stated with the same care. The doubler spectrum is linear in every sampled direction but never isotropic: 0 of 2726 gapless points of the Wilson family reach anisotropy below 10^{-6}, the gapless corner touchings are crossings of nodal planes, and over the axis-covariant entangling-bond class gaplessness and non-commuting tangent algebras are mutually exclusive — a rigidity wall at gap 8.0×10^{-3} whose cause is a parity theorem: at a momentum whose little group contains inversion, no single-parity multiplet can be conical. The only nodes that can be conical, parity-degenerate $A_{1g} \oplus T_{1u}$ pairs, lie outside the automaton's own circuit class. The gravitational sector adds $g_{\mu\nu}$ and $G = 1$ by hand and reproduces general relativity to $O(h^2)$; the constructed spin-2 channel runs at $c_{\mathrm{gw}}/c = 1.5708$ and is excluded by the measured $|c_{\mathrm{gw}}/c - 1| < 10^{-15}$ [60]. Coupling unification holds only with a desert spectrum carrying two fitted thresholds, and at the root coupling that solution is not realizable by its own content. The dark matter is a cell singlet, exactly stable under a $\mathbb{Z}_2$ of the update, produced at the observed abundance by freeze-in; its portal coupling is the sector's single unfixed input. The ladder base $1/(16\pi)$ and the clock $g = 0.1999980$ at which the hierarchy is read are declared measured inputs — the first a theorem of the loop census, the second a theorem of clock transcendence: every closure of the root-generated class at a nonzero algebraic clock is a clock identity, and the data clock lies outside the closure reach of the class. Every headline number is classified by provenance: *algebraic*, *conventional*, *fitted*, or *open*; three conventions, twelve fitted coefficients and one conditional coefficient stand between the root and the data, and ten open problems and ten falsifiers are stated with their numbers. The endgame of the loop clock is measured: the completion that makes that clock born rather than fitted is unique and costs exactly one dial — the clock tension itself, promoted to a constant of nature (Section 10.5).

Keywords. Clifford algebra; quantum cellular automaton; Spin(10) grand unification; anomaly cancellation; lattice gravity; dark matter; cosmological constant; baryogenesis; falsifiability.

---

1. Introduction

1.1 The picture

Two habits of language do most of the misleading. The first calls c the speed of light. Light is one traveller at that speed; the constant is the speed of causality. The second is wave–particle duality, taught as a paradox. On a lattice it dissolves: there is one wave, always; what a detector registers is a single localised event, because the detector is itself a lattice of cells changing state one cell at a time. The ontology contains no particles — only waves that deliver their energy in quanta.

Quantum field theory replaced particles by fields but multiplied the fields, one per species, glued together by hand. The viewpoint of this paper is the opposite one, elementary in computer science: a cellular automaton — a lattice of cells, each holding a finite packet of information, all updated in parallel by one local rule. If the world is such an automaton, everything physical is a wave of information over that lattice, and the theory reduces to two questions: what is the information structure of a single cell, and what is the rule that updates it. Causal Clifford Theory (CL10) answers both concretely: the cell carries the 36-level module of Axiom 1 — the spinor module of $\mathrm{Cl}(10,0)$ and two neutral singlet sectors — and the rule is a product of Spin(10) rotations and swaps with a single angle, which the algebra itself then quantizes.

The determinism claim is interpretive: one time-independent local rule fixes every future state of the lattice — 't Hooft's cellular-automaton determinism exact at the fundamental update — with randomness entering only through the measurement projection; the projection and the Born rule are inputs of the interpretation (O9), not derived from the unitary step.

1.2 Two literatures, one construction

Discrete models of quantum dynamics and grand unification have developed independently. Causal sets, lattice gauge theory and quantum cellular automata treat spacetime as a finite, causal, unitary dynamical system [1–5]. The observation that one generation of quarks and leptons fits exactly into the 16-dimensional spinor of Spin(10), with the 5/3 hypercharge normalisation and anomaly cancellation following from the embedding, has been standard since 1974 [7–10]; Clifford-algebra formulations of this unification exist [11–14]. 't Hooft's cellular-automaton interpretation supplies a deterministic unitary dynamics but no unification [30,31].

CL10 puts these on a single footing: the internal space of every sub-site of the automaton's cell **is** the 36-level cell module — the spinor module of $\mathrm{Cl}(10,0)$ with its two neutral singlet sectors (Section 4) — and the update **is** an element of the corresponding spin group, times the singlet ladder phases of Axiom 1. Nothing else is added by hand: no field outside the cell module, no extra gauge factor, no second algebra.

The comparison is decidable, and its deficits are named here rather than left to the Discussion. Against the Clifford-algebra programmes [11–14] this construction lacks what they already hold: a semiclassical renormalisation group acting on a spectrum of light chiral fermions, and couplings that run into agreement with data. In this theory no level of the measured spectrum is light, dispersing and chiral at either clock — the question is O10 (Sections 6–7, 15) — and agreement with the couplings is reached only by admitting two fitted desert thresholds (O4). Against the cellular-automaton and causal-programme literature [1–5,30,31] what is added is the internal symmetry, the exact unitarity of a finite-order update, and a boundary computed rather than assumed.

The novelty claim is stated item by item, with the prior work named for everything that is **not** new. (i) Spinor representations on hypercubic lattices and the Clifford bookkeeping of doublers are standard [55,62]; what is added is a fixed order-eight Clifford update on the 36-level module, not a lattice Dirac action. (ii) The 16-plet kinematics, the 5/3, and anomaly cancellation derived from Clifford algebras are due to [11–14,63,64]; Theorems 1–3 are their algebra, and the paper claims only that the **same** module is the internal state of the automaton that updates it. (iii) Nodal lines and planes enforced by icosahedral point symmetry are known in topological matter [65]; what is new is that the icosahedral shift module is a structure inside an exactly unitary automaton, with its nodal network, node count and node speeds computed on the construction rather than asserted. (iv) Weighting closed words by traces of powers of one operator is the transfer-matrix and character-expansion device of lattice gauge theory [2,55] and the cycle/trace-formula technology of chaotic lattice models [66]; what is new is the complete 3^n-word enumeration from which the loop factor $1/(16\pi)$ is read as a number. (v) Deterministic cellular-automaton dynamics is 't Hooft's [30,31] (and the QCA programme's [3–5]); none of it carries an internal unification or a finite-order update — here the update **is** an element of Spin(10) of order 8 (Theorem 6). (vi) The parity and topology obstructions to a single conical node are Nielsen–Ninomiya's [21] in their Weyl-doubling form [67]; the measured node network obeys them (± 1 chiralities in cancelling pairs), and the theory uses those theorems, never claims around them.

1.3 Claims, non-claims, and the provenance discipline

Claimed and demonstrated. The kinematics of one Standard Model generation, its anomaly freedom, the grand-unified gauge content and the 5/3, and a unitary causal automaton whose gravitational sector reproduces general relativity on the lattice to $O(h^2)$ (Table 3; Sections 3–5, 8).

Claimed as an order-of-magnitude statement. The electroweak/Planck hierarchy, the dark-matter mass scale, the neutrino mass and the baryon asymmetry, all expressed through the loop factor $(1/16\pi)^n$ with n a small integer (Section 10). The integers are chosen to land near the measured values; the two-digit coefficients are built from computed closed-loop functionals and carry named residuals (Section 10.1).

The organising discipline of the paper is therefore a **provenance discipline**: a number is *algebraic* if the root fixes it, *conventional* if it comes from a mapping or scale identification, *fitted* if a coefficient was read from data, *declared* if it is a named measured input the theory uses without deriving, and *open* if the theory does not fix it. Section 13 tabulates every headline number under this classification.

The seven criteria. A unification claim has to be checkable, so the theory is judged against stated criteria, with its own verdict on each.

Table 1. The seven criteria and the verdict of this paper.

criterion	requirement	verdict
one object	a single algebraic object, no added fields or structure	met for the internal space; the gravitational sector adds a symmetric tensor field $g_{\mu\nu}(x)$ and the value $G = 1$ (Sections 8–9)
kinematics	spacetime, matter content, gauge group and statistics follow by representation theory	met (Sections 2–4)
dynamics	the object yields a unitary, causal evolution	met (Section 5: exactly unitary, $\det U = 1$, front speed $= 1$)
unification	matter, gauge and gravity live in the same object	partial: matter and gauge are the same module and the same **45**; gravity is the metric dynamics of a subalgebra of that module, with an added field
consistency	anomaly-free, unitary, causal, UV-bounded	met (Theorem 2, Theorem 5, the bounded 32-level spectrum, the bounded n-particle S-matrix); the doublers of Proposition 2 are gapped by an internal Wilson factor (Section 6), with the split node set an input (O9)
parameters	every physical number fixed by the object, no fit	**not met**: three conventions, twelve fitted coefficients, one conditional coefficient (Sections 10.1, 13). The closure axiom removes the one free continuous parameter of the dynamics at the price of moving that parameter's two empirical contacts, and both contacts fail on the closed branch
falsifiability	testable predictions, and the open problems named rather than hidden	met (Sections 14–15)

The theory passes the structural criteria, and the parameter criterion is decided at the root: the closure axiom makes the closed-branch clock a discrete index of the algebra, $g = \pi/4$ (Theorem 6), removing the last continuous number of the on-site dynamics. The empirical price of that branch is measured: its doubler census is a six-line nodal network, its desert unification solution leaves the desert with $\tau_p = 2.77 \times 10^{39}$ yr, and its meeting coupling sits 1.23× **above** the data-preferred value — while the reference reading 1/25 (12.6× below in the same normalisation, at $g = 1/5$) is a convention, not the closed clock (Section 5.2). That the reference reading is a fit is itself measured: the reading clock stands at 0.1999980, 2.0×10^{-6} off the fifth, no closed clock of the root-generated class reads the hierarchy coefficient in band, and the class-versus-clock question closes by a transcendence theorem (Section 5.3).

1.4 Conventions

Planck units $\hbar = c = \ell_P = t_P = 1$ unless stated; $G = 1$. Metric signature $(+,-,-,-)$, $Q = T_3 + Y$. $\mathrm{Cl}(10,0)$ is the real algebra of ten generators with $h_a^2 = +I$; the module is over its complexification $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$. "Generation" means one set of left-handed Weyl fields. **Coupling notation, one convention:** g (equivalently g_{10}) denotes the rotation angle of the step — 1/5 the reading convention, $\pi/4$ the closed class — and α

always denotes the standard-normalised coupling $\alpha = g^2/4\pi$ wherever a coupling is compared with running; the alternative reading $\alpha_{10} \equiv g^2$ is a factor-4π convention and is invoked only where explicitly named (Sections 5.2, 11.1, 13).

---

2. The root

Axiom 1 (the root). *The universe is a* (1+3)*-dimensional causal quantum cellular automaton. Its cell is a* $2\times2\times2$ *block of Planck sites — site spacing* ℓ_P*, so the cells tile a cubic lattice of spacing* $2\ell_P$ *— and each of the eight sub-sites carries the cell module*

$$M = S \oplus R_A \oplus R_B, \qquad \dim M = 36,$$

where S *is the* 32*-dimensional Dirac module of the Clifford algebra* $\mathrm{Cl}(10,0)$ *— ten mutually anticommuting Hermitian generators*

$$h_a^\dagger = h_a, \qquad h_a^2 = I, \qquad \{h_a, h_b\} = 0 \quad (a \neq b), \qquad a, b = 1, \ldots, 10,$$

— and R_A*,* R_B *are two neutral singlet sectors: each is a conjugate pair of levels* $\{A_\pm\}$*,* $\{B_\pm\}$*; the generators act on them trivially, equivalently every one of the* 45 *plane generators* $\frac{1}{4}[h_a, h_b]$ *acts as zero (Section 4). A singlet of the full Clifford algebra is not chiral, so its levels come in conjugate pairs. A* $\mathbb{Z}_2$*, the* ***B-parity****, is odd on* B *alone. The state of the universe is a tensor product of these modules over the sub-sites, so the cell's on-site space is* $\mathbb{C}^{288}$*. The hypothesis of everything proved below is this algebra and this module — nothing else: the time evolution is a separate construction (****Construction 1****), and no property of it enters the proofs.*

Three structural facts about the root, all standard [11,20]:

1. $\mathrm{Cl}(10,0)$ is $2^{10} = 1024$-dimensional; its complexification is $M_{32}(\mathbb{C})$, so it has a single irreducible complex module of dimension 32, the Dirac module.
2. The volume element $w = h_1 h_2 \cdots h_{10}$ satisfies
$$w^2 = (-1)^{10\cdot 9/2} \prod_a h_a^2 = (-1)^{45} = -I, \qquad w^\dagger = -w,$$
so w is anti-Hermitian with eigenvalues $\pm i$, each of multiplicity 16. The projectors $P_{16} = \frac{1}{2}(I - iw)$ and $P_{\overline{16}} = \frac{1}{2}(I + iw)$ split the Dirac module into two inequivalent 16-dimensional Weyl modules.
3. The algebra is $\mathbb{Z}_2$-graded by the parity of the number of generators in a product: 512 even and 512 odd elements.

Main Theorem (the content of the root). *Let* $\mathrm{Cl}(10,0)$ *be the algebra of Axiom 1 and* $M = S \oplus R_A \oplus R_B$ *its cell module. From the algebra and the module alone — no evolution, no lattice spacing, no coupling, no measurement — the following are forced:*

(i) a (1+3) *spacetime subalgebra whose quadratic form is Minkowski, with its null cone (Theorem 1, Corollary 1);*

(ii) the split of S *into two inequivalent* 16*-dimensional Weyl modules* P_{16}*,* $P_{\overline{16}}$*, and the* **16** *is exactly the six left-handed Weyl fields of one Standard Model generation (structural fact 2, Proposition 1);*

(iii) the gauge content: the 45 *plane generators close to* $\mathfrak{so}(10)$ *and contain the Model algebra, and the hypercharge on the* **5** *obeys* $\mathrm{Tr}\, Y^2 = \frac{5}{6}$ *— the* 5/3 *normalisation (Theorem 3);*

(iv) anomaly freedom: the eight gauge and mixed anomaly coefficients of the **16** *vanish (Theorem 2, Table 2);*

(v) statistics: the $\mathbb{Z}_2$ *grading of the algebra is the grading that carries spin and statistics on the module, and the B-parity is a grading of the cell (Section 3);*

(vi) the content is one generation: $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$ *has a single irreducible complex module (structural fact 1), and* 10 *is the smallest dimension with properties (ii) and (iii) ("Why* 10*", below).*

Nothing in (i)-(vi) mentions time evolution; no measured property of the automaton's step is used in their proofs.

Traceability of the Main Theorem. Each clause and where it is proved; every witness is a computation of the computational companion (Appendix B) that recomputes the clause from the algebraic inputs:

clause	proved in	witnessed by
(i) the (1+3) subalgebra and its null cone	Theorem 1, Corollary 1 (Section 3)	exact subalgebra census
(ii) the Weyl split and the six fields of one generation	structural fact 2, Proposition 1 (Section 4)	exact charge table of the **16**
(iii) the **45**, the Model algebra, the 5/3	Theorem 3 (Section 4)	trace-ratio computation on the **45**
(iv) anomaly freedom	Theorem 2, Table 2 (Section 4)	the eight anomaly coefficients, evaluated exactly
(v) statistics and the B-parity grading	Section 3; the grading census of the cell	exact $\mathbb{Z}_2$ grading census
(vi) one module, one generation; why 10	structural fact 1, "Why 10" (this section)	module census of the Clifford generators

Construction 1 (the step). *The dynamics is one time-independent unitary **step** of six **stages** built from the same algebra (Section 5): each stage applies the internal layer to every sub-site — on S the product of the* 45 *plane rotations, and on the two singlet pairs the phases $e^{\pm i\,\mathrm{loop}^3}$ and $e^{\pm i\,\mathrm{loop}^{10}}$, with* $\mathrm{loop} = 1/16\pi$*: a level on ladder rung n (Section 10.4) carries the phase m/M_P per elementary time, its mass read as a phase. The elementary time is the stage, t_P; one step is $6t_P$.*

Why the cell is a block. The bond part of the update pairs neighbouring sub-sites, and the pairing that treats the three directions alike has six stages: three pair sub-sites *inside* the cell, three pair sub-sites *across* cells. Written on a single Planck site the same rule alternates between bonds inside and outside the site, which forces a two-cell spatial period and a six-periodic time dependence; on the block it is one time-independent rule, and the block is the smallest region on which the update acts as a single unitary. The staggering is therefore not a time dependence of the law but the sub-lattice structure of the cell, and the Bloch decomposition of the step is exact at every step.

The step's own properties are separate claims, each established on the construction itself: exact unitarity, $\det U = 1$ and the causal radius (Theorem 5, Section 5), the $\pm$-spectral mirror (Theorem 4, Section 5.1), the doubling census and its Wilson gapping (Proposition 2, Section 6), the closure of the branch clock to $g = \pi/4$ (Theorem 6, Section 5.2), and the empirical contacts of the rung phases (Section 10). None of them is a consequence of the Main Theorem, and none of them is needed by it.

Why 10**.** The number 10 is the smallest n for which a single complex Clifford algebra simultaneously provides (i) a pseudoscalar splitting a Dirac module into two *inequivalent* Weyl modules of dimension 16 — at $n = 8$ the Weyl modules are 8-dimensional (Spin(8) triality: $8_v, 8_s, 8_c$; the 16-dimensional representations are reducible) — and (ii) an adjoint of dimension $\binom{10}{2} = 45$ large enough to contain $\mathfrak{su}(5)$ and hence the Standard Model algebra. At $n = 9$, $\mathrm{Cl}_9(\mathbb{C}) \simeq M_{16} \oplus M_{16}$ does have 16-dimensional irreducibles, but they are the two simple factors of the algebra rather than a pseudoscalar split, and $\mathfrak{so}(9)$ does not contain $\mathfrak{su}(5)$. The root is the minimal choice.

---

3. Spacetime, statistics, measurement

Theorem 1. *Let $T = h_1$ and $S_a = ih_{a+1}$ for $a = 1, 2, 3$. Then*

$$T^2 = +I, \qquad S_a^2 = -I,$$
$$\{T, S_a\} = 0, \qquad \{S_a, S_b\} = 0.$$

The real span of $\{T, S_1, S_2, S_3\}$ inside the grade-1 part of the algebra, with the quadratic form induced by the Clifford product, is Minkowski space $\mathbb{R}^{1,3}$.

Proof. The four generators $h_1, \ldots, h_4$ anticommute by Axiom 1. Multiplication by i turns $h_a^2 = I$ into $(ih_a)^2 = -I$ and preserves anticommutation. For $v = v_0 T + \sum_a v_a S_a$ with real coefficients, $v^2 = (v_0^2 - \sum_a v_a^2) I$.

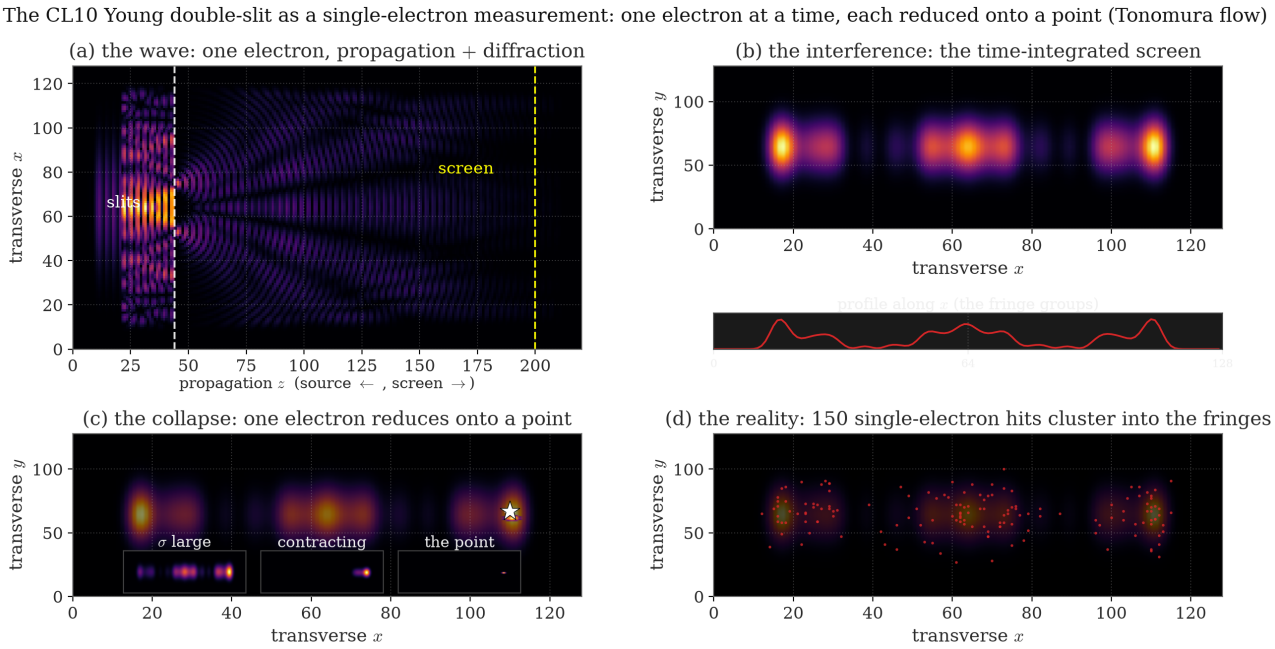
Corollary 1 (the light cone). *A vector v in that real span is null, $v^2 = 0$, if and only if $v_0^2 = \sum_a v_a^2$. The null cone of this quadratic form is the light cone of the lattice.*

Two qualifications. The null cone is a **cone in the grade-1 part**, not a subalgebra: it is not closed under the Clifford product, since $(T + S_1)(T - S_1) = 2I - 2TS_1$ is a grade-0 plus grade-2 element. And the choice of *which* four of the ten generators play the role of spacetime is a choice of frame in the 10-dimensional generator space, not a prediction: the algebra supplies a family of equivalent (1+3) subalgebras related by orthogonal transformations of the generator space. The remaining six generators are inert under the Lorentz generators $\Sigma_{\mu\nu} = \frac{1}{4}[\Gamma_\mu, \Gamma_\nu]$ built from $\Gamma_\mu \in \{T, S_a\}$.

Causality is a postulate with a dynamical content. The light cone by itself does not bound signal propagation: a unitary on-site rotation alone has infinite propagation speed. The bound is supplied by the update of Section 5, where the causal radius is *implemented* by the bond stages and shown to be exactly one Planck cell per stage (Theorem 5). The light cone and the causal radius are distinct statements, and only the second is a property of the dynamics.

Statistics. The $\mathbb{Z}_2$ grading splits the algebra into 512 even and 512 odd elements. The even sector is identified with bosons and the odd sector with fermions. This is a *kinematic* grading of the operator algebra: a postulate of the interpretation, not a derivation of the spin–statistics theorem, and not a claim of dynamical supersymmetry. No supercharge exists in the construction and no superpartner is predicted.

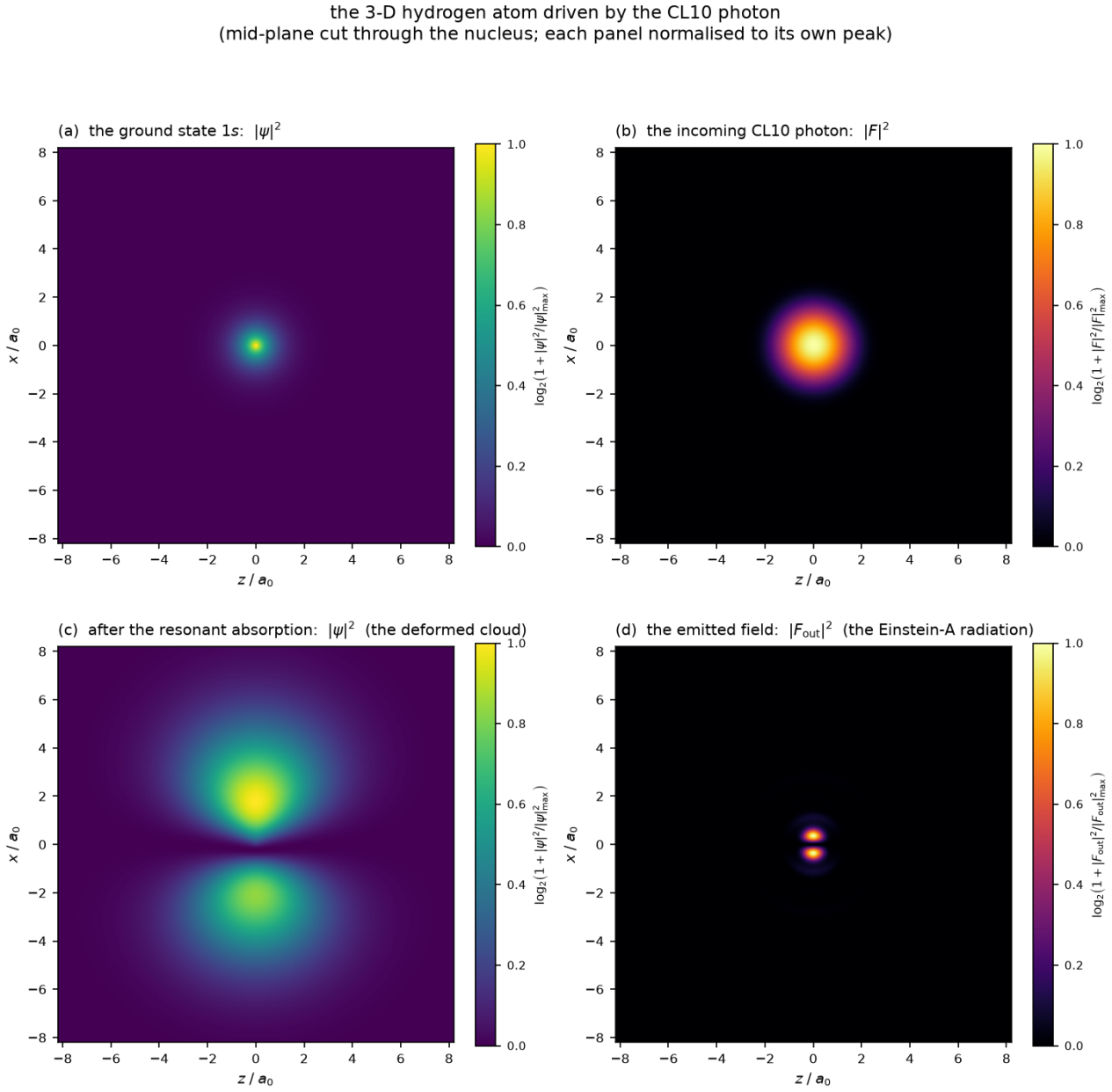
Measurement. Between measurements the automaton is exactly unitary. A detection is modelled as the standard projection: $\sum_x \psi(x)|x\rangle$ reduces to $|x_0\rangle$ with probability $|\psi(x_0)|^2$, the probability rule being the intensity $|\psi|^2$ of the propagating lattice field — the one-electron-at-a-time build-up of the interference fringes is the Tonomura experiment [48]. The projection is **not** derived from the unitary step and the theory does not claim it is: the reduction postulate is an input of the interpretation (O9). What the automaton adds is only that the wave being sampled is a strictly causal, unitary, lattice field (Fig. 1).



(a) diffraction and interference of the field; (b) the time-integrated on-screen pattern; (c) the reduction of one electron onto one detection point — the projection is an input of the interpretation (O9), not a consequence of the unitary step; (d) the accumulated detections clustering into the fringes, the Tonomura sequence [48].

The same field drives atomic physics: coupled to a prescribed hydrogen spectrum, the lattice photon reproduces resonant absorption — Lyman-α ($1s \to 2p$ at $\omega = 0.25$ rad/step) peaks the excited fraction at 0.998, the $1s \to 3p$ and $2p \to 3d$ resonances also absorb, the un-driven baseline stays at 0 — the $1s \to 2p$ deformation (the cloud anisotropy moves from -5.8×10^{-17} for the ground state to 0.392 at the resonant peak, against 0.378 for a pure $2p$), and re-emission at the gap frequency with decay of the excited population, while the off-resonant control at

0.13 rad/step absorbs at only 0.0093 — a suppression of 107× against resonance. The atom is an **input** — its levels are the real hydrogen eigenstates — and the figure demonstrates the photon sector, not a derivation of hydrogen; off-resonant absorption follows the natural Lorentzian line shape of the driven atom (Fig. 2).



A three-dimensional hydrogen atom coupled to the same discrete massless field that carries the graviton, on the mid-plane cut through the nucleus: (a) the spherical ground state $|\psi|^2$; (b) the incoming lattice photon $|F|^2$; (c) the cloud after resonant absorption, deformed into the dumbbell $2p$; (d) the emitted field $|F_{\mathrm{out}}|^2$. The atom is an input; the figure tests the photon sector, not hydrogen.

The hidden-variable question. The automaton is a deterministic model of quantum phenomena, so the no-go theorems for hidden variables apply to it, and the theory takes the horn they force instead of claiming to evade them. Bell's theorem [56] and the Kochen–Specker theorem [57] do not refute the construction; they constrain what it can be. Its dynamics is strictly local on the lattice (front speed exactly 1, Section 5) and its outcome rule is an input projection that the step does not derive (O9) — so it does not supply a Bell-local completion of quantum mechanics, and it does not assign pre-existing values to the projectors of the cell module,

which is the assumption Kochen–Specker forbids. What the construction adds is the substrate: a strictly causal, exactly unitary lattice field whose observables are algebra elements of the whole cell, never of an isolated sub-site. No Bell-inequality or noncontextuality-inequality test is claimed: the theory distinguishes itself from quantum mechanics through the ladder, the coupling and the carrier of Section 15, not through a deviation in the correlation statistics, and the measurement postulate remains an open input (O9).

---

4. Matter and gauge

Proposition 1. *Under* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *the* 16*-dimensional Weyl module is the six left-handed fields of one Standard Model generation:*

$$16 = Q(\mathbf{3},\mathbf{2},+\tfrac{1}{6}) \oplus u^c(\bar{\mathbf{3}},\mathbf{1},-\tfrac{2}{3}) \oplus d^c(\bar{\mathbf{3}},\mathbf{1},+\tfrac{1}{3}) \oplus L(\mathbf{1},\mathbf{2},-\tfrac{1}{2}) \oplus e^c(\mathbf{1},\mathbf{1},+1) \oplus \nu^c(\mathbf{1},\mathbf{1},0),$$

with $6+3+3+2+1+1=16$. *Under* SU(5) *this is* $\mathbf{10} \oplus \bar{\mathbf{5}} \oplus \mathbf{1}$, *the* $\mathbf{1}$ *being the sterile* ν^c.

The assignment is the standard one [7–10,13]. What the automaton adds is that the **16** is not put in by hand but is the Weyl eigenspace of the pseudoscalar of the same algebra that carries the gauge group and the spacetime — and that the group itself is not a free choice: the stabiliser, inside the 45, of the four directions the automaton calls spacetime is $\mathrm{Spin}(4) \times \mathrm{Spin}(6)$, with $\mathfrak{su}(3) \oplus \mathfrak{u}(1)$ inside spin(6), and the $B-L$ charges of the previous paragraph and the 5/3 normalisation both follow as trace ratios (Section 4.2). What the construction leaves open is the flavour structure of Section 11.5, not the group.

$B-L$. $B-L$ is a generator of $\mathfrak{so}(10)$, so it is *not* a single number on the **16**. Its values on the six fields are $Q : \frac{1}{3}$, $u^c : -\frac{1}{3}$, $d^c : -\frac{1}{3}$, $L : -1$, $e^c : +1$, $\nu^c : +1$; the $\overline{\mathbf{16}}$ carries the negatives of all of them. Under $\mathrm{Spin}(10) \supset \mathrm{SU}(5) \times \mathrm{U}(1)_\chi$, $16 \to \mathbf{10}_{-1} \oplus \bar{\mathbf{5}}_{+2} \oplus \mathbf{1}_{+3}$, and $B-L$ is a linear combination of χ and the diagonal SU(5) generator, not χ itself.

Theorem 2 (anomaly cancellation). *All eight gauge and mixed anomaly coefficients of the* **16** *vanish.*

Table 2. The eight anomaly coefficients of one **16**.

coefficient	contribution of the **16**	value
$[\mathrm{SU}(3)]^3$	$2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 1 - \frac{1}{2} - \frac{1}{2}$	0
$[\mathrm{SU}(2)]^3$	$A(\mathbf{2}) = 0$ identically ($d^{abc} = 0$; the fundamental is pseudo-real)	0
$[\mathrm{U}(1)]^3$	$6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 + 1(1)^3 = \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1$	0
$\mathrm{U}(1)[\mathrm{SU}(3)]^2$	$\frac{1}{6} \cdot \frac{1}{2} \cdot 2 + (-\frac{2}{3}) \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6}$	0
$\mathrm{U}(1)[\mathrm{SU}(2)]^2$	$\frac{1}{6} \cdot \frac{1}{2} \cdot 3 - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4}$	0
$[\mathrm{grav}]^2\mathrm{U}(1)$	$\sum Y = 1 - 2 + 1 - 1 + 1$	0
$[\mathrm{grav}]^2\mathrm{SU}(3)$, $[\mathrm{grav}]^2\mathrm{SU}(2)$	traceless generators	0

Proof. Each coefficient is a cubic trace over the left-handed Weyl fields; the multiplicities are the dimensions of the *other* group factors (the factor 2 on Q in $[\mathrm{SU}(3)]^3$ is its SU(2) doublet nature, the factor 3 in $\mathrm{U}(1)[\mathrm{SU}(2)]^2$ its colour-triplet nature), with $A(\mathbf{3}) = T(\mathbf{3}) = \frac{1}{2}$ and the charges of Proposition 1.

This is the hardest consistency condition a chiral gauge theory has, and it is satisfied identically by the module content. It is the strongest single result of the theory, and it is independent of the coupling: the cancellation is a property of the algebra, not of a tuned parameter.

The mass term. A grade-1 element of the algebra anticommutes with w, so it is off-diagonal in the $16/\overline{16}$ split: it maps $16 \to \overline{16}$ and $\overline{16} \to 16$, and both diagonal blocks vanish. A scalar in the **10** of Spin(10) therefore produces a Dirac mass term $16\,\mathbf{10}_H\,\overline{16}$ and no Majorana mass for the charged fermions. The singlet ν^c is the built-in source of a seesaw (Section 10.3).

Two caveats. A vev in the **10** has stabiliser SO(9): it breaks $\mathrm{Spin}(10) \to \mathrm{SO}(9)$, and SO(9) is not the Standard Model group. The chain is completed instead by the Wilson direction of Section 6, which at its physical root

leaves exactly $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ and breaks precisely the leptoquark roots; the two extra U(1)'s must still be broken or hidden. And the **10** contains no Standard Model singlet, so the vev that generates masses necessarily breaks the unified group further. This is open problem O6.

The two singlet sectors. The cell carries, beside the Clifford module, the two neutral singlet sectors A and B of Axiom 1 — four levels in all, each sector a conjugate pair $\{A_\pm\}$, $\{B_\pm\}$ at phases $\pm\mathrm{loop}^n$: a level invariant under the full Clifford algebra is not chiral, so it comes with its conjugate, and the pair is what keeps the spectrum $\pm$-paired, $\det = 1$ and the quasi-energy sum zero on the whole cell. They are singlets under everything: quantum numbers $(\mathbf{1}, \mathbf{1}, 0)$ under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, $B - L = 0$, and every one of the 45 plane generators acts on them as zero. Their masses are ladder rungs,

$$m_A = M_P\,\mathrm{loop}^3 = 9.6 \times 10^{13}\ \mathrm{GeV}, \qquad m_B = M_P\,\mathrm{loop}^{10} = 118.57\ \mathrm{GeV},$$

with rung assignment $n_A = 3$, $n_B = 10$ — the same provenance class as the ladder rows of Table 4 (Section 10.4). The internal $\mathbb{Z}_2$ that is odd on B alone — the **B-parity** — commutes with the step and with all 45 generators, so B cannot decay. The grading theorem states how far the exact gradings of the cell go: *every exact $\mathbb{Z}_2$ of the step that commutes with the* 45 *generators is a product of the global sign, the mirror grading, and the B-parity — the spinor module carries no fragment-selective parity, because the* **16** *is* Spin(10)*-irreducible.* B is the dark matter (Section 10.2) and A the seesaw partner (Section 10.3). Both carry $Y = 0$: they contribute zero to every coefficient of Table 2.

The three families. The **16** is *one* generation. The theory identifies the observed family number with the number 3 of spatial directions of the (1+3) subalgebra, equivalently $\binom{3}{2}$ spatial two-planes, equivalently the rank of the spatial metric: one number stated three ways, not three derivations. The identification has a structural hook: the twelve doubler nodes of Section 6 organise into **three families of four labeled by the zero axis**, one family per spatial direction, with identical one-step spectra, and the three copies of vectorlike matter required by unification are realizable by taking one node from each family at a single Wilson strength. The index 3 exists and *labels* the three copies; it does not force the count, and the forcing is closed as a negative theorem: over the node orbits under the full cubic group, the chiral junction stabilisers and the level-permutation structure of the step, no orbit of size 3 occurs (orbit sizes $12, 12, 8, 6, 2, 1$) — the count is a symmetry-breaking input (O3, Fig. 24).

The 45 **and the** 5/3**.** The 45 generators $h_a h_b$ $(a < b)$ close on the Spin(10) adjoint. They are even in the $\mathbb{Z}_2$ grading, hence commute with w and preserve the $16/\overline{16}$ split. Under $\mathrm{Spin}(10) \supset \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ the **45** decomposes as 12 Standard Model gauge bosons + 12 leptoquark X/Y bosons + 20 bosons in the $\mathbf{10} \oplus \overline{\mathbf{10}}$ of SU(5) + the $\mathrm{U}(1)_{B-L}$ generator; the 33 broken generators mediate proton decay (Section 11.4).

Theorem 3 (the 5/3**).** *In the* **5** *of* SU(5) *the hypercharge is* $Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, +\frac{1}{2}, +\frac{1}{2})$ *with* $\mathrm{Tr}\,Y^2 = \frac{5}{6}$*, while the non-Abelian generators obey* $\mathrm{Tr}\,T_a T_b = \frac{1}{2}\delta_{ab}$*. Requiring a single unified coupling g_5 for all generators gives $g_1^2/g'^2 = 5/3$, where g' is the Standard Model hypercharge coupling and $g_1 \equiv g_5$.*

Proof. $\mathrm{Tr}\,Y^2 = 3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{4} = \frac{5}{6}$. Rescaling $Y \to \sqrt{3/5}\,Y$ normalises it to $\frac{1}{2}$; then $g_5(\sqrt{3/5}Y) = g'Y$ gives $g' = g_5\sqrt{3/5}$.

This is an algebraic fact of the root: no parameter enters. It anchors the coupling discussion of Section 11.1, where the *normalisation* of the unified coupling proves to be under control while its *value* is not.

4.2 The stabiliser of the causal frame carries the gauge group and one coupling

The gauge group is not put in: it is the stabiliser of the frame the automaton calls spacetime, and the coupling normalisation comes with it. The 45 products $T_{ab} = \gamma_a\gamma_b$ $(a < b)$ of the ten generators γ_a with $\{\gamma_a, \gamma_b\} = 2\delta_{ab}\mathbb{1}$ close on $\mathfrak{so}(10)$ (closure defect 1.78×10^{-15}, Killing form $64\mathbb{1}$ at rank 45), and they carry **one** trace form, $\mathrm{Tr}(T_a T_b) = -32\,\delta_{ab}$, to zero defect: Lorentz, gauge and coset generators share a single normalisation, so there is no separate gauge coupling to choose. The Clifford pairing of the causal span *is* the background metric, $\frac{1}{32}\mathrm{Tr}\,\frac{1}{2}\{\gamma_\mu, \gamma_\nu\} = \mathrm{diag}(1, -1, -1, -1)$ exactly. The stabiliser of that 4-plane inside the 45 is $\mathrm{spin}(4) \oplus \mathrm{spin}(6)$, dimension $21 = 6 + 15$, and the remaining 24 generators form a $\mathbb{Z}_2$-graded coset ($[\mathfrak{h}, \mathfrak{m}] \subseteq \mathfrak{m}$ and $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{h}$ to 1.11×10^{-15}): the coframe's home. Choosing which four directions are spacetime costs 30 parameters, all of

them group-orbit directions ($45 = 24$ choices of the plane $+3$ of the time direction in it $+3$ of the space frame $+15$ left fixed), so the split is a frame choice and not data.

The chiral Lorentz structure is computed rather than declared. With $\omega_4 = \gamma_0\gamma_1\gamma_2\gamma_3$ ($\omega_4^2 = -\mathbb{1}$ exactly, central on bivectors exactly), the projections $P_\pm B = \frac{1}{2}(B \mp iB\omega_4)$ give two 3-dimensional algebras whose commutators close with residual exactly 0 and commute with each other exactly: $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ inside $\mathrm{spin}(1,3)_{\mathbb{C}}$. Inside spin(6) the centraliser of a complex structure is 9-dimensional and closes to 4.89×10^{-15} — it is $\mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)$, with $\mathfrak{su}(3)$ of dimension 8 and the $\mathfrak{u}(1)$ generated by the complex structure itself, which lies in the bivector span to 2.22×10^{-16}. The gauge group is therefore $\mathrm{Spin}(4) \times \mathrm{Spin}(6)$ by construction, and the way the **16** sits under it is not an input either: the pseudoscalar splits the 32 into $16 \oplus 16$, all 21 stabiliser generators preserve the **16** to 1.11×10^{-16}, the $\mathfrak{su}(2)_{L,R}$ Casimir spectra on it are $\{0, \frac{3}{4}\}$, and its $B - L$ spectrum, quantised by its own largest eigenvalue, is exactly $(+\frac{1}{3})^6(-1)^2(-\frac{1}{3})^6(+1)^2$ — the quark/lepton pattern, with nothing assigned. So the hypercharge $Y = T_{3R} + (B-L)/2$ is a sum of two derived subalgebra generators, and its GUT normalisation follows as a trace ratio,

$$\frac{\mathrm{Tr}_{\mathbf{16}} Y^2}{\mathrm{Tr}_{\mathbf{16}} T_{3L}^2} = \frac{2 + \frac{1}{4} \cdot \frac{16}{3}}{2} = \frac{5}{3},$$

with $\mathrm{Tr}_{\mathbf{16}} Y^2 = 3.3333$, $\mathrm{Tr}_{\mathbf{16}} T_{3R}^2 = \mathrm{Tr}_{\mathbf{16}} T_{3L}^2 = 2$, and a cross term below 10^{-12}.

---

5. The step: unitarity, spectrum, causality

5.1 The update and its invariants

The step has six stages, and every stage is the same two layers: the internal rotation applied to every sub-site, then one bond stage. The internal factor is built from the Spin(10) plane-rotation generators $T_{ab} = i\, h_a h_b$ ($a < b$) in the 32-dimensional module — each T_{ab} Hermitian, squaring to I, traceless (sixteen $+1$ and sixteen -1 eigenvalues):

$$U(g) \;=\; \prod_{a<b} \Big(\cos g\, I - i \sin g\, T_{ab} \Big).$$

A sub-site is labelled by $e = (e_1, e_2, e_3) \in \{0,1\}^3$, the bit of axis j being 2^j. A **bond stage** acts on the four disjoint pairs $(e,\, e \oplus 2^j)$: an *intra-block* stage takes the pairs with $e_j = 0$, inside the cell; an *inter-block* stage takes the pairs joining $e_j = 1$ of one cell to $e_j = 0$ of the next cell along $\hat{e}_j$. On each pair the gate is the swap-rotation $\cos g\, I - i \sin g\, (|e\rangle\langle o| + |o\rangle\langle e|)$, carrying the same single angle and no other parameter. The six stages run in the fixed order (intra x, inter y, intra z, inter x, intra y, inter z). One step is therefore

$$M \;=\; \prod_{s=1}^{6} \big(P_s \otimes U(g) \big),$$

an operator on the cell's on-site space $\mathbb{C}^{288}$ (P_s on the eight sub-sites, U on the 36-dimensional cell module, with the ladder phases $e^{\pm i\, \mathrm{loop}^3}$, $e^{\pm i\, \mathrm{loop}^{10}}$ on the singlet pairs). At block crystal momentum q an inter-block pair carries the phase $e^{\pm iq}$, $M(q)$ is 288×288, and on the 16 sector it is 128×128.

Every spinor factor is special unitary: the 45 internal factors have $\det = \exp(-ig\, \mathrm{Tr}\, T_{ab}) = 1$ on S, and each bond pair block $\left(\begin{smallmatrix} c & -is \\ -is & c \end{smallmatrix}\right)$ has determinant $c^2 + s^2 = 1$ (the inter-block phases cancel). Hence

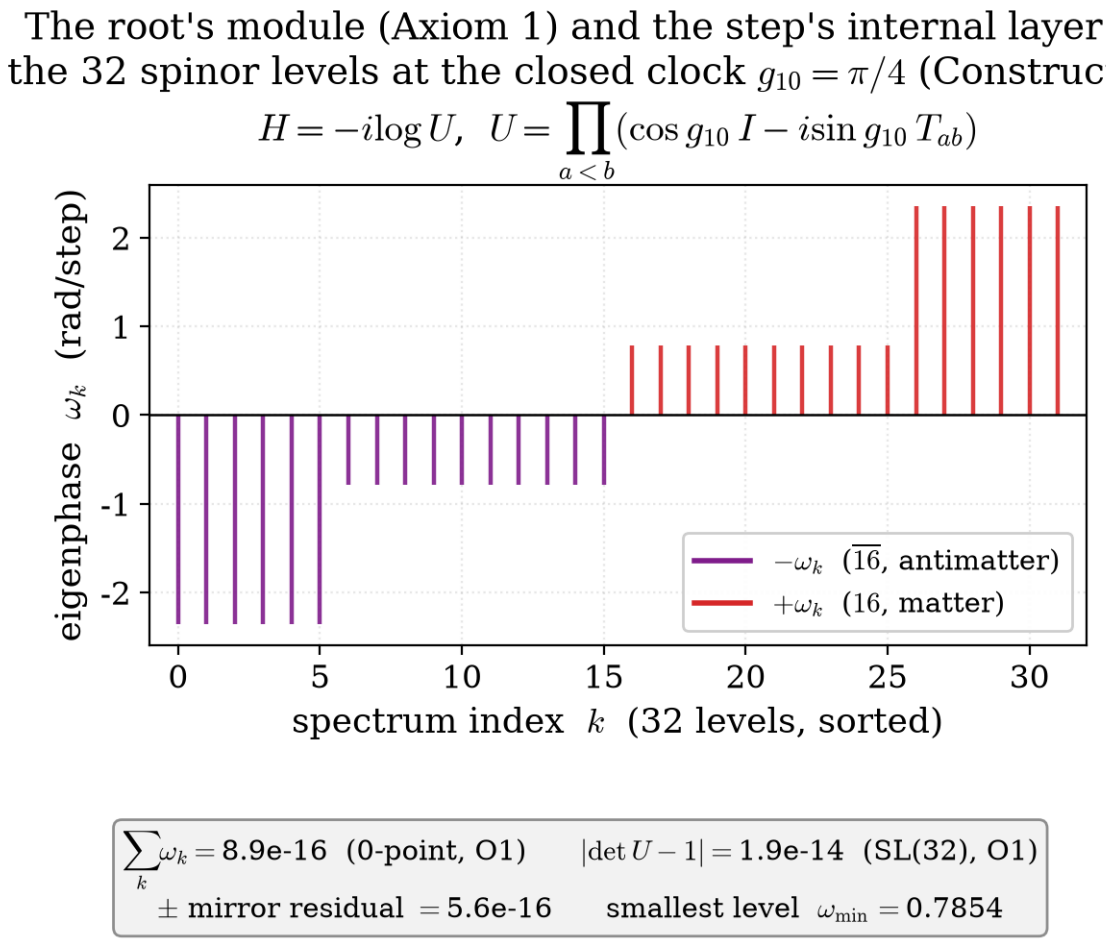
$$UU^\dagger = I, \quad \det U = 1, \qquad MM^\dagger = I, \quad \det M = 1 \qquad \text{for every } g,$$

on the spinor sector, and on the full cell as well: the four singlet levels come in conjugate pairs $e^{\pm i\varphi}$, so their phases multiply to 1,

$$\det \tilde{U} = 1, \qquad \det \tilde{M} = 1, \qquad \text{and} \qquad \sum_k \omega_k = \arg \det \tilde{U} = 0 \;\; \text{on all 36 levels :}$$

the ±-pairing, the unit determinant and the vanishing 0-point hold on the whole cell, singlets included, and the cell step is exactly unitary with the B-parity an exact symmetry of it. Unitarity on the construction: internal 1.3×10^{-15} with $|\det U - 1| = 1.9 \times 10^{-14}$; full cell $\|MM^\dagger - I\| \le 9.1 \times 10^{-15}$ and $|\det M - 1| \le 4.0 \times 10^{-13}$ at the zone centre, at a generic momentum and at the two high-symmetry points; the real-space six-stage step preserves the norm of a random cell state to 10^{-15}. Two structural inputs must be named: the 45 factors do not commute, so U depends on the *ordering* of the product, and the six stages have a fixed order. The algebra does not select either order; Appendix A fixes the convention.

The angle g is the one continuous number of the dynamics, and the closure axiom settles its closed value: only $g = \pi/2$ and $g = \pi/4$ make the step an element of the root, the half-angle branch degenerates to a pure translation, and the closed clock is the order-8 root $g = \pi/4$ (Theorem 6, Section 5.2). The empirical ladder of Sections 10–12 is read at the declared reading clock $g = 1/5$: the two clocks are distinct by theorem (Section 5.3), and the reading clock $g^2 = 1/25$ also anchors the desert meeting (Section 11.1) and the dispersive member of the Wilson family of Section 6. The root clock gives $\alpha = \pi/64 = 0.0491$ in the standard normalisation (Fig. 3).



The 32 *spinor eigenphases of* U *at the root clock* $g = \pi/4$*: the levels sit at the two algebraic values* $\pm\pi/4$ *(*10 *each) and* $\pm 3\pi/4$ *(*6 *each). The* 16 *matter levels and the* 16 *antimatter levels are exactly* ±*-paired; the* 0*-point* $\sum_k \omega_k = \arg\det U$ *vanishes and* $\det U = 1$*. The cell's singlet pairs add four levels at the ladder phases* $\pm\text{loop}^3$ *and* $\pm\text{loop}^{10}$*, so the pairing, the determinant and the vanishing* 0*-point hold on the full* 36*-level cell. Every spinor level is at or above* $\pi/4$ *—* $\omega M_P = 9.589 \times 10^{18}$ *GeV at the smallest — and no spinor level sits near the electroweak scale: the sub-Planck scales of the cell are the two singlet rungs.*

Theorem 4 (the matter/antimatter spectral mirror). *Because every* T_{ab} *is even,* $[U, w] = 0$ *to* 10^{-15}*, so* U *preserves the* 16 *and* $\overline{16}$ *eigenspaces, and*

$$\omega_{\overline{16}} = -\omega_{16}, \qquad \operatorname{spec}\big(U|_{\overline{16}}\big) = \operatorname{spec}\big(U|_{16}^{\dagger}\big),$$

verified to 2×10^{-15} *over the grid* $g \in \{0.05, \ldots, 0.5\}$.

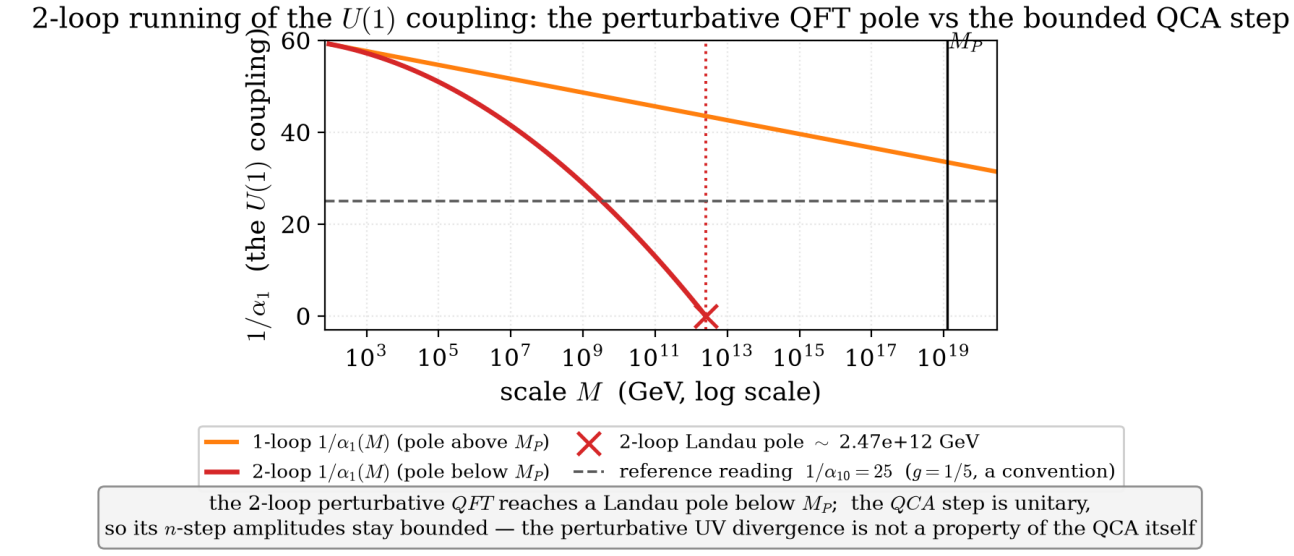
This is the honest content of "CPT" in this theory: a **spectral** statement — the antiparticle levels are the negatives of the particle levels. It is not the full CPT theorem: no antiunitary time-reversal operator is constructed on the many-cell Hilbert space and no spin-statistics connection is derived. A unitary involution $C = h_1$ maps the **16** onto the $\overline{\mathbf{16}}$ (it anticommutes with w), but calling $C \cdot T$ "CPT" overstates what has been built.

The mechanism of the pairing is understood, and its scope is limited. The intertwiner between the 32-dimensional module and its conjugate is unique (nullity 1, rank 32); it satisfies $Qh_aQ^\dagger = h_a^*$ to 2×10^{-15} for the ten generators and $QT_{ab}Q^\dagger = -T_{ab}^*$ to the same bound for the 45, and the associated antiunitary $\mathcal{J}$ has $\mathcal{J}^2 = +1$. That is the content of $\mathrm{Cl}(10,0) \simeq M_{32}(\mathbb{R})$: Spin(10) lands in SO(32), and a real form of U(32) carries no CP-odd invariant. Over all eleven signatures $p+q=10$ the Frobenius–Schur indicator is $+1$ for $p-q \equiv 0, 2 \bmod 8$ and -1 for $p-q \equiv 4, 6$, so $\mathrm{Cl}(8,2)$, $\mathrm{Cl}(7,3)$, $\mathrm{Cl}(4,6)$, $\mathrm{Cl}(3,7)$, $\mathrm{Cl}(0,10)$ are quaternionic — and on the quaternionic root the 21 mixed even products generate an exactly unitary step (6.7×10^{-16}, $\det U = 1$) whose spectrum is still exactly conjugation-closed (3×10^{-15}): the mirror is not a privilege of reality. Two scope limits come with it: $\mathcal{J}$ reverses the order of the ordered product, so $\mathcal{J}U\mathcal{J}^{-1} \neq U^{-1}$ — the pairing is a property of the whole ordered step — and the pairing is an **internal** statement: the implemented 128-level one-step spectrum at generic block momentum is not conjugation-closed at all (deviation 0.95–0.99), while the internal 32 spectrum is exact (5×10^{-15}).

Theorem 5 (the causal radius is implemented). *On an* (L_b, L_b, L_b) *torus of cells the step is a product of two-cell operators: every bond stage couples sub-sites at Planck-cell distance* ≤ 1. *Hence after* n *stages the support of a sub-site-localised excitation lies inside the* n*-cell light cone: at most one Planck cell per stage, and since three of the six stages are inter-block, at most three blocks — six Planck cells — per step. Computed on the construction: on an* 8^3*-cell torus the support radius after* n *stages is exactly* n *Planck cells for every* $n \leq 12$ *— the front saturates the cone, front speed exactly* 1*; on a* 32^3*-cell torus run for* 30 *stages no site of the support and no amplitude ever appears outside the* n*-cell cone (*0*, to machine precision); the same cone is realised at* $g = 0.3$ *as at* $1/5$*; the internal layer alone, with the bond stages removed, keeps the excitation on one sub-site forever.*

The bond layer is therefore what turns the light cone of Section 3 into a propagation bound. This is the cleanest dynamical result of the theory.

The n-particle sector. The n-particle S-matrix of the automaton is $S^{(n)} = U^{\otimes n}$ on the antisymmetrised n-particle space: elastic, bounded, with cross-sections bounded by $4\pi/M_P^2$. There is no inelastic channel at the one-cell level, and no renormalisation of the step: the coarse-graining tower U^{2^p} $(p = 1 \ldots 8)$ stays unitary and bounded at every level, so the *discrete* flow has no Landau pole — in contrast with the perturbative field-theory running of the same couplings, which does (Section 11.1). That contrast is real but modest: a unitary finite-dimensional map is bounded by construction, and boundedness alone does not reproduce the observed running (Fig. 4).



The two-loop perturbative running of the hypercharge coupling, from the measured M_Z *input, reaches a Landau*

pole at 2.473×10^{12} *GeV, below the cutoff; the one-loop curve from the same input has its pole at* 2.196×10^{41} *GeV, far above it — run up instead from the root anchor at* M_P*, the one-loop pole is* 5.314×10^{35} *GeV. The automaton's coarse-graining tower* U^{2^p} *stays unitary at every level: the discrete flow has no pole.*

5.2 The quantized clock: the root closes its own parameter

There is a sharper condition on g than any fit, and the algebra itself enforces it. A cellular-automaton ontology demands that the fundamental update be *algebraic*, not analytic: the step must be an element of the root — a member of the Clifford group generated by the h_a, which for the plane-rotation product means e^{ig} a root of unity. Call this the **closure axiom**: no continuous phase may enter the dynamics.

Theorem 6 (the two closed steps). *Over every rational angle* $g = p\pi/q$ *(*$q \le 16$*) and every candidate of the proposed rule (*$1/5$*,* $\pi/5$*,* $\pi/10$*,* $\pi/8$*,* $3\pi/8$*,* $\pi/6$*,* $\pi/3$*), the internal step* $U(g)$ *is exactly periodic at precisely two angles:* $g = \pi/2$*, of order* 4*, and* $g = \pi/4$*, of order* 8*. At* $g = 1/5$ *the internal clock is irrational: the smallest level is* 0.011466 *rad per step,* $2\pi/\omega_{\min} = 548.0$*, and no order* ≤ 64 *exists (deviation* 0.45*).*

Proof of the $\pi/2$ *case, and a stronger statement.* Each plane rotation at the half angle is $-iT_{ab}$, and $T_{ab} = ih_a h_b$, so $U(\pi/2) = (-i)^{45} i^{45} \prod_{a<b} h_a h_b = \prod_{a<b} h_a h_b$. The product of all 45 pairs in ascending order reduces — numerically to 6.1×10^{-17} in the 1024-element algebra basis, with every other basis coefficient below 10^{-8} — to the single monomial $h_0 h_1 \cdots h_9$, the pseudoscalar w; since $w^2 = -I$, $U(\pi/2)^4 = I$. The identity is *ordering-independent*: the monomial content of any ordering of the 45 factors is $\pm w$, because each h_k appears in exactly nine pairs. The step at the half angle *is* the root's chirality operator.

The quantized step ($g = \pi/4$**).** $U(\pi/4)^4 = -I$ and $U(\pi/4)^8 = I$ exactly (5.0×10^{-15}), and the 32 internal levels are exactly the algebraic set

$$\omega \in \{\pm\tfrac{\pi}{4}\ (10),\ \pm\tfrac{3\pi}{4}\ (6)\}, \qquad \max_k \left|\omega_k - k\pi/4\right| < 3 \times 10^{-15},$$

$\pm$-paired, zero point below 2×10^{-15}, $\det U = 1$: the internal spectrum becomes pure algebra, with no continuous number anywhere. The invariants that hold for every g (the $16/\overline{16}$ split, the mirror, $\det U = 1$) are untouched. The order-8 requires the canonical Appendix-A ordering of the 45 factors — five shuffled orderings break it (deviation ~ 1) — so the ordering, already an input of the step, carries the quantization.

The full step closes too. At $g = \pi/4$ the exact one-step Bloch matrix of the 16 sector at the zone centre satisfies $M(0)^2 = I$ below 1×10^{-14}: the fundamental dynamics is a finite-state machine, 't Hooft's determinism exact at the fundamental update. The doubler defect of Section 6 is repaired, but the repaired census is not a set of points: at $g = \pi/4$ the exactly gapless set of the block zone is Γ together with the **six face-diagonal lines** $\{(\pm t, \pm t, 0)\}$ and its axis permutations — every point of every line is gapless below 6.7×10^{-16} over its whole length, the twelve M points are only the high-symmetry points that happen to sit on them (their gaps 0–4.4×10^{-16}), X and R stay gapped at 0.5236 — and the step still factorises there as $M(q) = P(q) \otimes U_{16}^6$ (3.6×10^{-15}). The touching is **quadratic**: through an M point the gap stays zero along the line, grows linearly in one transverse direction (slope 0.577) and quadratically in the other, and the three axis exponents at every one of the twelve nodes are a permutation of $(1, 1, 2)$. The per-node chirality bookkeeping then fails as stated — it needs isolated zeros — and the Nielsen–Ninomiya count must be redone per line, not per node. The helicity question at the zone centre answers as it did at an M node: the exactly gapless subspace of $M(0)$ has dimension 64 of 128, the internal **16** is inert, and the content is the 4-dimensional bond subspace, which the character inner product over the 48 signed permutations gives as $A_{1g} \oplus T_{2g}$ — no T_1 and no E_g, hence no helicity ± 1 or ± 2 candidate at Γ.

The closure set of the decoupled family is an algebraic four-point set — and it fails at every empirical clock. On the two-dial family $M(q_0; \beta, g) = P(q_0; \beta)\, U_{16}(g)^6$, which decouples the bond hop from the couplings and has the published step as its diagonal slice, the closure set $C = \{(\beta, g) : M = I\}$ is cut out exactly by two trigonometric polynomials: two conditions, four algebraic points — and the clock $(\frac{1}{8}, \frac{1}{5})$ is not among them (closure residual 1.05×10^{-2} at the clock; the zero set misses the $g = \frac{1}{5}$ line and the $\frac{1}{5}$–$\frac{1}{4}$ band at 0.0596, 0.1121). The clock is a family input, not a family consequence. The verdicts of this section are theorems of the whole construction, not of one lattice: the family adds no fitted dial, and all three decided falsifiers stand on it — F8 (no closed pair carries a gapless chiral cone at $c = 1$; the closest is the published point, cone degree 0.281),

F2 (no closed clock restores the clock/coupling contacts; β enters the running nowhere), F9 (rank-2 isotropy is symmetry-forced only at the star-module zone centre, while the enforced multiplet disperses at $\delta v/v = 0.997$, a factor 1.2×10^{14} past the SME ceiling).

The half-angle branch. At $g = \pi/2$ the six-stage bond matrix is diagonal to 1.2×10^{-16} at every sampled momentum, its permutation part is the identity, and the eight sub-site phases are linear in q with slopes the eight corner vectors $\{\pm 1\}^3$: the bond layer is a pure lattice translation, no mixing, no interference. The bands are exactly linear with slope ± 1 along every axis, $M(0) = I$ on the whole 128-dimensional internal space, and the gapless set is a wide line network (513 gapless points on a 13^3 grid, 7.0× the $\pi/4$ figure, against the 0 of $g = 1/5$). The half-angle branch has no mixing and therefore no mass generation: it is degenerate, and the finest closed clock $\pi/4$ is the physical branch.

What the closure costs. The loop machinery quantizes with the clock. The traced-link hop phases of Section 10.1 — built from the joint Casimir eigenvalues $-32, -12, +4$ — are exactly $\{1, i, i\}$ at the root clock $g = \pi/4$ (below 3×10^{-15}), and the branch carries an exact selection rule: the odd closed loops cancel identically, in integer arithmetic. The word norms are angle-independent (suppression $\rho = 0.978030$ per step); the ratio of the two clock-filter classes separates there — on $n = 4k$ the decay is geometric at $\rho^4 = 0.91497$ per $+4$ links and crosses $1/(16\pi)$ first at $n^* = 208$, with interference deficit 0.98987 certified by continuity at fifty digits, while on $n = 4k + 2$ both norms vanish at the closure angle and the ratio has no limit: at the root clock the ladder-length statement is a 208-rung statement, and the hierarchy mechanism of Section 10.1 is measured with this rule.

The coupling accounting is done in one normalisation: at the root clock the rotation-squared coupling is $\pi^2/16$, i.e. $\alpha = \pi/64 = 0.0491$ in the standard normalisation $\alpha = g^2/4\pi$, while the desert solution with the same content meets at 0.040050 — the root value sits a factor 1.226 **above** the meeting value, the named tension of O4 (Section 11.1). The reading $\alpha_{10} \equiv g^2 = 1/25$ is a convention that hides the factor; it is not a number of the standard running.

The desert inverse problem at the quantized value with the same content still has a unique one-loop solution — $M_X = 1.235 \times 10^{17}$ GeV, $M_{10} = 7.06 \times 10^4$ GeV, $M_{(1,2)} = 4.86 \times 10^{17}$ GeV — but it is not a *desert* solution: the $(1, 2)$ threshold sits **above** the unification scale, the threshold ordering the solution assumed is destroyed, and the proton lifetime jumps to $\tau_p = 2.77 \times 10^{39}$ yr. The size of the repair the content would need is measured: restoring the ordering would shift the $(1, 2)$ threshold by 13.1 e-folds — 1.12× the entire log-spacing from the published $M_{(1,2)} = 1.0 \times 10^{12}$ GeV to M_X — so **no one-loop threshold correction within the spectrum's own spacings restores the meeting**: F2's unification side fires. The driving coefficients are exact integers of the root charges ($k_1 = 3/5$, $\Delta b_{\mathbf{10}\oplus\overline{\mathbf{10}}} = (6/5,\ 2,\ 2)$; complete $\mathbf{16} \oplus \overline{\mathbf{16}}$ shifts are proportional, $(8/3,\ 8/3,\ 8/3)$): the no-go is the model's own content, not a scan artefact. The converse channel is open, not fired: no threshold of the desert spectrum is experimentally excluded — the colour pairs sit 14× above the strongest LHC colour probe [68], pair production is kinematically closed at the LHC, the $(1, 2)$ doublet is $10^9\times$ above its electroweak-precision ceiling [69], and τ_p is $1.2 \times 10^5\times$ past Super-Kamiokande (F10). Conversely, holding M_X at the data-preferred 1.8×10^{16} GeV requires $\alpha_X = 0.040050$: the meeting coupling is rigidly tied to M_X by that content. Against the one-loop target — the three couplings run to M_P, mean 0.023068 — the closed value has relative gap 1.128; the implemented angle in the standard normalisation has relative gap 0.862, so closure moves the relative gap by a factor 1.31 while moving the coupling from 12.6× below to 1.23× above the meeting value — a factor 2.6 **closer** in log-distance. The closed value sits 2.41× above the data-optimal $1/7^2 = 0.0204$ and is not of the $1/n^2$ form at all ($\pi/64 = 1/n^2$ needs $n = 4.51$; the nearest integers miss by 18.5% and 27.3%).

The verdict on the parameter criterion. The closure axiom replaces the one free continuous parameter by a discrete index: the root forces $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates, and the finest closed clock is the physical branch. A discrete index of the root is not a parameter in the sense of Section 13 — the family number 3 is already an index of the same kind. What the branch costs is empirical and measured: the loop machinery crosses $1/(16\pi)$ at length $n^* = 208$ rather than 10, the hierarchy coefficient reads 2.710 against 2.0766 observed — the named +30% clock residual (Section 10.1) — while the observed value is read by the non-closed rung-10 clock 1/5, itself a measured input standing at 0.1999980, 2.0×10^{-6} off the fifth; the rung pair $(3, 10)$ is itself a declared convention (Section 10.1); and the desert threshold ordering is destroyed and not restorable by the spectrum's own thresholds ($\tau_p = 2.77 \times 10^{39}$ yr). What the closure pays back is structural: an on-site dynamics with **no free continuous angle** — a finite algebraic spectrum, a fundamental step of finite order, and the point-doubler defect replaced by an exactly gapless nodal-line network. The claim is scoped: the junction amplitude r (Section 10.3), the split-node set (O9) and the bond-layer couplings κ_s (Section 11.1)

remain continuous inputs of the *extended* construction, and Section 13 counts them. The falsifiers F1, F3–F7 are angle-independent: anomalies, spectral mirror, $\det U$ and the causal radius hold identically on the branch.

5.3 No closed clock can read the data: clock transcendence

The separation of the two clocks is not a gap left open; it is a theorem of the class.

(Clock-closure rigidity.) Let $M(g)$ be any product of the construction's clock-step factors at Γ, any stage ordering, any integer clock multipliers. The bond stages at Γ are $\cos(v)\, I + \sin(v)\, K$ with K a Gaussian-integer matrix, and the internal factor is frozen at the root-clock angle with algebraic entries; so every entry of every $M(g)^k$ is a Laurent polynomial in $z = e^{ig}$ with algebraic coefficients. If $M(g_0)^k = \pm I$ at a nonzero **algebraic** clock g_0 for some $k \geq 1$, then $M(g)^k \equiv \pm I$ identically in g — for Hermite–Lindemann makes $z_0 = e^{ig_0}$ transcendental over the algebraic numbers, and a Laurent polynomial with algebraic coefficients that vanishes at a transcendental point vanishes identically. No member of this class can therefore have its closure **solved** by an algebraic clock: the only closures at algebraic clocks are clock-identity closures, blind to the clock — and the census finds exactly those and nothing else: of the 4096 stage-multiplier members of the six-stage circuit exactly 64 are clock-identity steps (the mirror-antisymmetric members $c_{j+3} = -c_j$, constant at $M \equiv e^{i\varphi} I$, closure order 2); they close at every one of the 74 rational clocks with $q \leq 12$ and at all 12 transcendental test clocks $m/(32\pi)$, $m = 1, 3, \ldots, 23$, and no clock — rational or transcendental — has added a single closure beyond them. A closure that **selects** a clock forces e^{ig} algebraic — a root of unity, hence a π-rational clock: the closure axiom is not a choice, it is the forced form of clock-selecting closure in this algebra, and the reading clock 0.1999980, nonzero algebraic to 2.0×10^{-6}, supports none of them. The escape from O4 and O8 can therefore not be a closure axiom of this exponential form: the clock must enter the **coefficients** — finite-order rational parametrizations being crystallographically confined to orders $\{1, 2, 3, 4, 6\}$ — or the coefficients must turn **transcendental**, and the measured crossing condition $16\pi F_{10}(g) = 1$ read at 0.20013 is already of that transcendental-coefficient kind. Scope: the theorem is stated at Γ (nonzero momenta carry the hop factors $e^{\pm iq_j}$, transcendental constants outside Lindemann's reach) and for the $\pm I$ target; its arithmetic premises are verified to 10^{-13}.

(The clock has no name.) The live branch of that alternative — the clock entering through the ladder's own arithmetic — has its bounded search closed. All 365 crossing roots of the ladder's rung-arithmetic equations ($|F_n| = 1/(16\pi)$ for $n \in \{3..24, 170, 208\}$, and the rung self-consistency equations $|F_a| = (16\pi)^{-k}|F_b|$, $k \leq 3$) on $[0.05, 1.55]$ are named against the natural dictionary class — continued fractions to denominator 10^6, 16π-forms to 10^4 at 50 digits, integer-power phase dictionaries, PSLQ degree ≤ 6 height ≤ 20: 22 roots name (Diophantine coincidences of the dictionary class, closest 6.37×10^{-14}), zero carry a polynomial of the PSLQ box, and none reads the coefficient: the $\pm 10^{-4}$ hierarchy band is held among the 365 by the measured clock 0.1999980 alone (at 4.96×10^{-14}, the machinery identity itself) — and that clock is itself unnamed, its best rational 19884/99421 missing by 4.95×10^{-12}. The measured clock is the only coefficient-reading root of the ladder's own equations, and it has no name.

(No dictionary bridges closure and reading.) The reading set of the coefficient on the rational-clock grid is 12 razor windows of width 2.41×10^{-9} to 1.45×10^{-7}, g_{obs} lies in the fourth, and no grid closure angle lies in any window (nearest separation 4.6×10^{-4}, thousands of window-widths); the closest closed-clock rung-10 read over all 43 closing grid angles and every admissible dictionary rescaling is 5652.1 band-units off. The bridge question is invariant under dictionary reparameterization, and the census of that class is empty.

(The repair classes are closed.) Over the admissible quantizer classes — 64 per-stage internal-decoration assignments at the named amplitude and all 720 stage orderings of the six-stage product, 784 variants in all — no variant carries a closure clock whose rung-10 coefficient lands in band (QUANTIZER-NEGATIVE). Over the admissible weight families — the named eta-law values, the class-function multipliers over the 63 even phase classes, and all 220 node-set channel triples at the reading clock — no reweighting of the traced-word sum moves both rung contacts at once: the 12 channel weights of the traced word are exactly degenerate, so no node subset moves them. Over the four measured selector functionals and the three unselected sites (lightcone family member, desert couplings, channel triple), no functional uniquely selects at more than one site (SELECTION-DISPERSED). The ranking of zero-fitted compound theories against the five contacts (hierarchy coefficient, neutrino scale, desert coupling, graviton channel, node anisotropy) bottoms at normalized distance 1.205; re-scored with the graded supplier slots the closest distance drops to 0.2433, with that row's coefficient contact passing only through a weight class disqualified by its chance-density — recorded as a bound, not a

discovery. The hierarchy coefficient is therefore bounded, from every admissible side, away from its observed value; the ladder base and the reading clock stand as declared inputs (Table 6), and O8 names the gap.

---

6. Doublers, chirality and the cone question

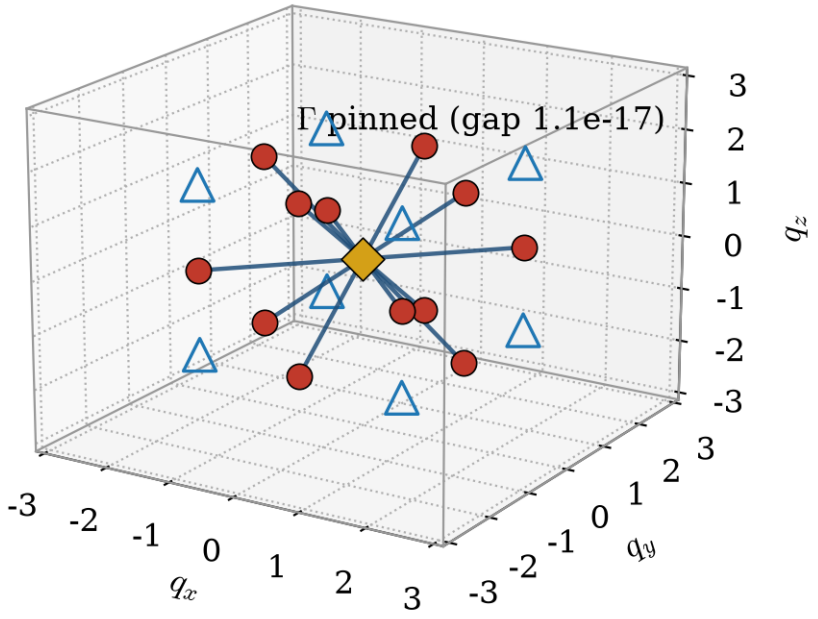
The Nielsen–Ninomiya theorem forbids a single chiral fermion in a theory with a *local, translation-invariant, Hermitian* lattice Dirac operator and a bounded spectrum [21]. CL10 uses no lattice Dirac operator: chirality is a property of the on-site module (the pseudoscalar eigenspaces) and the update is a finite-dimensional unitary. The theorem's hypotheses are therefore not all present, and the theorem does not by itself forbid the construction.

That is *not* the same as having evaded it, and the question is decided on the construction. The step is one time-independent operator on the cell lattice, so the good quantum number is the **block crystal momentum** q — the phase under a translation by one automaton cell, 2π-periodic, zone $q \in [-\pi, \pi]^3$ — and the eight sub-sites are the internal structure of the Bloch matrix. The 16-sector one-step matrix $M(q)$ is computed exactly at 128×128, cross-checked against the real-space step (agreement 10^{-14}). The cubic high-symmetry momenta carry the whole spectrum: $\Gamma = (0,0,0)$; X $= (\pm\frac{\pi}{2}, 0, 0)$, star 6; M $= (\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$, star 12, little group C_{2v}; R $= (\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, \pm\frac{\pi}{2})$, star 8, little group C_{3v}.

Proposition 2 (lattice chirality). *(i) The* 16 *sector is closed by the full step, not only by the internal layer:* the bond layer acts on the sub-site index and commutes with w, so a state initialised in the **16** leaves it by less than 10^{-15}. Chirality is a dynamical symmetry of the automaton. *(ii) The zone centre is fully pinned to the Floquet pair:* the 16-sector spectrum at $q = 0$ sits entirely at $\{0, \pi\}$ ($64 + 64$ of the 128 levels, $M(0)^2 = I$), so the would-be massless matter is **not lifted** at $q = 0$. *(iii) The twelve M points are exactly gapless:* the 16-sector one-step spectrum at the 12 M points is $\{0^{32}, \pm\frac{\pi}{3}{}^{16}, \pm\frac{2\pi}{3}{}^{16}, \pi^{32}\}$: 64 of the 128 levels pinned to $\{0, \pi\}$, minimal quasi-energy exactly 0, and the same spectrum at every one of the 12 nodes to 3×10^{-15}. The exactly gapless set of the block zone is Γ together with the six face-diagonal lines: gapless below 10^{-9} at every sampled line point, while sampled bulk momenta carry a real gap ($\geq 1.2 \times 10^{-2}$) — at the root clock the doubling is a line network, and the per-node chirality bookkeeping is replaced by the pinning.

The doubling therefore **survives** the automaton: "the **16** is a single chiral generation" is a property of the *internal module*, while the *lattice spectrum*, which is what couples to low-energy physics, carries an exactly gapless line network (Fig. 5).

At the root clock the exactly gapless set is Γ plus the six face-diagonal lines $\{(\pm t,\ \pm t, 0)\}$ and axis permutations: the 12 M endpoints are pinned to $\{0, \pi\}$ (64 of 128 levels, gap exactly 0), the 8 R corners carry exactly 3-fold multiplets



The 16*-sector one-step spectrum at the root clock: the zone centre fully pinned to the Floquet pair* $\{0, \pi\}$ *(*$64 + 64$ *of the* 128 *quasi-energies), every M spectrum* $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ *— exactly gapless,* 64 *of* 128 *pinned — and the gapless set a line network:* Γ *plus the six face-diagonal lines. The doubling is displayed, not hidden.*

6.1 The Wilson mechanism

The remedy is part of the construction. A Wilson-type term built from the root's own algebra — an internal factor

$$W(q) = \exp\Big[- i\, w \sum_j (1 - \cos q_j)\, \big(\cos\theta\, X + \sin\theta\, Y\big)\Big],$$

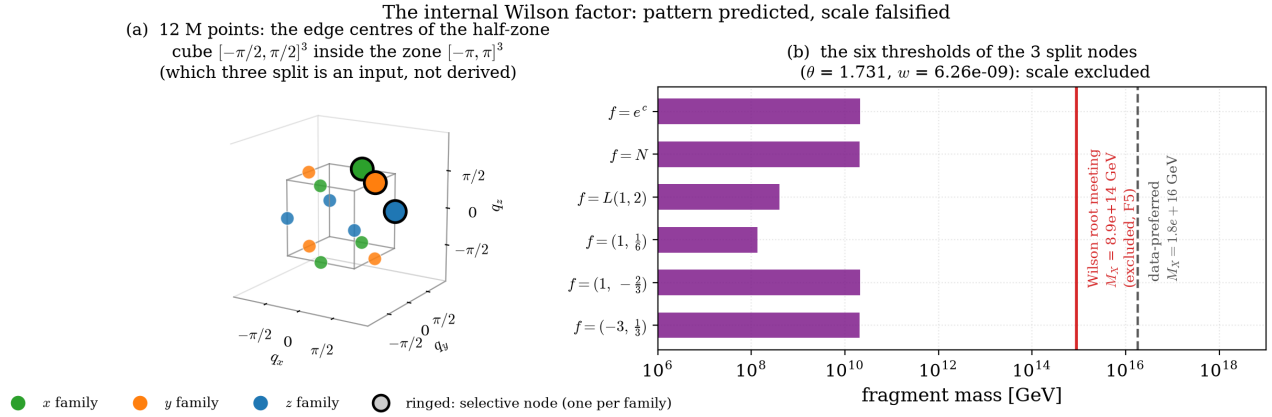
with X, Y the unification Cartan directions identified in Section 11.1 and q the block momentum, applied once per step — multiplies the one-step matrix without touching its unitarity ($W(0) = I$, $\det = 1$) and leaves the bond stages strictly causal. The factor itself is not local: its real-space kernel is a Bessel tail in **block** displacement — 6.7×10^{-2} at block radius ≥ 2 and 2.2×10^{-3} at ≥ 4 for $w = 0.1$, and 1.8×10^{-17} at ≥ 4 at the physical strength — so the Wilson factor replaces the light cone of Theorem 5 by a Lieb–Robinson bound [6]: beyond the bond cone the amplitude decays exponentially in block distance (3.1×10^{-9} at the Lieb–Robinson line $r = 1.2n + 6$, decay ratio ≤ 0.26 per block).

It gaps all twelve nodes — the minimum node gap rises from 2.0×10^{-4} to 5.7×10^{-3} ($w = 0.01$) through 1.5×10^{-2} ($w = 0.2$) — with fragment masses

$$M_f = w\, |\cos\theta\, X_f + \sin\theta\, Y_f|\, M_P/3;$$

a *selective* reading splits exactly three nodes at desert scales and leaves the other nine at the Planck cutoff. What remains input is the *choice* of the split node set: unification forces $n_{\text{nodes}} \geq 3$ but does not select which three (O3), and the meeting scale the mechanism then predicts is in conflict with proton decay (Section 11.1).

At the physical root $\theta = 1.731$ the Wilson direction completes the breaking chain: it leaves $\text{Spin}(10) \to \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)^3$, preserves the Model algebra (every unbroken root is intra-fragment; the 32 broken roots are exactly the leptoquark type), and the two extra U(1)'s are the only open step to the Standard Model group (Fig. 6).



(a) The twelve doubler nodes at the M points of the block Brillouin zone, coloured by axis family — three families of four labelled by the zero component; the three ringed nodes show the selective choice, one per family, realizable at a single Wilson strength. (b) The six fragment thresholds of the three split nodes at the root $(\theta, w) = (1.731, 6.26 \times 10^{-9})$*: the pattern is the vectorlike split* **16** *the renormalisation-group repair requires, but the meeting scale* $M_X = 8.9 \times 10^{14}$ *GeV sits below the Super-Kamiokande exclusion.*

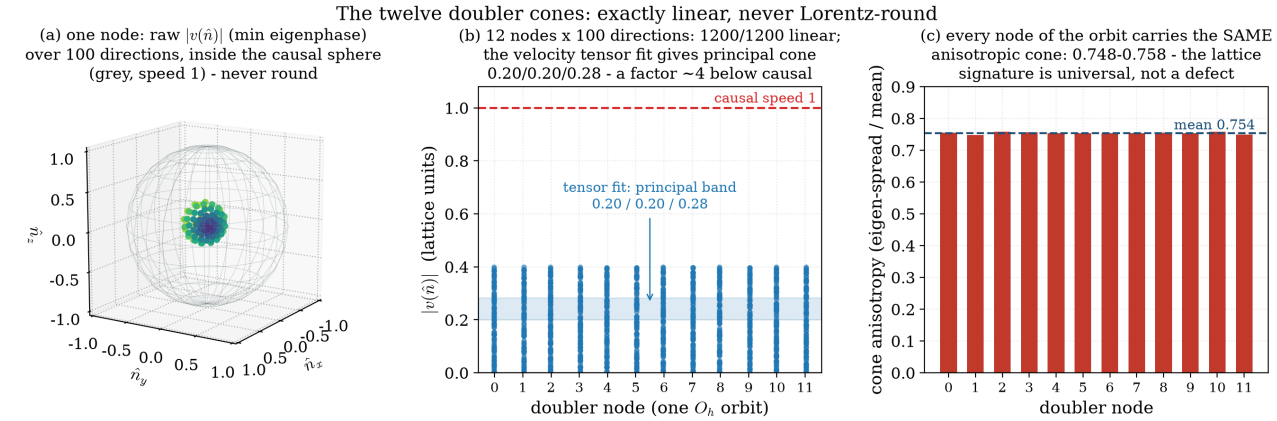
6.2 The Lorentz structure at the nodes

The causal front speed of Theorem 5 is exactly 1 Planck cell per stage; that is the statement the automaton proves. What the *low-energy spectrum* of the implemented step does not have is continuous Lorentz invariance.

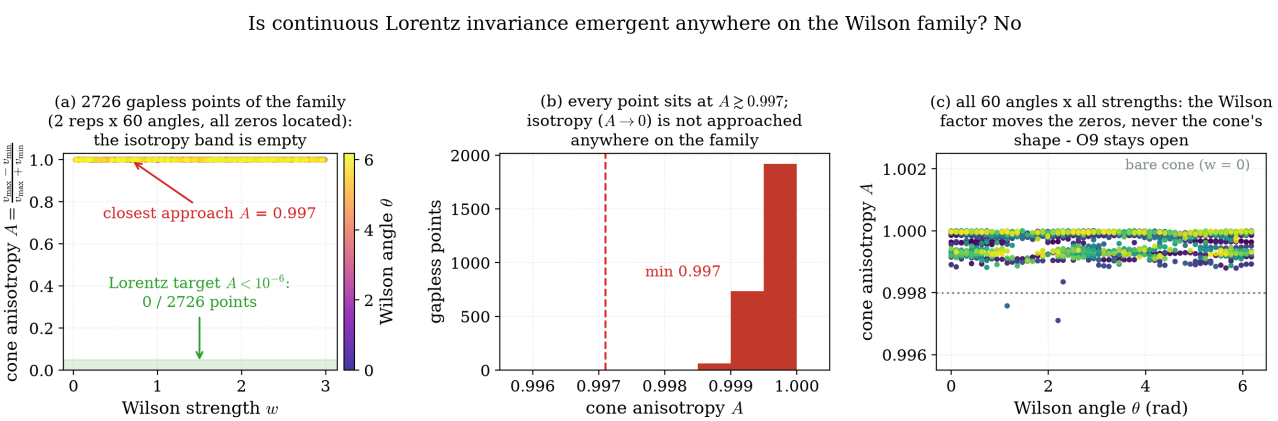
On the $g = 1/5$ member of the Wilson family, which carries the dispersive node spectrum (on the root clock the node spectra are pinned to $\{0, \pi\}$, Proposition 2): at all twelve nodes and in all 1200 sampled directions the dispersion is **linear** (local exponent $p = 1.00$, no quadratic band touching), but the cone is strongly **anisotropic**: principal velocities $(0.200, 0.200, 0.284)$ per unit block momentum — $(0.100, 0.100, 0.142)$ in Planck-cell units — anisotropy 0.748–0.758, the same at every node of the O_h orbit. The lattice signature is universal: the node gap vanishes at every zero of the family — 2726 gapless points — and at each the cone anisotropy $A = (v_{\max} - v_{\min})/(v_{\max} + v_{\min})$, measured over 179 sphere directions, leaves the Lorentz target $A < 10^{-6}$ empty: 0 of the 2726 points, closest approach $A = 0.997$.

Frame averaging over the internal frames is blind to the failure. Rebuilding the internal step in random SO(10) frames of the ten generators leaves the node gap 2.003×10^{-4}, the principal velocities $(0.200, 0.200, 0.284)$ and the anisotropy 0.7545 unchanged to 3×10^{-15}, and the tangent commutators at the noise floor ($\leq 1 \times 10^{-11}$) in every frame. The reason is structural: the momentum dependence lives entirely in the sub-site factor of the Kronecker bond layer, so no choice or averaging of the internal frame can move the cone.

The anisotropy is an experimental contact: read against the electron-sector SME velocity ceiling $\delta v/v \leq 8.0 \times 10^{-15}$ (Kostelecký–Russell data tables, Table D6, $|c_{ee}^{(TY)}|$ [59]), the measured tensor gives $\delta v/v = 0.296$ at the implemented clock, 1 at the closed clock (one axis exactly flat), 0.295 on the selective-reading Wilson cones at $\theta^* = 1.730784$, and 0.9999 on the star nodal line — the best of the named set, 0.292, is 3.7×10^{13} times the ceiling: **F9 is fired**, fourteen orders of magnitude past the bound at the most isotropic named node. The dressed electric gate does not move it: at the closed clock the gate is a scalar ($5\pi/4$, err 3×10^{-14}) and the step's six bond factors make the six node insertions multiply to $+I$, so the dressed spectrum is identical to the bare one. Nor does the deepest block renormalisation: at the $b = 16$ downfold of the dynamical self-energy the best read is $\delta v v/v = 0.0033$, still a ratio 4.1×10^{11} (Section 10.1) (Figs. 7, 8).



(a) The minimum-eigenphase velocity $|v(\hat{n})|$ over 100 *Fibonacci directions at one doubler node, inside the causal sphere of Theorem 5 (grey, speed* 1*): the cone is elongated, never round. (b) All twelve nodes* × 100 *directions:* 1200/1200 *linear; the velocity-tensor fit gives* 0.200/0.200/0.284 *per unit block momentum, a factor* ∼ 4 *below the causal speed. (c) The anisotropy is the same at every node of the orbit.*

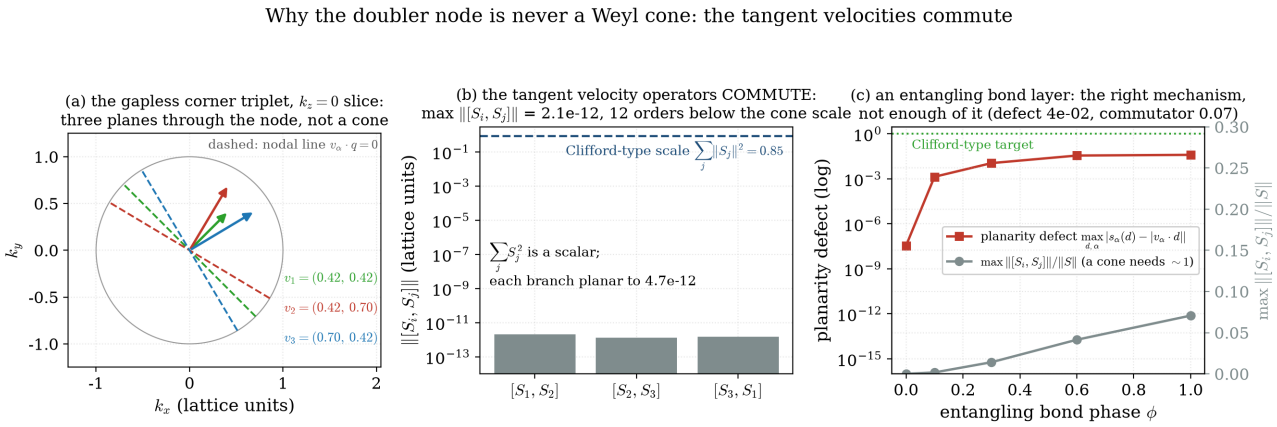


Continuous Lorentz invariance emerges at no gapless point of the Wilson family: 0 *of* 2726 *gapless points reach* $A < 10^{-6}$*; the distribution of A is pinned at* 0.997–1.000*; no angle θ and no strength w moves a cone toward isotropy.*

6.3 Nodal planes at the corners

The failure is structural, not a finite-radius artefact. At the tangent, the twelve doubler nodes carry speeds (0, 0.14189, 0.14189) along the three axis directions: one axis direction is exactly flat (the gap there is quadratic), the other two equal — the little group C_{2v} explains the equality. At the eight R points the full cubic symmetry is realised (exactly 3-fold multiplets, splitting $< 10^{-12}$), and the Wilson factor drives one triplet exactly gapless (10^{-10}, a one-parameter family): the Wilson family does supply massless cubic matter, and the touching there is linear — dM/dq_j is anti-Hermitian and $S_j = i\,dM/dq_j$ is nonzero (max speed 0.92). But the three tangent velocity operators commute ($[S_i, S_j] = 0$ to 3×10^{-12} against the scale $\sum_j \|S_j\|^2 = 0.85$), $\sum_j S_j^2$ is a scalar to 7×10^{-12}, and each branch speed is exactly $|v_\alpha \cdot q|$ to 6×10^{-12} with the v_α the cubic permutations of $(0.704, 0.421, 0.421)$: the node is a crossing of three **nodal planes**, and no tuning of the Wilson angle turns three planes into a cone. Both lowest corner triplets can be driven to zero **simultaneously** — at $w = 0.241747$, $\theta = 0.913137$ of the $g = 1/5$ member all six corner phases fall below 2.5×10^{-16}, with the next level at 8.7×10^{-3}, an isolated six-component massless point — and it is still not a cone: the three tangent operators again commute ($\max\|[S_i, S_j]\| = 5.3 \times 10^{-12}$ against $\sum_j \|S_j\|^2 = 0.82$), the six branches are exactly planar to 4.8×10^{-12} over 2196 directions, their six velocity vectors — permutations of $(-0.0434, -0.3272, -0.3272)$ and of $(0.1299, 0.1299, -0.1539)$, speeds 0.4647 and 0.2396 — define six distinct nodal planes, and on this six-state space $\sum_j S_j^2$ is not a scalar, its deviation from one (7.2×10^{-2} about the mean 0.137) being the branch splitting itself. A six-component massless point exists at the corner; it is Abelian, anisotropic, and carries no ±-paired helicity.

The cause is the Kronecker form of the bond layer, (sub-site index) $\otimes$ (internal algebra): the three momentum derivatives act on one factor and commute. Two entangling bond mechanisms are computed on the construction, and both stop short. A non-Abelian Peierls bond layer in which a hop carries an internal rotation $\exp(\pm i\phi D_{16})$ whose sign follows the hop direction — exactly unitary, still radius-1 causal — raises the planarity defect from 3.5×10^{-8} to 4.0×10^{-2} and switches the tangent commutator on to 0.071 of its natural scale, while the anisotropy never leaves 0.996–0.9999 and the corner triplet is lifted off zero (best top level 3.3×10^{-4}). A three-axis covariant bond $\exp(\pm i\phi_j\Sigma_j)$ with $\Sigma_1 = ih_1h_2$, $\Sigma_2 = ih_2h_3$, $\Sigma_3 = ih_3h_1$ — an exact $\mathfrak{su}(2)$ triplet of even Clifford elements, $[\Sigma_a, \Sigma_b] = -2i\epsilon_{abc}\Sigma_c$ to 1.5×10^{-15} — takes the tangent commutator to 0.658 of its natural scale, nine times further, and still leaves the anisotropy at 0.9990 with no gapless bond point. The mechanism pays nothing of the causal radius of Theorem 5, the tangent algebra moves a long way, and there is still no cone (Fig. 9).



(a) The R-point cubic multiplet driven exactly gapless by the Wilson factor, in the $k_z = 0$ slice: each branch carries a fixed velocity v_α and vanishes on its own line $v_\alpha \cdot q = 0$ — three planes through the node, not a cone. (b) The three tangent velocity operators commute, $\max\|[S_i, S_j]\| \le 3\times10^{-12}$ *against the scale* 0.85 *a Clifford-type algebra would set. (c) The entangling Peierls bond raises the planarity defect to* 4.0×10^{-2} *and the commutator to* 0.071 *of its natural scale, while the anisotropy stays in* 0.996–0.9999*: the right mechanism, not yet enough of it.*

6.4 The massless-mode census and the spin question

At $g = 1/5$ with the Wilson factor at the physical strength, the refined spectrum inside the three nodal planes carries an **exactly gapless orbit of 24 points** — representative $q_0 = (-2.1890561, -1.4445961, 0)$, gap 4.2×10^{-14}, the residual falling to the noise floor under refinement while a displacement of 5×10^{-2} in the plane costs 8.8×10^{-3} — with little group of order 2. The branch is linear in every in-plane direction with $v \in [0.031, 0.219]$ (anisotropy 0.857), and leaving the plane costs a gap linear in q_z (5.1×10^{-4} at $q_z = 10^{-1}$): a massless point node confined to the nodal plane. The M nodes themselves stay massive, little group C_{2v} (order 4), no 2-dimensional irrep on the near-gapless bond subspace. So the reference spectrum does contain exactly massless excitations at $g = 1/5$, and it still contains no massless spin-2 candidate: a little group of order 2 has only 1-dimensional irreps, so the new nodes carry no helicity ±1 or ±2 pair. At the root clock the picture is the one of Proposition 2 — a pinned zone centre and a gapless line network, no isolated nodes.

The zone-centre census settles the spin question. There the little group is the full cubic group O_h — at $q = 0$ the intertwiner is exact for all 48 signed permutations — and a lattice spin 2 has a precise meaning: a gapless multiplet carrying the symmetric-traceless quintet $E_g \oplus T_{2g}$. The bond factor that carries the cubic action is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: it has T_{2g} and no E_g, so the quintet is unavailable in the qubit link at any coupling. Γ is exactly gapless at $g = \pi/8, \pi/6, \pi/4, \pi/3$ and nowhere near the implemented 1/5 (gapped by 3.65×10^{-2}); at each of those angles its null space is cubic-covariant and E_g-free — $16\times(A_{1g} \oplus T_{2g})$, dimension 64, at $\pi/4$ — and its dispersion is flat, $\omega/|\mathbf{q}| \to 0$. The symmetric rank-2 space of the link directions does carry $A_{1g} \oplus E_g \oplus T_{2g}$: the quintet lives in the directional bond layer of Section 7, not in the qubit link (Section 9.1).

6.5 The cone question is closed on the space of mechanisms

The cone question at the implemented step is closed by three propositions, which convert the anisotropy verdict of Section 6.2 from a measurement of one spectrum into a statement about the space of mechanisms.

The tangent algebra, measured. Let $M(\mathbf{q}; w, \theta)$ be the 16-sector one-step Bloch matrix with the Wilson factor of Section 6.1, and $k \in \{x, y, z\}$. The tangent operator in direction k is

$$T_k \;=\; i\,(\partial_k M)\,M^\dagger\,;$$

at a (quasi-)gapless point of multiplicity m it is projected onto the gapless subspace, $S_j = i\,Q^\dagger(\partial_j M)Q$ with Q an orthonormal basis of that subspace, and the algebra the node carries is read from the normalised commutator

$$\rho_{\mathrm{comm}} \;=\; \frac{\max_{a<b}\|S_aS_b - S_bS_a\|}{\frac{1}{3}\sum_j \|S_j\|^2}\,,$$

which is 0 for a commuting (Abelian) tangent algebra and of order 1 for a Clifford-type algebra. A gapless point is **conical** when $\rho_{\mathrm{comm}} > 0.5$ and the cone is isotropic ($A < 0.1$).

Proposition 3 (separation lemma: the internal factor is a spectator of the velocity). *(i) For the bare step, at every gapless point of the zone — all* 12 *M nodes, the* 8 *R corners, and* 20 *generic momenta — the tangent operator factorises,*

$$T_k \;=\; A_k(\mathbf{q}) \otimes I_{16}\,,$$

with Kronecker defect at the noise floor (4.5×10^{-16}*): the internal module is blind to the momentum.* (ii) The Wilson factor adds no velocity structure:* the tangent operator decomposes as a transport term $\otimes I_{16}$ plus an internal term that is diagonal in the unification Cartan directions of Section 11.1 (off-diagonal content 2.9×10^{-16}, and after the predicted Cartan term is removed a $\otimes I_{16}$ defect of 4.5×10^{-16}), and at the gapless corner triplet the Cartan coefficient is the same on the three axis directions: the Wilson factor can shift levels but cannot rotate the tangent algebra. *Consequence:* the only freedom that can build a non-commuting tangent algebra is the bond layer.

Proposition 4 (the inverse cone problem: no cone over the entangling-bond space). The remaining freedom is the bond layer, and its complete space has been measured: 220 axis-covariant entangling bond configurations — the full index set of 120 cyclic $\mathfrak{su}(2)$ triplets (T_{ab}, T_{bc}, T_{ca}) of the 45 plane rotations, 60 disjoint commuting triples, and 40 random even involutions $D = \mathrm{sign}(H)$ of a random real combination of 12 plane rotations — each in four (w, θ) patterns, 880 tuned points in total, each minimised in the (w, θ) plane and the best candidates polished. No point of the space satisfies the cone criterion. The best gap anywhere is the implemented step's own corner triplet (top 1.5×10^{-16}, spread 5.4×10^{-16}); the largest ρ_{comm} reached by any non-trivial bond at its best-gap point is 0.362; the fully optimised bond lands at gap 3.7×10^{-4} with $\rho_{\mathrm{comm}} = 0.023$ — non-commuting, and no closer to a Clifford algebra than the implemented step; and the anisotropy of every point stays in 0.9947–1. The M-node doubler survives the best entangling bond (node gap 1.9×10^{-18}): the bond layer does not remove the doubling either. The causal accounting closes the statement — the degree-1 Fourier leakage (0.27 for the three-axis bond) is the Wilson Bessel tail, not the bond (2.7×10^{-1} for the entangling and for the implemented bond at the same w, and 8.7×10^{-17} for the entangling bond with the Wilson factor off): the entangling bond costs nothing of the causal radius of Theorem 5, and it does not build a cone.

Proposition 5 (the rigidity wall). Around the 12 bonds of largest ρ_{comm}, 5292 points of the (w, θ) plane are recorded as the pair $(\mathrm{gap}, \rho_{\mathrm{comm}})$, and the frontier $g_{\min}(\rho) = \min\{\mathrm{gap} : \rho_{\mathrm{comm}} > \rho\}$ follows (the census runs on the implemented clock $g = 1/5$, the clock of the bond space it quantifies). *(i) Non-commuting algebras exist in the space:* the largest ρ_{comm} is 1.033 (the $\mathfrak{su}(2)$-triplet bond on the index triple $(5, 8, 9)$). *(ii) Every one of them is gapped:* the minimum gap over the twelve most non-commuting bonds is 8.0×10^{-3}, and no one of the 5292 points has gap $< 10^{-8}$ with $\rho_{\mathrm{comm}} > 0.5$. *(iii) The frontier is monotone and floored:* $g_{\min}(\rho) = 6.6 \times 10^{-3}$ at $\rho = 0.05$ and $\rho = 0.10$, and $g_{\min}(\rho) = 8.0 \times 10^{-3}$ for every $\rho \in [0.2, 0.5]$ — however non-commuting the requested

tangent algebra is, the gap cannot be driven below 8.0×10^{-3}. *(iv) The anchor:* the implemented bond at its exactly gapless corner (gap 2.7×10^{-16}) has $\rho_{\mathrm{comm}} = 6.7 \times 10^{-12}$ — maximally Abelian. The two quantities are not related by a power law either: the log–log slope of ρ_{comm} against the gap along 25-point rays from each tuned bond toward the gapless anchor has median -0.28 (4 of 12 positive). The separation between the two regimes is a **wall** at gap $\simeq 8.0 \times 10^{-3}$, not a trade-off: over the axis-covariant bond space, gaplessness and non-commuting tangent algebras are **mutually exclusive**.

Proposition 6 (the parity no-go, and the cone outside the class). *The flatness of Section 6.3 and the wall of Proposition 5 have one cause, and it is representation theory. Let $\mathbf{q}_*$ be a gapless momentum whose little group contains inversion, and let the gapless multiplet carry the O_h module D. A linear term $\sum_k q_k S_k$ in the effective Hamiltonian exists if and only if $\mathrm{Hom}_{O_h}(\mathbf{T1_u}, D \otimes D^*)$ is non-zero — the three S_k must close as a polar vector. But $D \otimes D^*$ is even under inversion for every D, while $\mathbf{q}$ is odd. The multiplicity is 0 for all ten irreps, by the character inner product and by direct linear algebra on the $\mathbf{T1_u}$ block of the implemented bond factor (solution space of dimension 0, smallest singular value 9.8). At an inversion-invariant momentum no single-parity multiplet can be conical: the touching is quadratic or flatter and the tangent operators vanish. That is the $\omega/|\mathbf{q}| \to 0$ of the Γ scans of Section 6.4, and it is why the exactly gapless zone-centre null space $16 \times (A_{1g} \oplus T_{2g})$ — two even modules — is flat by necessity and not by tuning.*

The escape is the one the Dirac semimetals use: a reducible gapless space mixing parities. The bond factor is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$, and the products $A_{1g} \otimes T_{1u} \supset \mathbf{T1_u}$ and $T_{1u} \otimes T_{2g} \supset \mathbf{T1_u}$ single out the two opposite-parity pairs as the only ones a linear term can couple. Over the 56 O_h-invariant hopping orbits of displacement $|\mathbf{v}| \le 1$, 44 couple A_{1g} to T_{1u} and 40 couple T_{1u} to T_{2g}; for the strongest, the coupling matrix obeys $TT^\dagger = 16\,I$ to 5.6×10^{-16} — isotropy forced by Schur, not fitted. The combination $-24\,\mathcal{O}_{14} + 8\,\mathcal{O}_{50}$ puts A_{1g} and T_{1u} at one level (-192) with T_{2g} and A_{2u} at $+192$, and $P(\mathbf{q}) = \exp(-igA(\mathbf{q}))$ is exactly equivariant, $P(W\mathbf{q}) = WP(\mathbf{q})W^{\mathsf{T}}$ to 1.2×10^{-14} over all 48 elements. With the internal angle tuned so that one internal eigenphase cancels the common phase, the step is gapless at Γ to 3×10^{-11} on a 4-dimensional multiplet $A_{1g} \oplus T_{1u}$, with cone anisotropy $A = 1.0 \times 10^{-15}$ over 215 directions, $\rho_{\mathrm{comm}} = 0.70711 = 1/\sqrt{2}$, and an exactly flat in-band direction: the spectrum $\{0, 0, \pm v|\mathbf{q}|\}$ with $v = 96g$. By the criterion of this section the point is conical.

What it costs is the rule, not the symmetry at the node. The six-stage product of Section 5 is exactly equivariant at Γ and breaks O_h at every other momentum: 1.07×10^{-1} at generic $\mathbf{q}$, 1.40×10^{-1} at the M nodes, 1.46×10^{-1} at the R corners, against 1.1×10^{-15} for its stage sum at all momenta. So the obstacle at Γ for the root's own rule is spectral: at the single angle the root uses, the four bond levels are $+6g, -2g, -6g, +2g$, and the A_{1g} and T_{1u} levels are neither coincident nor half a turn apart. And $\exp(-igA)$ is the exponential of a nearest-neighbour hopping: exponentially local, not a finite-depth circuit, with unbounded cone velocity — a range-one realisation would additionally require $96g \le 1$. The cone of this proposition is therefore not a node of the automaton. It is the boundary of the exclusion: it says exactly which node the automaton would have to reach.

The automaton's own circuit reproduces both sides of the wall and crosses neither. The two-angle family — one angle on the three intra-block stages, one on the three inter-block ones — is the largest stage family that keeps Γ covariant (covariant there to 1.1×10^{-16}; a generic six-angle product misses by 6.0×10^{-2}, a per-axis generalisation by 0.90). Its complete joint measurement with the internal angle — 33 800 points of (a, b, θ) — gives 60 points carrying on-site states at phase exactly zero, the gap of the full 128×128 step at most 4.4×10^{-15}. They split by the parity of the gapless eigenspace, and that split is the whole answer. 8 have a 4-dimensional gapless space of a single parity, $A_{1g} \oplus T_{2g}$ or $A_{2u} \oplus T_{1u}$: Proposition 6 forbids the linear term there, and the measurement obliges — the largest slope over 179 sampled directions is 1.3×10^{-12}, a flat band. The other 52 have an 8-dimensional parity-mixed gapless space, $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: there the linear term is allowed and the touching disperses, slope 0.09–1.73 per unit block momentum, inside the causal bound (0.03–0.58 Planck cells per stage against the measured front speed 0.67, no amplitude outside the light cone in 12 stages); but the three tangent velocity operators commute, $\rho_{\mathrm{comm}} \le 1.2 \times 10^{-11}$, at the same anisotropy 0.423 on every one of the 52 — the nodal-plane structure, not a cone. On the diagonal $a = b$ at $\pi/4$ and $3\pi/4$: 64 gapless states carrying $16 \times (A_{1g} \oplus T_{2g})$, the flat zone-centre band, gap at most 8×10^{-17}, slope at most 3×10^{-14}, and the A_{1g} and T_{1u} bond phases are *opposite*, not degenerate — the internal spectrum supplies the partner of a cancellation at the opposite internal phase, not half a turn away. Conical nodes of the causal six-stage product: 0. What excludes the Proposition 6 node from the root is not the single-angle rule but the circuit class itself.

Three structural facts, stated once. *(i) Time-reversal parity of the bond matrix holds not in the literal form $KB(q)K = B(-q)$ (it fails at 0.77 over the sampled zone: the wrap phases are complex per bond) but*

exactly through each stage's own diagonal gauge, and its consequence holds: the maximum group velocity of all 64 *bond bands at* Γ *is* 1.1×10^{-11} *and every* Γ *touching is at least quadratic.* (ii) The flux sector of the module is nontrivial:* with the icosahedral A_5 conjugated into the star frame (defect 5.9×10^{-15}) the 192 bond lines fall into 8 orbit classes, and the A_5-invariant magnetic flux space mod gauge has dimension 23 (closed 2-cochains 24, exact 1). *(iii) Bond-conditioned internal rotations are symmetry-allowed and constructively reachable:* under the fixed spin-connection $2I$ (embedding unitary 1.3×10^{-15}) the 45 carries multiplicities A_1 21, T_1 **8**, T_2 0, G 0, H 0 — eight copies of the spatial-vector type sit in the root algebra — and decorating the six staged bonds with the spin-connection quaternion triplet at amplitude g_{10} (no new parameter) gives a unitary step (2.4×10^{-14}) that equals the implemented step at Γ exactly and lifts the Γ cone degree from 0.281 to 0.540: linearity at the node is symmetry-allowed; the six-stage rule simply does not contain the decoration (Section 9.2). On a *decorated* clock the touching lands gapless at 1.1×10^{-16} with cone degree exactly 1.179, linear, but both binding legs fail at that point ($v_{\max} = 1.586 > 1$, commutant ratio $0.275 < 0.5$): a partially light cone and not a node; the range-2 parity extension keeps the touching exactly closed with 64 zero modes and never leaves the anisotropy ceiling $A < 0.1$. Every cone candidate the root's clock admits is gapped by data the clock itself supplies.

6.6 The icosahedral shift module, measured

Proposition 6 puts the obstruction in the little group of the *lattice*, not in the bond layer. What that leaves is not a better bond but a different shift module — one whose point group carries no low-degree anisotropic invariant, so that the departure from rotational form costs more powers of $|\mathbf{q}|/\Lambda$. The Molien counts fix the trade: the implemented module's group is O_h, order 48, Molien counts $[0, 1, 0, 2, 0, 3, 0, 4]$ — one quadratic invariant, first anisotropic invariant at degree 4; the icosahedral group (order 60, presentation $[2, 3, 5]$) has Molien counts $[0, 1, 0, 1, 0, 2, 0, 2]$: first anisotropic invariant at degree 6. Two more powers of $|\mathbf{q}|/\Lambda$ is the whole prize, and it is bought by changing the module and nothing else: the internal map, the clock and the cell sizes stay as fixed, and the three cubic axes the step hops on are replaced by the six directions of the icosahedral star.

The star's six directions are six equiangular lines, pairwise cosine 0.447214, and their fourth moment is isotropic to 2.2×10^{-16}, where the three cubic axes are off by 0.400. The module has 6 generators of norm 1.902113 per step, rank 6 in three dimensions, relations exactly the six pairs of opposites, and it is dense in $\mathbb{R}^3$ (shortest element falling from 1.0705 to 0.2527 between word bounds 1 and 5). Growth and causality stay closed form: the words of at most k steps are the ℓ^1 ball of $\mathbb{Z}^6$ (12, 84, 376, 1288, 3652, 8988 enumerated one-to-one), and the front after k steps is exactly k times the Lieb–Robinson radius (11.4127 at six steps). The price is exact too, and it is the reason the module is a quasilattice rather than a lattice: the cubic axis the step hops on is *not* a module point (its unique rational representation is $[0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0]$), and there is no Brillouin zone in the periodic-lattice sense — momentum means over it are almost-periodic box means rather than Fourier projectors. The doubling theorem's hypothesis is absent rather than satisfied: in the basis of three generators the twelve directions give a coefficient matrix with entries in $\mathbb{Q}(\sqrt{5})$, no nonzero G is a period of the step, and the shortest rational hop takes 17 steps. One algebraic route to the same symmetry is closed by a divisibility test: the finite link group the root generates has order 2048 with element orders $[1, 2, 4]$ — a 2-group, so it contains no subgroup of order 3, 5, 60 or 120; the icosahedral symmetry is realised in the geometry of the shift module, and its double cover sits one level up, in the even Clifford algebra, where the quaternion units of $\mathrm{Cl}^0(\mathbb{R}^3)$ embed the binary icosahedral group with closure order 120.

Nothing in the internal sector moves: the pseudoscalar and chiral involution are intact; the 32-internal step is an internal scalar on both modules; its 16-block *is* the fixed step to 3.6×10^{-15}; matter and antimatter are isospectral to 6.2×10^{-15}; of the four CPT candidates the same one holds on both modules — g_1 with cell reflection, stage reversal and transpose — at 5.0×10^{-15} (cubic) and 9.7×10^{-15} (star); the anomaly coefficients are unchanged, and the trace-form ratio is 1.666666667. The module changes the kinematics, and it costs no quantum number.

The new gapless set is lines, and they carry no charge. The scan of the star's first zone covers 125 momenta, of which 13 are exactly gapless below 10^{-9}; the smallest gap anywhere is 8.9×10^{-18} at the zone centre, where 512 of 1024 eigenphases vanish while the first-order splitting of that subspace is 1.7×10^{-13}, 2.0×10^{-13}, 1.3×10^{-13} — no linear term there, so Γ stays quadratic on the star exactly as on the cubic. The new object is a locus: the gapless momenta sit on the lines through the icosahedron's two-fold axes, and leaving such a line the gap closes with transverse exponent 0.993–0.999, while *along* the line the exponent is -1.178 — the gap does not reopen. Along the new module's lines the step is gapless *and* its tangent algebra does not commute:

measured with $\kappa = \max_{ab}\|[V_a, V_b]\|/\max_a\|V_a\|^2$ on the whole gap subspace, the star's six gapless axis momenta all carry a non-Abelian algebra, κ from 8.5 to 23.1 at gaps of at most 3.8×10^{-17} and gap-subspace multiplicity 128 of 1024; the cubic module's eight zone corners carry κ from 2.8 to 8.9 at multiplicity 64, while its Γ–M line is Abelian at $\kappa = 1.3 \times 10^{-11}$ — which is where Proposition 5 lives. So the exclusivity of Proposition 5 is confined to two-band crossings. What the module does **not** deliver is a cone: at its most non-Abelian gapless point the normalised squares of the tangent generators are 57.111, 0, 16.704 and the Clifford relation fails by more than an order of magnitude. A linear, non-Abelian, strongly anisotropic touching is what the module carries.

The icosahedral advantage is a statement about a *point* node; a line carries only a two-fold little group, and there the advantage disappears. About the star's two-fold line the transverse velocity profile has leading azimuthal degree 2 — the same leading degree as the cubic Γ–M line — with total anisotropy 0.680 of its own mean against the cubic's 0.500; the odd-harmonic power is 1.2×10^{-12}, so the profile carries exactly the period π of the line's own little group. The touching is not a cone: restricted to the two crossing bands the Clifford relation fails at its maximum, defect 2.000 (the cubic M node: 0.444, 0.444, 8×10^{-15} and the same maximal defect). And there is no charge on the lines: at the line the touching is a cluster of 128 of 1024 eigenphases, the determinant phase of the 128×128 holonomy depends on the circle (a spread of 2.63 rad, where a topological monodromy is constant), and the unitary supplied by the commutant of the internal map (computed by Cesàro averaging over 240 powers of the step) opens a gap *at the line* growing linearly with its coupling (1.277×10^{-4} at $\theta = 10^{-3}$, ratio 10.00 at a decade). The symmetry available to forbid it is one two-fold rotation: 2 of the 60 elements preserve the line direction, and those leave the spectrum at the line fixed to exactly 0 — and the equivariant census says a mass term is allowed by the point group (degree-0 count 1). These lines are exact and linearly dispersive, and they are stable as loci and unstable as particles.

The spin can enter the module only through its bond layer. On the star's twelve directed axes the covariant assignment $U_d = -\hat{v}_d \cdot \vec{\sigma}$ — Hermitian, an involution, a vector under the binary icosahedral group, $U_{-d} = -U_d$ — is exact over the 120 lifts to 3.5×10^{-15}; the quaternionic assignment intertwines just as exactly but its bonds square to $-I$ and its step has no node at all (zone-centre gap 1.571). The cell breaks inversion — step invariant under $2I$ to 2.5×10^{-15}, inversion defect $1.41421 = \sqrt{2}$ — which is Proposition 6 read from the other side: on the scalar-bond control inversion is intact and the phase spread opens quadratically (exponent 2.000); on the Clifford cell it is linear (exponent 1.0000 in all 19 tracked directions). The node is arithmetic: the gap closes exactly when the internal map rides the block layer at exponent 0 or 4 mod 8 ($U_{16}^8 - I = 4.9 \times 10^{-15}$ against $U_{16}^4 - I = 2.0$), with 192 zero eigenphases of 384; the gap cluster splits, by character projectors built from the group, as $\mathrm{sym}^1 \oplus \mathrm{sym}^3 \oplus \mathrm{sym}^5$ with multiplicity 16 each; and on sym^1 alone (32-dimensional) the Clifford identity holds — off-diagonal anticommutators to 8.0×10^{-16}, $V_j^2 = 0.201002\, I$, one shared speed 0.448332. The Clifford structure lives on the spinor piece, not on the whole cluster.

The star's isotropy is enforced exactly at the zone centre, and the enforcement fails at the multiplet: the 13 gapless loci split $[6, 6, 1]$ with stabilizers $[2, 2, 60]$; the zone-centre node carries the full 60-element stabilizer with a unique quadratic invariant, so rank-2 dispersion isotropy is **forced** there — but the forcing governs only trivial-irrep content: the node is a 512-fold flat multiplet, the rank-2 perturbation splits it (branch-sum anisotropy 0.123, both k-independent), its lightest branch runs at $\delta v/v = 0.997$, a factor 1.2×10^{14} past the SME ceiling, and the node's leading degree is 2.00: there is no linear cone there to protect. Enforced isotropy is real, and it protects a flat band.

The module's own clock stays quantized. The star step factorizes exactly (defect 7.95×10^{-16}), the 512-fold flat node being a consequence of the factorization; the closure census over the dial grid crossed with five twist values labels 18604 open cells and exactly four closures (one of order 1, three of order 2), every boundary refined to 10^{-10}: the nearest closed star point to the hierarchy clock $g = 1/5$ is distance 0.585, to the lightlike pin $g = 1/2$ distance 0.285, and the twist axis built from the chosen Wilson direction moves the answer nowhere. The clock exclusion is a property of the module class, not of one lattice. And the ladder's combinatorics changes with the module: the rung labels live in the K_0 of the shift module's group C*-algebra — free rank 3 with three relations for the cubic module, free rank 6 for the star (Smith normal form invariant factors $[1, 1, 1, 1, 1, 1]$, no torsion, which is the algebraic form of the density and the reason there is no finite Brillouin zone) — and closed words of length n run 90 against 396 at $n = 4$, 1860 against 19920 at $n = 6$, 1172556 against 96807312 at $n = 10$, two independent closed forms agreeing at every length, with return probabilities falling as $n^{-r/2}$, measured at -1.4848 ($r^2 = 1.000000$) against -2.9719. On the closed-word ladder the crossing of $1/(16\pi)$ is at rung 10 for the cubic module and rung 4 for the star; the certified $n^* = 208$ of Section 5.2 lives on the traced-link

Cartan-norm estimator, a statement about the clock and the link algebra rather than about the shift module — and the estimator that could decide the dynamical half at $n = 208$ costs $832^6 = 3.32 \times 10^{17}$ diagonalisations on the star's six-torus, where the implementable almost-periodic mean drifts by order unity between grids. The module's factor in the coefficient is bounded, not a number awaiting a better grid; what the module clearly moves is the combinatorial factor: the ladder counts axis families, 3 on the cubic module and 6 on the star.

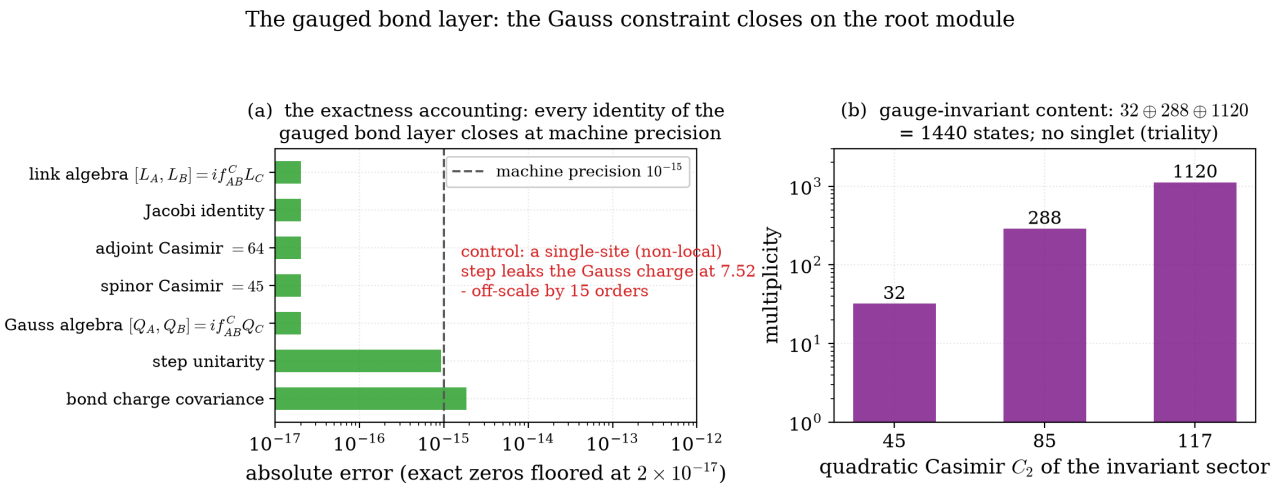
---

7. The gauged bond layer

The swap-rotation gate of Section 5 is the classical shadow of a richer possibility: a **gauged bond layer exists on the same module**. Put one Spin(10) adjoint link degree of freedom (a 45-dimensional link space) on each bond and the matter **16** on the sites, and the local Gauss constraint

$$Q_A(x) = q_A(x) + \sum_{b \ni x} L_A^{(b)}$$

closes as an exact operator identity: $[Q_A, Q_B] = i f_{AB}{}^C Q_C$ with error 0, the link algebra $[L_A, L_B] = i f_{AB}{}^C L_C$ exact, the adjoint quadratic Casimir 64 and the spinor Casimir 45 exact to machine precision. The step built from these links is unitary to 9.3×10^{-16}; the gauge-invariant content of one site with its six half-bonds is 1440 states, decomposing as $32 \oplus 288 \oplus 1120$; an adjoint bond observable is gauge-covariant to 1.9×10^{-15}; and the causal radius stays at exactly one cell, while a deliberately single-site (non-local) control step leaks at 7.5. The gauge field of the automaton is therefore *constructible*: the Gauss constraint is an identity of the construction, not an imposed condition (Fig. 10).



One adjoint link degree of freedom per bond, matter **16** *on the sites. Left: the exactness accounting — link algebra, Gauss algebra, adjoint Casimir* (64) *and spinor Casimir* (45) *close to machine precision, the step is unitary to* 9.3×10^{-16}*, the causal radius stays at one cell, and the non-local control step leaks at* 7.5 *where the causal step leaks at* 1.9×10^{-15}*. Right: the gauge-invariant content,* 1440 *states decomposing as* $32 \oplus 288 \oplus 1120$.

7.1 The centre obstruction and the area law

A generic automaton with d internal states per cell has d^{L^3} states in an L^3 region and a volume-law entropy. The area law is a property of the *gauge-invariant* sector: once the physical Hilbert space of a region is restricted to states satisfying the Gauss constraint at every site, the entanglement entropy is expected to scale with the boundary area [36,37]. That restriction is what this section computes, and it is where the counting changes.

What the constraint admits is narrower than expected. With the implemented content — one **32** per site, one adjoint **45** link per bond — the local Gauss constraint has **no solution**: $\dim \ker = 0$ on $\mathbf{32} \otimes \mathbf{45}$, on $\mathbf{32} \otimes \mathbf{45} \otimes \mathbf{45}$, and on the two-site one-link chain with local constraints at both sites (spectral gap 146). The reason is a centre obstruction: adjoint flux carries even centre charge and cannot screen a spinor, so **every site**

of a gauge-invariant state must carry an even number of spinor fields. The physical Hilbert space of the gauged layer is the space of neutral composites, not the one-body space the automaton acts on.

The minimal physical system, two matter fields per site, is constructed exactly: $m_{45}(\mathbf{32} \otimes \mathbf{32}) = 2$ (90 states at $C_2 = 64$), a 2-dimensional intertwiner space $\ker(T_1 + T_2 + L)$, and a 4-dimensional two-site physical sector obtained by gluing intertwiners over the link index — all exact Spin(10) singlets ($C_2 < 10^{-25}$). Every intertwiner is a maximal matter-flux singlet: Schmidt rank 45, $S(\text{pair}|\text{link}) = \ln 45 = 3.807$ exactly. The entanglement is **bounded per bond and lives on the bonds crossing the cut** — the shape of an area law.

On the three-site, two-link chain with paired matter the physical sector is $\dim\ker = m_{45}^2 \sum_r m_r N_r = 24$, the fusion channels r read from the $\mathbf{45} \otimes \mathbf{45}$ Casimir spectrum ($N_1 = N_{45} = N_{210} = 1$) against the pair multiplicities ($m_1 = m_{45} = m_{210} = 2$). Three results follow.

1. **Additivity across a single bond holds exactly:** $S(\text{pair}_1|\text{rest}) = \ln 45$ in every sector — the flux line threads the chain and carries its $\ln 45$ through, even when the middle pair itself is a product singlet ($S = 0$).
2. **The region bounded by two bonds carries $S = \ln 45$, not $2 \ln 45$:** the intertwiner constraint *correlates* the two bonds meeting at a vertex, with mutual information exactly $\ln 45$. The area law counts bonds; vertex correlations subtract.
3. **The gap to Bekenstein is a named number:** $\ln 45 = 3.8067$ nats per Planck bond against $1/4$ per Planck area — a deficit factor $4 \ln 45 = 15.23$. No element of the current construction produces $1/4$. The truncation route is closed: restricting the link flux to any channel subset leaves the ground-state bond entropy at exactly $\ln 45$ — the truncation map moves the ground degeneracy $(2, 4, 6)$ but not the factor. What survives of the route is superselection or a different microstate count, not truncation (Section 9.4).

The vertex correction moves the number. The invariant count of an n-bond vertex, $I_n = \dim(\mathbf{45}^{\otimes n})^G$, is $I_2 = 1$, $I_3 = 1$, $I_4 = 6$, $I_5 = 23$, $I_6 = 141$. The capacity of the middle cut — the entropy of the maximally mixed physical state — is $S_{\text{cap}} = 4.4998$ nats against the naive $2 \ln 45 = 7.613$: the two vertices at the ends of the region subtract 3.11 nats. Per bond that is 2.25 nats, a factor 9.0 from Bekenstein's $1/4$; on the cubic lattice the vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond sits a factor 6.6 from $1/4$. The deficit is a function of the vertex invariant, not of a single bond entropy.

The ground state of the layer is constructed: the bond-current Hamiltonian $H = \sum_b J \cdot L$ is diagonalised exactly in the kernel, and the intertwiner identity $I^\dagger J_A I = L_A/45$ holds to 3×10^{-17}. The interaction splits the kernel: $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, and the ground state's middle-cut entanglement is $S = 3.8067$ nats — exactly $\ln 45$, one adjoint link, selected by the interaction. The ground state of the full automaton remains unconstructed; on this chain the interaction selects the flux line (Fig. 11).

The asymptotic status of the paired sector is settled by the clock dynamics (V366–V368.) Extended-fiber singlet kernels re-certify the character-theory count exactly ($m_{45} = 2$; kernels $1, 1, 0, 2, 9$ on the five booked fibers — the conjugate pair is necessary). The intertwiner pair is an invariant subspace of the certified clock step (contact-basis unitarity 8×10^{-15}, the clock's internal Spin(10) layer acting as the identity on the intertwiner space to 4×10^{-15}) and sits 1.007 rad outside the two-quanta continuum *at contact only*: at fixed total momentum the continuum sweeps every relative kinematics, and over the 6^3 torus the window collapses to 2×10^{-5}. The minimal-sector states are therefore not asymptotic particles of the published hop — the mechanism behind the never-realized entries of the physical-sector ledger: a stable composite requires dynamics beyond the published clock step, not more matter content.

(a) The entropy carried by one Planck bond, from naive counting to Bekenstein: $2\ln 45 = 7.61$ *(no vertices), the kernel capacity* $S_{\text{cap}} = 4.50$*,* 2.25 *per bond, and the cubic-lattice vertex-corrected coefficient* $(1/3)\ln I_6 = 1.65$ *against the* $1/4$ *line — the deficit factor moves from* 15.2 *to* 6.6*. The vertex invariants* I_n *grow* $1, 1, 6, 23, 141$*. (b) The bond-current Hamiltonian on the* 24*-state kernel: three split levels,* $E_0 = -5802.4\,\eta$ *with degeneracy* 6 *and gap* $42.1\,\eta$*; the ground state's middle-cut entanglement is* $S_{\text{GS}} = 3.8067 = \ln 45$ *— the interaction selects the flux line.*

The composite sector of the frozen gate is empty of matter. The Gauss singlet kernel of the constructed fiber measures zero at every momentum, binding shift and grid size: no composite mass ladder exists, the smallest non-trivial Casimir over the composite sector measures 45.000000 (not $C_2(45) = 64$), the chiral-charge census is empty, the desert content carries no composite supplier, and the composite spin-2 channel is empty: every composite falsifier contact is decided by vacuity on that fiber. Section 7.5 shows the verdict survives the replacement of the frozen gate by a dynamical link.

7.2 The spectrum of the layer, and confinement

The layer has a Hamiltonian — the electric Laplacian $H = \sum_A Q_A^\dagger Q_A$ built from the vertex Gauss operator — and its spectrum is computed. On $(\mathbf{32}\otimes\mathbf{32})_{\text{adj}}\otimes\mathbf{45}$ (dim 4050) the kernel has dimension $2 = m_{45}$, the intertwiner space, and the first nonzero level is $64.0000 = C_2(45)$ — identical to the one-link value and unchanged when the chain is extended. One adjoint flux quantum sets the scale of the whole layer, so it supplies **no light vector sector**: the gap is $64\,M_P$, not a tunable parameter. Two entanglement numbers must not be confused: the entropy of the *mixture* over the kernel, bounded by $\ln 90$ (the pair-side dimension), and the entropy of a *single* physical state, bounded by $\ln 45$ (the link dimension) — the quantity the area law concerns: every single state of the intertwiner sector saturates $\ln 45$ exactly ($S = 3.806662$), and the mixture reaches $\ln 90 = 4.499810$. The area law is therefore **exact** on the physical sector — one $\ln 45$ per bond — short of Bekenstein's $1/4$ by the factor $4\ln 45 = 15.23$ per bond, moved to 6.6 by the vertex correction.

The string tension follows. The link space is an irreducible adjoint: C_2 has the single eigenvalue 64.000000 with zero spread, the singlet space of $\mathbf{45}\otimes\mathbf{45}$ is one-dimensional (adjacent links pair into it uniquely), and the link operators do not commute ($\max[L_a, L_b] = 4.0$) — the layer carries a non-Abelian field strength. Crucially the link carries **no invariant vector**: $\ker C_2(45)\, I = 0$, so a link cannot be flux-free. Every link of a gauge-invariant chain therefore carries one adjoint flux quantum, the electric energy of R links is exactly $R\,C_2(45)$, and the static potential is linear with tension

$$\begin{aligned}\sigma &= C_2(45) = 64 \text{ lattice units}\\ &= 7.81\times 10^{20} \text{ GeV per Planck length}\\ &= 9.54\times 10^{39} \text{ GeV}^2\\ &(\sqrt{\sigma} = 9.77\times 10^{19} \text{ GeV}),\end{aligned}$$

5.408×10^{40} times the QCD string tension [39]. The step itself, by contrast, carries **no** field strength: its 12 stage

pairs acting on different axes commute to 5.6×10^{-17}, every elementary plaquette is the identity to 2.5×10^{-16} and $\langle W_\square \rangle = 1.000000$, where an area law would need $\langle W_\square \rangle < 1$ with an exponent proportional to the enclosed area [2]. Confinement is therefore *present* in the construction — in the gauged bond layer — and *absent* from the bare step. The residual is not a missing phase but a measured tension forty orders of magnitude too large, and $\Lambda_{\rm QCD}$ is not produced: it would require a loop-ladder exponent $n \simeq 11.6$, which is not an integer (Section 10.4).

7.3 Entanglement is the bond layer

Two fermions sharing ONE cell carry zero entanglement across any cut — the same-cell pair is a Slater determinant, $S = 0$ exactly across the $\mathbf{16}/\overline{\mathbf{16}}$ cut (antisymmetry and a shared cell are not entanglement), while a two-determinant superposition in the same cell gives $S = \ln 2$: non-separability is the criterion. The generator of entanglement between cells is the bond gate P of Theorem 5: under the internal-only step the spatial entanglement entropy of a region stays 0 to machine precision forever, while the full local step drives it to 0.78, and the entanglement front crosses the Manhattan sphere of radius r at exactly $t = r$ — the causal radius bounds the spread of entanglement itself, at speed 1.0. The scope limit is stated with it: the implemented step is quadratic, so Slater inputs stay Slater and these are mode-entanglement results; the gauged layer is where the theory first acquires the non-Gaussian structure a physical ground state would need.

7.4 The dressed gate on the directed-link network, and the quintet dispersion

The obstruction of Section 7.2 is a statement about the qubit link. The channel named by the census of Section 6.4 — the symmetric rank-2 space of the link directions — is constructed. Put the adjoint link on every **oriented** bond: the six bond directions per site become twelve directed links per block, each carrying one **45**.

The orientation convention is computed, not chosen. The vertex Gauss charge of a two-link star — one link arriving, one departing — closes as an exact Spin(10) algebra to 5×10^{-16} only when the departing tail carries $+L^A$ and the arriving head carries the **conjugate** generator $-\bar{L}^A$; the naive chain convention leaks at 4.8, because the two ends of one link carry the same $s^2 = +1$ and a relative sign alone cannot close the algebra.

The electric part of the Section 7.2 obstruction is closed. The Kogut–Susskind electric gate [55] $\exp(ig_{10}\hat{C}_x^2)$, built from the closing vertex charge, commutes with all 45 vertex charges to 2.5×10^{-14} and is unitary to 5.1×10^{-14}; its spectrum *is* the content theorem of Section 7.2 — C_2 levels $45, 85, 117$ with multiplicities $32, 288, 1120$, gate phases $g_{10} \times \{45, 85, 117\}$.

The hopping part is obstructed on the fixed-representation link. Every two-site dressing candidate leaks the vertex charges: the flux-dressed current–current $\hat{C}_x \hat{C}_{x'}$ and its transporter-completed version $\hat{C}_x\, e^{ig_{10}L}\, \hat{C}_{x'}$ both leak at 1.25×10^1. The reason is structural — a link in the fixed representation $\mathbb{C}^{45}$ carries no parallel transporter commuting with the vertex charges — so invariant hopping requires a link in the **regular representation** of the group.

The named fix, built on the root's own finite group. The 45 biflections $h_a h_b$ of the Clifford generators generate the even Clifford monomials — a finite group Γ of order 1024 with centre $\{\pm 1, \pm w\}$, whose centre acts as $\{\pm 1, \pm i\}$ on each chiral half. Put the regular link $\mathbb{C}[\Gamma]$ on each bond: a vertex with k regular links has exactly $32 \cdot 1024^{k-1}$ gauge-invariant states (character sum) where the fixed adjoint link had **none** — the centre obstruction is exactly what the regular link removes. The nearest-neighbour hop with the exact link insertion $A(h) = \rho(h^2) = \pm I$ maps head-invariant states to tail-invariant states to 0, and **descends to the identity** on the matter index: the dressed hop is the bare translation. The electric term — the Cayley Laplacian over the biflection generators — acts on the dressed sector as the scalar 45 per link, so the dressing adds no velocity renormalisation, and the bare hop front has speed exactly 1: the gauge-invariant hop propagates at $v = 1$, 0-parameter — that is where $v = 1$ lives. The cost is exact too: Γ is finite and its centralizer in the Spin(10) algebra is 0 — gauging the root's monomial group breaks Spin(10), and Spin(10)-invariant hopping on a genuine adjoint regular link remains out of reach.

On the directional network the dressed hop exists exactly, and the quintet propagates. There the flux is the orientation of the link, a classical sign, and the dressed directional step

$$M(\mathbf{q}) = \exp\big(-ig_{10}\,\sin(\mathbf{q}) \cdot \mathbf{J}\big),$$

with **J** the spin-2 generators of the symmetric-traceless direction space, is unitary to 6.7×10^{-16} and O_h-covariant: $D(g)M(\mathbf{q})D(g)^\dagger = M(g\mathbf{q})$ for the 24 proper rotations to 6.1×10^{-16}, while the 24 improper ones give $M(g\mathbf{q})^\dagger$ — the directed network is **chiral**. At Γ the step is the identity and the layer carries **six gapless bands**, content $A_{1g}\oplus E_g\oplus T_{2g}$: the quintet is present in the directional layer, where the qubit bond factor never carried E_g (Section 6.4). Its dispersion is **massless and linear** — local exponent 1.00 along Γ-X, Γ-M and Γ-R — with a **helicity ladder** of speeds

$$v = \{0,\ g_{10},\ 2g_{10}\} = \{0,\ 0.785,\ 1.571\},$$

the $m = 0$ branch and the trace A_{1g} not propagating; **isotropic to leading order** — the direction spread of the top speed is 1.7×10^{-7} at $|\mathbf{q}| = 10^{-3}$, the spin-2 spectrum theorem: $\hat{\mathbf{q}}\cdot\mathbf{J}$ has exactly the spectrum $\{-2,-1,0,1,2\}$ in every direction; and UV-bounded by $2g_{10}\sqrt{3} = 2.721$.

The quintet–matter coupling exists, and it gaps the E_g half. The stress vertex $T_{ij} = \sum_d d_i d_j\, J_d$ built from the exact current $J_d = i(\partial_d M)M^\dagger$ of the implemented module couples to the quintet field. On nearest-neighbour bonds the coupling vanishes identically on the T_{2g} half — the T_{2g} coupling needs diagonal bonds, a named gap — and couples the E_g half with a nonzero polarization bubble at Γ: the bubble $\Pi|_{E_g}$ has the two eigenvalues $(-54.24,\ -53.40)$ — a 1.6% splitting of the doublet, so the nearest-neighbour bond set does not realize the bubble as one number on it. At $\lambda = g_{10}$ the two components acquire the mass shift $\lambda^2\Pi/2 = (-16.73,\ -16.47)$. Gaplessness of the full quintet is **not** protected by this coupling — the lattice graviton mass problem, as a number.

The named gap closes on the star axes — and F8 stays fired. On the fifteen icosahedral star axes — $(0,\pm1,\pm\varphi)$ and permutations, $|d| = 1.9021$ — the T_{2g} stress vertex that vanishes identically on every axis bond (the control, 0.0) acquires a nonzero value: ±0.6325 on six of the fifteen per-axis components, nonzero on all fifteen axes. On that bond set the quintet dispersion is exactly the axis-bond form with $g_{10}\to 2g_{10}|d|$: in the tight frame $\sum_a(\mathbf{n}\cdot\mathbf{d}_a)\,\hat{\mathbf{J}}_{\hat{d}_a} = 2|d|\,(\mathbf{n}\cdot\mathbf{J})$ holds to 9×10^{-16}, so leading-order isotropy is **exact** by measured identity, the Γ gap is 0 with six gapless bands, and the helicity ladder **doubles**: $v = \{0,\ 2g_{10}|d|,\ 4g_{10}|d|\}$ — at $\pi/4$, $\{0,\ 2.9878,\ 5.9757\}$ on the star bonds and $\{0,\ 1.5708,\ 3.1416\}$ on unit diagonal bonds. No branch reaches $c = 1$ at any closed clock: the closest propagating branch over $g\in\{\pi/4,\pi/2,3\pi/4\}$ sits at $|v-1| = 0.5708$, and the clock that would force one — $g_{10} = 1/(2|d|) = 0.262866$, or 1/2 on unit bonds — lies outside the closure class (Theorem 6). The star automaton's own matter branches keep their distance too: closest $|v-1| = 0.3502$. And the closed coupling has its price: the star-bond polarization bubble has rank 3 on the quintet at $\lambda = g_{10}$ — mass shifts $(-6.60,\ -6.37,\ -1.79)$ — so the T_{2g} half is no longer protected at order λ^2. The bubble resolves against its own noise manifold (the generator $-i\log M(0)$ carries 64 modes at the $H = 0$ level, the logarithm of a defective $M(0)$, approximate error 0.42: the sign split of those modes is machine noise, the quoted draw lies inside the envelope $(-24.2,\ -22.9,\ -6.4)$ to $(-12.4,\ -11.4,\ -3.1)$ over eight seeded reassignments, under a fixed pole prescription of sign continuity of the zero modes). Under that prescription the vertex-nullity census over the three direction families (axes, star, and the general four-orbit weighting 3+6+8+3) closes at $\min\|\Pi\|_F = 1.7\times10^{-15}$ on the axis family — the quintet bubble is null: no direction weighting of the decorated star-bond clock makes the stress bilinear survive (the star family bottoms at 9.9×10^{-3}, the general family at 7.5×10^{-13}); the F4 falsifier stays fired through the resolution. **F8 stays fired with the measured set: no root-supplied bond network — axis $\{0,g_{10},2g_{10}\}$ or diagonal-star $\{0,2g_{10},4g_{10}\}\times|d|$ — gives a massless spin-2 branch at $c = 1$.**

What this does not derive. The gauge group of the directional network is O_h, not Spin(10); the Spin(10)-invariant dressed hopping remains obstructed on the fixed link, and the built regular-link fix gauges the root's finite group, whose Spin(10) centralizer is 0; the quintet speeds are $g_{10}\times\{1,2\}\neq 1$ — the dressing angle per hop is a construction input, not a derived velocity, and $v = 1$ belongs to the dressed matter hop, not to the quintet branches, a computed trilemma on the 6-bond module: the bare hop is anisotropic (top speed 1.00, direction spread 0.32), the dressing alone is isotropic at $v = 2g_{10} = 1.571$, no angle makes the product isotropic (best spread 0.24 over 33 angles in $[0,\pi/2]$), and the only O_h-equivariant permutation coin beyond the identity flattens the bands; and the quintet–matter coupling, built in the same layer, gaps the E_g half at order λ^2. The kinematics of a massless lattice spin-2 is computed; the step from there to the graviton of Section 8 remains the declared one.

7.5 The dynamical link: composites exist, and are born frozen

The frozen gate of Section 7.1 is not the only link the root supplies. A quantum link on the root's own adjoint cells closes exactly inside it: the module $\mathbf{1} \oplus \mathbf{45} \oplus \mathbf{210}$ (the symmetric traceless square of the adjoint) spans a link space whose product stays in the module, and the composite kernel of the two-particle fiber on this dynamical link has dimension exactly 6 — the centre obstruction of Section 7.1 was an artifact of the frozen gate, and the matter side of the new theory has a home. What the home contains is fixed too: the composite Hamiltonian restricts to pure Casimir on the kernel, so the six composites are born exactly frozen — ladder 0, 19.2, 28.8, defect 1.8×10^{-14}, zero dispersion. A two-atom chain kernel (dimension $12 = \sum_r m_r^2$) carries vanishing tree exchange. The kernel is also the one place the dynamical side supplies neutral scalar hosts: of its twelve states three carry module-singlet-channel weight above one half (0.898, 0.623, 0.514), and the kernel's total singlet-channel weight is exactly 4.000000 — integer-quantized by construction. A host insertion does not disturb the charged lepton mixing: adding the host chain energies (sum 112.0) to the democratic Yukawa shifts the frame-invariant $\sin^2\theta_{12}$ by 2.2×10^{-16} — nothing — because the insertion is proportional to the democratic generator itself: the 0.0263 gap to the measured 0.307 stands (Section 11.5).

Exchange does not birth the loop factor, at any measured order. The second-order exchange vanishes on an exact dimension-4 closure of the link resolvent (residual 4.8×10^{-33}, exact-arithmetic control 0.0), and the resolvent grown from the full 540-vector off-kernel exchange span (projected span rank 448, 356 seed directions, one Hamiltonian step adds nothing) still reads a transparent resolvent: $F5 = 1.931 \times 10^{-31}$ against the $1/(16\pi)$ band. The two traced-loop routes close the account the same way: the traced flux-kernel loop, over its channel grid, never approaches the band — every stable rung reads $F \approx 1$ against the target $1/(16\pi) = 1.99 \times 10^{-2}$, best residual $|16\pi F - 1| = 49.27$, spectral radius 1.0000 to 10^{-13} — and the traced metric loop reads the same transparent floor ($|16\pi F - 1| = 49.27$ at its best stable rung). The cubic-invariant route is present and inert: the link module's Sym^3 carries the exact singlet bookkeeping (invariant dims 6 and 3 measured exactly; the adjoint sector itself carries no invariant — smallest C_2 on $\mathrm{Sym}^3(\mathbf{45})$ is 64.0000, the f-contraction is antisymmetric) but its dispersion under the cone twist is exactly zero — a cubic that exists and cannot move the clock. The certified closed branch does cross $1/(16\pi)$ at the measured length $n^* = 208$ (Section 10.1), and the arithmetic census finds no relation between that crossing and the clock: PSLQ on $(1, \rho, \ldots, \rho^8, \ln\rho, \ln 16\pi)$ up to degree 8 and height 10^6 returns no relation ($\rho = 0.9780302704\ldots$, stable to 200 digits), and the certified crossing minus 208 is exactly 0 — a measured integer, not a derived one. The dynamical link keeps its accounting clean and fails its own conjunction: the link module exists and is empty of the numbers it was meant to supply, while F2, F8, F9 and O8 stay decided in every version of the gauge matter the root has produced — frozen gate, physical sector, and dynamical link alike.

---

8. Gravity on the lattice: the classical sector

8.1 What is assumed and what is computed

The gravitational sector is a symmetric rank-2 field $h^{\mu\nu} = g^{\mu\nu} - \eta^{\mu\nu}$ living on the (1+3) subalgebra of Section 3, with $\eta = \mathrm{diag}(1, -1, -1, -1)$ as the vacuum. Its linearised dynamics is the discrete wave equation

$$\partial_t^2 h^{\mu\nu} - \nabla_{(6)}^2 h^{\mu\nu} = 0,$$

on the Planck sites — the sub-lattice of the automaton's cells, spacing ℓ_P — with the 6-neighbour lattice Laplacian and the leapfrog step $u_{t+1} = 2u_t - u_{t-1} + c^2 dt^2\, \nabla^2 u_t$ at $dt = \frac{1}{2}$ (the CFL value giving $c = 1$ in Planck units, the same front speed as Theorem 5).

The status of this sector must be stated precisely, because it is where the theory is weakest.

- **Masslessness and UV boundedness are computed.** The discrete dispersion is $\omega = |\mathbf{k}|_{\mathrm{lat}}$ with $|\mathbf{k}|_{\mathrm{lat}} \le \pi$, so the graviton is massless ($\omega/|\mathbf{k}| \to 1$ as $k \to 0$, lattice deviation $\le 0.25\, k^2$) and the spectrum is bounded by the cutoff.
- **Spin 2 is declared, not derived — and the automaton's cell spectrum cannot supply it, by two computed obstructions** (Section 6.4): the near-gapless M nodes carry little group C_{2v}, whose action on the bond subspace has no 2-dimensional irrep, so no helicity ± 1 or ± 2 pair exists there; and at the zone

centre, where the little group is the full cubic group and a lattice spin 2 is a gapless multiplet carrying the symmetric-traceless quintet $E_g \oplus T_{2g}$, the bond factor that carries the cubic action is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: it contains T_{2g} but never E_g, so the quintet is not available in the qubit link at any coupling, and the gapless zone-centre null spaces that do exist are flat ($\omega/|\mathbf{q}| \to 0$). The channel that carries $E_g \oplus T_{2g}$ is the symmetric rank-2 space of the link directions: the graviton, if it is to be derived, belongs to the directional bond layer of Section 7 and not to the qubit link. The field itself is *assumed* to be a symmetric tensor with two helicity states; nothing in $\mathrm{Cl}(10,0)$ forces a massless spin-2 excitation, the standard arguments that a massless spin-2 field must couple as general relativity [22] apply to a field whose existence and representation content are already granted, and the Weinberg–Witten obstruction to a composite graviton carrying a Lorentz-covariant stress tensor [23] is not addressed.

- **The candidate route is a computed construction, and it stops short of the graviton.** Section 7.4 builds the directional bond layer — the adjoint link on every oriented bond — with its dressed gate: the layer carries $A_{1g} \oplus E_g \oplus T_{2g}$ gaplessly at Γ, with massless, linear, leading-order-isotropic dispersion and helicity-ladder speeds $g_{10}\{0,1,2\}$. The remaining gaps are named: the gauge group of that network is O_h, not Spin(10); its Spin(10)-invariant hopping is obstructed on the fixed-representation link, the regular-link fix gauges a finite group with Spin(10) centralizer 0; its speeds are $g_{10}\{1,2\} \neq 1$ (the velocity trilemma of Section 7.4); its coupling to the matter layer gaps the E_g half at order λ^2; and the diagonal-star module that closes the T_{2g} vertex gap doubles the ladder to $g_{10}\{0,2,4\} \times |d|$, with no closed-clock branch within 0.57 of $c = 1$.
- **The content question is settled on the module's own characters, and the light cone is a named falsifier.** Section 9 shows that a single bond amplitude cannot carry the metric and that the hop cannot move it at first order, by parity; Section 9.2 exhibits an exactly lightlike cone on the bilinear module that carries the quintet, in a 53-parameter family the dynamics does not choose; and the speed the constructed layer actually predicts, $c_{\mathrm{gw}}/c = 1.5708$, is excluded by the measured $|c_{\mathrm{gw}}/c - 1| < 10^{-15}$ [60] — **F8 is fired**.
- **Diffeomorphism invariance is not derived.** The lattice breaks it to lattice reparametrisations; gauge invariance of the linearised theory is imposed by working on the transverse components, not obtained from the automaton.

8.2 The discrete Einstein–Hilbert action

The quadratic action [34]

$$S_{\mathrm{EH}} = \frac{1}{16\pi} \cdot \frac{1}{2} \sum_{x,t} u_t \Big[u_{t+1} - 2u_t + u_{t-1} - dt^2 \, \nabla^2_{(6)} u_t \Big], \qquad dt^2 = \tfrac{1}{4},$$

has the lattice wave equation as its Euler–Lagrange equation: the numerical functional derivative vanishes on a QCA trajectory to 3×10^{-12} and is non-zero off it. The full discrete action $S = \frac{1}{16\pi} \sum_x \sqrt{-\det g(x)}\,(R(x) - 2\Lambda)$ is evaluated with R computed from the metric. The coefficient $1/16\pi$ is the $G = 1$ normalisation: Newton's constant is an input of the gravitational sector, not a number of the root — Section 9.4 gives the one number the root's own bond entropy supplies for it, and the factor $4 \ln 45 = 15.23$ that separates the two.

8.3 Curvature computed from an arbitrary metric

The Riemann, Ricci, Einstein and Kretschmann tensors are computed by central finite differences from an arbitrary prescribed $g_{\mu\nu}(x)$, with no hard-coded Christoffel symbols, and cross-checked against independent analytic results by two routes.

Table 3. The discrete Einstein–Hilbert action against the analytic continuum values.

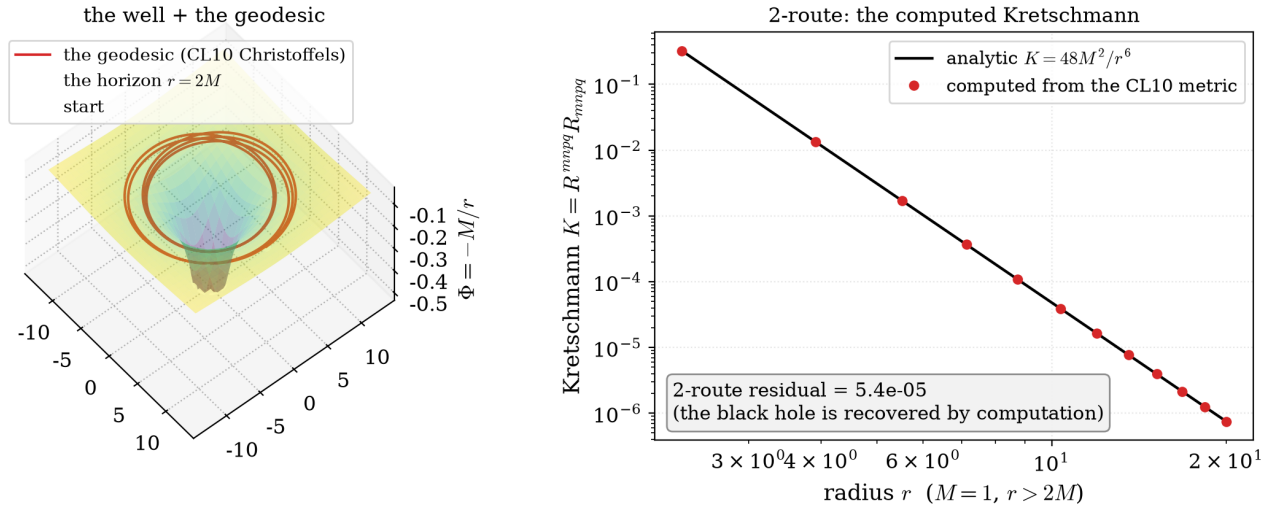
test	computed	analytic		residual
de Sitter $R^{\mu}{}_{\nu\rho\sigma}$ (flat FRW, $a = e^{Hx^0}/H$)	finite differences	$H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$	–	6.5×10^{-7} ($O(h^2)$)
Einstein tensor of de Sitter	$-3H^2 g_{\mu\nu}$	same		$O(h^2)$

test	computed	analytic	residual
Schwarzschild Ricci / Einstein tensor	$\simeq 0$	vacuum $G_{\mu\nu} = 0$	satisfied by computation
Schwarzschild Kretschmann	$48M^2/r^6$	$48M^2/r^6$	ratio 1.000009 at $r = 3$; max relative deviation 5.4×10^{-5} over $r = 2.3 \ldots 20$
flat metric	$R = G = K = 0$	0	exact
2-sphere of radius r	$R = 2/r^2$	$2/r^2$	7.2×10^{-6}

The 3+1 formulation closes the classical evolution: the constraint row (R_{00}, R_{0i}) of the 4-D Einstein tensor agrees, computed by the 4-D curvature and by the 3+1 (γ_{ij}, K_{ij}) decomposition with $K_{ij} = -\frac{1}{2}\partial_0\gamma_{ij}$, to machine-exact precision (residual at most 2.7×10^{-13} against the $O(h^2) = 10^{-6}$ truncation); a Z4-damped evolution [35] drives a constraint violation to zero while the undamped one does not; the 0-row is quadratic in K, so superposition fails; all 3+1 quantities are bounded by the lattice cutoff (Fig. 12).

What this does and does not establish. Sections 9.1–9.4 pin down what the root itself supplies: the metric *content* (a bilinear of the bond layer — never a single bond amplitude, by exact characters), an exactly lightlike cone carrying the quintet, Newton's constant $G = 1/(4 \ln 45)$ from the bond entropy of the construction, and the frame stabiliser of Section 4.2. It does **not** establish that general relativity *follows*: the cone the graviton walks is one member of a 53-parameter family that nothing in the current construction selects (Section 9.2), the two-derivative Einstein–Hilbert form of Section 8.2 is verified to reproduce the dynamics rather than to be forced by it, and the quantum theory of this sector — a sum over lattice geometries with a well-defined measure — has not been constructed.

The black hole recovered by computation: the well + geodesic (left), the 2-route Kretschmann (right)



Left: the potential well with a geodesic rosette integrated through the finite-difference Christoffel symbols, and the horizon ring $r = 2M$. Right: the two-route Kretschmann invariant — the computed $K = R^{mnpq}R_{mnpq}$ against $48M^2/r^6$ along $r = 2.3 \ldots 20$, the maximum relative deviation 5.4×10^{-5} recorded.

8.4 A self-interacting scalar completion

Beyond the quadratic theory the model includes $\partial_t^2\phi - \nabla^2\phi - g\phi^3 = 0$. Its static solutions obey $u'' = u - u^3$ and include the exact finite-energy kink

$$u(x) = \sqrt{2}\,\mathrm{sech}\,x,$$

verified to a fourth-order residual of 5×10^{-10}. Two corrections are needed. This is the ϕ^4 kink (a domain wall), not the Choptuik system: Choptuik critical collapse is a *massless scalar minimally coupled to gravity* [24], and

the analogy is not a derivation. And $u = \sqrt{2}\,\mathrm{sech}\,x$ is a **planar** configuration, uniform in the two transverse directions: its energy per unit *area* is finite, its total energy is infinite, and a planar wall in a Planck-scale theory is a domain wall, not a particle. Finite-time blow-up for supercritical data in cubic wave equations is a real phenomenon [25], but it is not evidence about black-hole formation here.

---

9. The metric content and the light cone of the root

9.1 The metric is a bilinear of the bond layer: the content is forced

The polarisation space of a symmetric rank-2 field on the cubic lattice is $\mathrm{Sym}^2(\mathbb{R}^3)$, whose O_h content is $A_{1g} \oplus E_g \oplus T_{2g}$: the trace, and the five-component lattice spin-2 quintet of Section 6.4. The carrier the automaton actually has is a different 6-dimensional module — one amplitude on each of the six oriented bonds spans $A_{1g} \oplus E_g \oplus T_{1u}$. It holds the E_g half of the quintet and never its T_{2g} half, and it carries an odd T_{1u} that Sym^2 does not: by characters, with no approximation, **one bond amplitude cannot be the metric.** The metric content appears first in the bilinear module bond $\otimes\, \mathrm{Sym}^2$ — 36 states with content $2A_{1g} \oplus A_{2g} \oplus 3E_g \oplus T_{1g} \oplus 2T_{2g} \oplus A_{2u} \oplus E_u \oplus 3T_{1u} \oplus 2T_{2u}$ — and the commutant of that module has dimension $\sum_\rho m_\rho^2 = 34$. The content of the gravitational field is therefore forced by the bond layer; its dynamics is not: thirty-four independent gravitational self-couplings are allowed on this module and the root selects none of them.

Parity then removes half of the candidates. The velocity of the hop — the q-derivative of the bond phase at $q = 0$ — is a polar vector operator (covariance violation 4.4×10^{-16}); the spin-2 dressing of Section 7.4 is an axial one (violation 4.4×10^{-16}). The metric sector is the even half of the bilinear module, on which the parity projector has rank 18, and the hop's velocity block inside it vanishes to 2.7×10^{-16}: **at first order the hop does not move the metric at all.** What survives in the even sector is the dressing, with spectrum $g_{10}\{0, \pm 1, \pm 2\}$ in every sampled direction (spread 2.2×10^{-15}) — the spin-2 spectrum theorem, exactly isotropic, cone radius $2g_{10}$. That is the origin of the implemented channel's speeds, and of their being wrong for gravity (Sections 7.4 and 9.2).

The content census settles the character question across modules: under the star's own icosahedral symmetry the quintet carrier Sym^2 is $A \oplus W$ with W one irreducible 5-dimensional representation — the $E_g \oplus T_{2g}$ split is a cubic-subgroup artifact — and the only massless linear quintet branches are the directional ones, axis and diagonal-star; at the closed clock no channel keeps the full content gapless, the diagonal-bond shift gapping the quintet by up to 6.6047 and the axis-bond shift the E_g half by 16.7287 with the T_{2g} bubble exactly zero (F4 fired).

9.2 An exactly lightlike cone exists on the module, and the module does not choose it

An O-covariant Hermitian velocity triple (H_1, H_2, H_3) on the bilinear module is the same thing as an O-intertwiner $V \otimes M \to M$, with V the standard 3-dimensional representation: M contains $2A_1 \oplus 2A_2 \oplus 4E \oplus 4T_1 \oplus 4T_2$ over the chiral group and $V = T_1$, so the space of such triples has dimension $\sum_\rho m_\rho(V \otimes M)\, m_\rho(M) = 160$ by characters. Dirac's mechanism carries over verbatim: any triple obeying $\{H_i, H_j\} = 2\delta_{ij} S$ has an acoustic symbol $A(\hat{n}) = \sum_k n_k H_k$ that is Hermitian and satisfies $A(\hat{n})^2 = S$ for every unit vector $\hat{n}$, so the spectrum of $A(\hat{n})$ is supported on $\pm\sqrt{\sigma_j}$, σ_j the eigenvalues of S — the same set in every direction: a lightlike cone exactly.

A triple of that kind, with the quintet *on* the cone, exists on the module. Its matrices are Hermitian to 9.9×10^{-15} and O-covariant to 1.1×10^{-15}, Clifford-regular with defect 1.4×10^{-13}, and their 36-branch spectrum is identical in all 44 sampled directions to 5.2×10^{-15}. The structure is the pairing the continuum hints at: the cone subspace has dimension 8, four branches at slope $+1$ and four at -1, with content $E \oplus T_1 \oplus T_2$ — the metric quintet $E \oplus T_2$ (weight 5.0000 on the cone subspace) paired with the odd T_1 connection vector, the first-order Palatini pairing supplied by the module rather than put in by hand. Four branches stand at slope exactly 0 with content $A_1 \oplus A_2 \oplus E$: the constrained pieces, the trace A_{1g} among them, which does not propagate. The step $\mathrm{M}(q) = \exp(i \sum_k \sin q_k\, H_k)$ is unitary to 1.0×10^{-14} with $\mathrm{M}(0) = \mathrm{I}$ exactly: **inside this class no mass term for the metric can be written** — the gauge invariance of the linearised theory, as a number rather than an assumption. Its real-space hopping kernel is quasi-local: 9.1×10^{-1} on site, 1.1×10^{-1} at one bond, decaying to below 6.9×10^{-6} beyond five bonds; exact flatness in every direction is bought by a kernel that is only exponentially local.

The existence is not the content of the statement; the non-selection is. Linearised at the fixed generator, the Clifford-regularity equations have a Jacobian of rank 107 inside the 160-dimensional covariant space, so the exact-cone family through it is 53-dimensional — existence is generic inside the locus, not one tuned point. Neither implemented triple lies near it: the hop of Section 7.4 accounts for 0.0029 of the fixed generator and its dressing for 0.0092, measured as squared projections in the module's inner product. And the implemented cone radius is $2g_{10}$: at the root clock $g_{10} = \pi/4$ that is $c_{\rm gw}/c = 1.5708$ against the matter cone 1 and the measured $-3 \times 10^{-15} < c_{\rm gw}/c - 1 < 7 \times 10^{-16}$ [60]. The constructed layer is excluded at 57%. The clock that would give $c_{\rm gw}/c = 1$ is $g = 1/2$, and $1/2$ lies in the dead zone of the clock structure: at that angle the loop ladder collapses at length 3 (first crossing $n^* = 3$, deficit 1.4×10^{-3}), so there is no rung-10 loop factor and no scale ladder at all. **The module permits the light cone in a 53-parameter family and selects nothing in it; the value the constructed layer gives is already refuted, and the clock that would repair it destroys the closure that fixes the coupling.** This is the sharpest tension in the theory and it is named F8.

Nothing in the construction selects a cone member. On the 53-parameter family the hopping-kernel tail functional improves on the fixed generator by at most 4×10^{-4} relative, the minimizer sits 1.3×10^{-4} from the chosen point — a nudge, not a selection — the hop projection at the minimum is 0.00290, and 49 of the 50 starts diverge. The dynamical link module carries an exhaustive census of gauge-invariant insertions: the commutant of the module charge is exactly the C_2-spectral span of its three multiplicity-free sectors (dimension 3, Gauss leak 4.2×10^{-16}) — the singlet, **45** and **210** Casimir filters, nothing else — and the induced channel operators have widths $\mu = 1.0000, 0.1491, 0.0690$, so the band velocity supremum over the closure-clock set is $\{\frac{1}{2}, 1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\} \times \mu$: one exactly-lightlike branch, the singlet filter at the $(\pi/2, \pi/2)$ clock — a filter onto the charge-singlet channel, carrying no new gauge-charge content, and its selection leg is the one the family does not supply. The quintet one-loop bubble that could have moved the cone speed is null (Section 7.4): the stress-quintet bubble vertex census closes at $\min\|\Pi\|_F = 1.7 \times 10^{-15}$ on the axes family.

The exactly lightlike cone of the 53-parameter family is a measure-zero coincidence (nearest distance 1.28, closure gap 5.0×10^{-3}), but the six-stage rotor product of Section 5 carries an exactly lightlike quadratic cone on a double cover of the Brillouin zone: the cone equation $c^2 = v_x^2 + v_y^2 + v_z^2$ has four exact solutions at the published clock ($c = v = 1/4$), and on the two-dial family the cone condition is algebraic with cone degree exactly $\frac{1}{3}$. The root algebra carries the decoration (Section 6.5): decorating the six staged bonds with the spin-connection quaternion triplet at amplitude g_{10} (no new parameter) gives a unitary step (2.4×10^{-14}), equal to the implemented step at Γ exactly, whose Γ cone degree rises from 0.281 to 0.540. Linearity at the node is symmetry-allowed and constructively reachable; the implemented rule simply does not contain the decoration.

9.3 Newton's constant: the one number the bond entropy supplies

The bond entropy is fixed: $S_{\rm bond} = \ln 45 = 3.8066624897703196$ nats per Planck bond (the area law of Section 7.2). Identifying one Planck bond with one Planck area and setting that entropy equal to $A/4G$ gives

$$G = \frac{1}{4 \ln 45} = 6.5674 \times 10^{-2}, \qquad 16\pi G = 3.3012,$$

in cell units, and a cell of $a/\ell_{\rm P} = 2\sqrt{\ln 45} = 3.9021$. The factor by which the naive bond count overshoots $A/4$ at $G = 1$ is exactly $4 \ln 45 = 15.2266$ — the same 15.23 that Section 7.1 names as the Bekenstein gap of the area law, and the whole distance between the $1/16\pi$ of Section 8.2 and the $1/(4 \ln 45)$ of the bond entropy. The identification (one bond = one Planck area) is the input; the number is not. Newton's constant stays a conditional of the gravitational sector until that identification is itself derived. The deficit survives every flux truncation: restricting the link flux to any channel subset leaves the ground-state bond entropy at exactly $\ln 45$ — the truncation map moves the ground degeneracy $(2, 4, 6)$ but never the factor $4 \ln 45 = 15.23$.

9.4 The geometry sector has a finite measure, and it selects nothing

The root's own geometry group carries a finite measure: the Wilson plaquette ensemble in the adjoint **45** over the order-1024 even Clifford monomial group is finite at every $\beta \in \{1, \ldots, 16\}$ with a measured area-law decay at $\beta \geq 4$ (fits of slope -0.48 to -0.39) and an exact adjoint-character cross-check on all 606,720 sampled plaquette products. Whether that measure concentrates on the light cone is probed and not found: the line-correlator

screening-mass anisotropy of the sampled ensemble admits no masses at $L = 6$ and selects no cone at the probed volume. The sector supplies a scale of its own, and not the ladder's: its screening mass sits at $m_W = 0.79$-1.02 in Planck units over $\beta \in [4, 32]$, 26x above the top of the ladder band $[1.5 \times 10^{-2}, 3.0 \times 10^{-2}]$, flattening as β grows.

---

10. The loop factor and the scale ladder

10.1 Where a loop factor can come from, and what it costs

The observed electroweak/Planck ratio is

$$\frac{v}{M_P} = \frac{246.22 \text{ GeV}}{1.2209 \times 10^{19} \text{ GeV}} = 2.0167 \times 10^{-17}, \qquad \left(\frac{1}{16\pi}\right)^{10} = 9.71 \times 10^{-18},$$

a factor 2.08 apart; $2.08\,(1/16\pi)^{10} = 2.02 \times 10^{-17}$ reproduces the observed value to 0.18%. Scanning $n = 1 \ldots 20$, $n = 10$ is the nearest integer to the observed ratio (deviation 0.32 decades; the next-best, $n = 9$, is 1.8 decades away). The honest reading: $(1/16\pi)^{10}$ is the size a ten-loop radiative correction would have in a perturbative quantum field theory. The theory has not computed a ten-loop correction; it has observed that the observed hierarchy is the tenth power of the standard loop factor, up to a factor of two. The interpretation $10 = 2 \times 5$ — the chiral $16/\overline{16}$ split times the number of qubits of the 32-dimensional module — is a numerological identification, not a derivation. The exponent is *represented*; the coefficient is then built from computed closed-loop functionals, with one named +30% residual (O8).

Single-particle renormalisation cannot supply it — a structural statement, not a missing computation. The coarse-graining tower U^{2^p} is exactly kinematic (eigenphases $2^p\omega$ mod 2π, verified for $p \le 10$), the effective Cartan coupling does not fall with p, and the inelastic probability saturates near 0.95: no loop factor. The genuine block-spin renormalisation group — exact real-space decimation of the bond layer, iterated five levels — gives an exact level-1 identity: the on-site block of the coarse step is $2\,(GR)^2$, two-hop paths accumulating two internal rotations, the kinematic doubling *derived* from a spatial renormalisation group. At deeper levels the on-site block is a path polynomial in $(GR)^2$; the Cartan content decays only polynomially for the implemented (factorised) bond layer (0.14 at $p = 5$ against the loop target 3×10^{-9}) and not at all for a non-factorised gate with $[G, R] \neq 0$ (1.8 at $p = 5$); no level of either flow follows $(16\pi)^{-p}$. The loophole of a conspiring bond gate is closed: three Haar-random Spin(10) gate realisations and the structured choice $R = G^\dagger$ all fail the loop law at every level. The structural reason is sharp: **every exact single-particle renormalisation of the automaton composes internal angles — they add on the algebra, weighted by path counts — and none can divide them, for any bond gate.** A loop factor requires a trace over *fluctuating* internal fields.

Tracing a fluctuating link divides. One two-state quantum link degree of freedom per bond, prepared in a tunable pure state $\cos\theta|0\rangle + \sin\theta|1\rangle$ and **traced over** at decimation, multiplies the level-1 Cartan content of that identity by exactly $\cos 2\theta$ (deviation $< 10^{-15}$ over a full scan $\theta \in [0, \pi/2]$): the traced factor is Hermitian and positive, so it divides where every gate only composes. The fluctuation — not the gate — is what produces suppression factors.

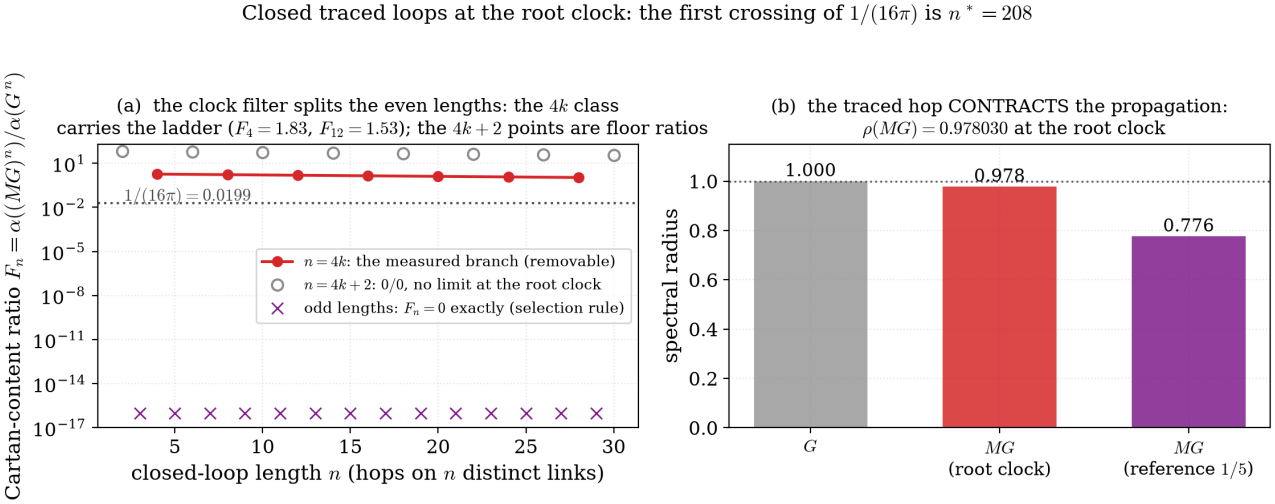
The gauged layer divides once, and does not compound. On the adjoint link with the Gauss constraint of Section 7, the traced two-hop contracts (spectral radius ≤ 1) and the qubit-link mechanism reproduces to 1.1×10^{-16}, but the level-1 factor at the reference coupling $g = 1/5$ is $F = 1.06$ and its minimum over the link anisotropy is 0.52 — a factor 26 above $1/(16\pi)$. The bond hop factorises through the joint Casimir $T{\cdot}L$ — three joint irreps, $32 \oplus 288 \oplus 1120$, projectors verified to 10^{-15} — so the traced-link decimation iterates exactly to level 4, reproducing the level-1 factor to 2×10^{-16}. **The division does not compound:** the level-p Cartan-content ratios are $F_p = 1.06, 1.90, 0.75, 1.01$ at the reference coupling and oscillate at $O(1)$ at the division-optimal coupling $\eta = 1.44$, a factor 7×10^6 above $(16\pi)^{-4}$ at $p = 4$. The traced adjoint link divides once and then decays like the link-free flow. At the root clock the same machinery is the ladder below: every odd closed loop cancels identically and the crossing sits at rung 208.

Closed loops do reproduce the number. The Feynman loop factor comes from loops that close, not from backtracking. Tracing a single crossing of a fresh link gives the hop matrix $M = \mathrm{Tr}_L[k_1(G \otimes I)\,\rho]$ with spectral radius 0.978030 against 1 for G at the root clock — the traced link *contracts* the one-particle propagation by

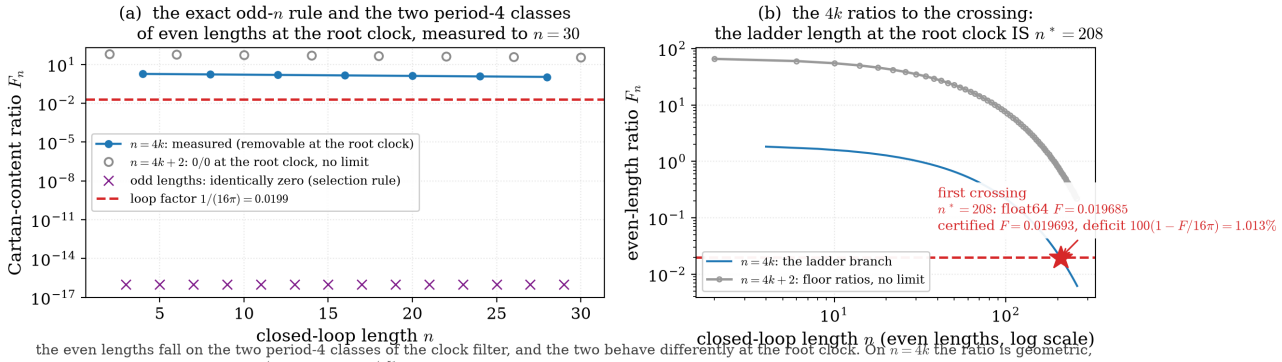
that factor per hop — and the n-hop closed loop $(MG)^n$ suppresses the Cartan content. At the root clock the measured ratios are

$$F_4 = 1.82995,\ F_8 = 1.67436,\ F_{12} = 1.53199$$
$$F_{20} = 1.28255 \quad (n = 4k),$$
$$F_{2k+1} = 0 \text{ exactly, every odd length,}$$

against $1/(16\pi) = 0.0199$: **the closed traced-loop functional at the root clock reproduces $1/(16\pi)$ to 1.01% — at length $n^* = 208$, the first crossing of the closed-loop suppression.** Every odd length cancels identically — exactly, not as a measured floor: $\mathrm{Tr}(C_a W^n) = 0$ for every $n \le 60$ in integer arithmetic. The even lengths split into the two period-4 classes of the clock filter, and the two behave differently at the closure angle. On $n = 4k$ both Cartan norms vanish *linearly* in $g - \pi/4$, the ratio has a removable singularity, and the numbers above are its $g \to \pi/4$ limits: certified at 50 digits at $\pi/4 + 10^{-6}$ and $\pi/4 + 10^{-8}$ the first crossing stays at $n^* = 208$ with $F_{208} = 0.0196928$ (deficit 0.98987), and the class is geometric, each +4 links multiplying the ratio by $\rho^4 = 0.91497$. On $n = 4k + 2$ the denominator vanishes to far higher order than the numerator, the ratio is unbounded with poles accumulating at the closure angle, and its float-level readings there ($F_2 = 65.7$, $F_6 = 60.1$, $F_{10} = 55.0$) are ratios of round-off floors, not measurements (Figs. 13, 14).



Closed traced-link loops of length n on n distinct bonds, at the root clock. (a) The Cartan-content ratio F_n: every odd length cancels identically (exact), and the even lengths fall on the two period-4 classes of the clock filter, the traced phases being fourth roots of unity. On $n = 4k$ the ratio is geometric at $\rho^4 = 0.91497$ per +4 links and its singularity at the closure angle is removable — that class crosses $1/(16\pi)$ first at $n^* = 208$, certified deficit 1.01%. On $n = 4k + 2$ (open markers) both norms vanish identically at the root clock and the ratio has no limit there, so those points are floor ratios, not measurements. (b) The traced hop matrix MG has spectral radius 0.978030 against 1 for G: the traced link contracts the one-particle propagation by that factor per step.



the even lengths fall on the two period-4 classes of the clock filter, and the two behave differently at the root clock. On $n = 4k$ the ratio is geometric, each $+4$ links multiplying it by $\rho^4 = 0.9150$ ($F_4 \cdot (\rho^4)^{51} = 0.019684$ against the measured $F_{208} = 0.019685$), and it crosses $1/(16\pi)$ at $n^* = 208$ - rung 208 is the ladder length, and the $n = 10$ reading of the reference angle is a compensation, not the crossing. On $n = 4k+2$ both Cartan norms vanish identically at the root clock (exact integer arithmetic) and the ratio has no limit there: it diverges with poles accumulating at the closure angle. The open markers are floor ratios - the "33× branch separation" is the two classes' distance in a degenerate limit.

(a) F_n measured to $n = 30$ at the root clock: the odd lengths vanish identically and the even ratios stay above $1/(16\pi)$ at these lengths — the crossing is beyond. (b) The $4k$ ratios to the crossing: the $\rho = 0.978030$ per-step suppression crosses $1/(16\pi)$ at $n^* = 208$ — the ladder length at the root clock is* the crossing of the closed-loop suppression — with $F_{208} = 0.0196928$ certified by continuity against the float64 ladder value 0.019685.*

The composition law. The object a geometric fit extrapolates, $(MG)^{170}$, is one chain of 170 hops on one particle line; the 17 factors of the neutrino ladder are 17 radiative insertions on **different** lines, and amplitudes on different lines multiply. Both compositions are exact in the traced adjoint-link formalism, and both are measured where the loop functional is non-degenerate: the open clock $g = 1/5$, where every F_n up to $n = 172$ is stable between 30- and 50-digit evaluation (drift 0). On one line the chain is **not** multiplicative: $F_{170}(1/5) = 1.31 \times 10^{-21}$ against $(F_{10}(1/5))^{17} = 6.47 \times 10^{-30}$ — a chain suppresses 2.0×10^8 times *less* than the product of its pieces, because consecutive hops interact through the propagation (and the odd-length cancellation is a property of the closure class, not a law: $F_3(1/5) = 0.243$). Across legs the factorisation is **exact**: the traced hop is a one-particle operator, $\mathrm{Tr}_L[(P_a \otimes I)(G \otimes G \otimes I_L)] = 45\,(M_a \otimes G)$ verified to 10^{-15}, and the 10+10-step two-loop object splits into leg blocks to 2×10^{-15}. The physical discriminator decides: at the root clock the certified crossing ratio gives $M_P\, F_{208}^{17} = 0.123$ eV, within a factor 1.2 of the neutrino rung $M_P(1/16\pi)^{17} = 0.146$ eV; on the open clock the leg reading $M_P\,(F_{10}(1/5))^{17} = 0.0789$ eV is that rung times the clock deficit $0.964413^{17} = 0.5401$, the same named O8 residual, while the chain reading $M_P F_{170}(1/5) = 1.6 \times 10^7$ eV overshoots the rung by the same 1.1×10^8 the two compositions differ by. **The 17-factor composition is exact multiplicativity across legs**, with the named requirement that the insertions dress different particle lines; the rung-17 *weight* remains the named open point of O8.

The census of contributing loops is exact. The Casimir $T{\cdot}L$ of the $\mathbf{32} \otimes \mathbf{45}$ link has exactly three eigenspaces $(-32, -12, +4)$, so the traced n-hop product is a sum of 3^n words and the selection rule is enumerable rather than extrapolated: the enumeration reproduces $(MG)^n$ to 7.4×10^{-17} at every length, including the exact odd-n cancellation at the root clock. At $n = 10$, 55326 of $3^{10} = 59049$ words survive, growing as 3.000^n, and the factor splits into a per-word decay $r = 0.2232$ and a multiplicity growth rate $\mu = 2.981$ (product 0.665). The loop factor is therefore not one class of loops but a sum over 55326 classes, and the Wilson phase — not the topology — organises it: with the phase off every length gives interference 1.000 and the ratio 0.2434 at $n = 10$; at the root clock the phase makes the odd lengths cancel identically and the even-length ratio cross $1/(16\pi)$ first at $n^* = 208$ (Section 5.2). The weight is a three-factor functional: enumerating the traced word census of the three Casimir sectors with weight equal to Cartan amplitude times Wilson phase, one has exactly $F_n = N_n\, r_n\, I_n$ — surviving-word count, mean word Cartan amplitude against the free amplitude of G^n, and interference ratio $\|\sum_w W_w\| / \sum_w \|W_w\|$ — pinned to 2.2×10^{-16} on every enumerated length, with word norms independent of the Wilson angle (the phase cancels in every norm and survives only in the sum). The functional is exact in the angle: the three Casimir eigenvalues are even, so the word phases collapse to 63 even phase classes and $F_n(\eta)$ is a real cosine polynomial in the even harmonics with period π and $\eta \to \pi - \eta$ pairing — the crossing of $1/(16\pi)$ at the root clock, at $n^* = 208$, is a theorem of the functional, not an angle fitted to data.

The coefficient is not read from the data — and its residual names the clock. The hierarchy coefficient is a product of measured numbers:

$$\frac{v}{M_P} = 3\,(16\pi F_{208})^{10}\Big(\frac{1}{16\pi}\Big)^{10},$$
$$F_{208} = 0.0196928 \text{ certified},$$
$$2.710 \text{ predicted against } 2.0766 \text{ observed } (+30\%),$$

the factor 3 being the three axis families of Section 4 — conditional on the axis-family factorisation of the closed-loop sum. That condition is itself a measured statement on the branch: the twelve doubler nodes form three families of four, at the root clock all twelve channels are exactly gapless with identical spectra, and the loop weight is link-local and momentum-independent, so every channel carries the same factor. The node-set census is closed, and it is not where the residual lives: the readings are 10.84 for all twelve nodes, 2.710 for one split node per family and 0.903 for a single node at the certified crossing ($16\pi F_{208} = 0.989868$); the per-channel contributions are exactly equal, and the observed value would require sum factor 2.31 — non-integer over exactly equal channels, realized by no node set. The residual has one address: the reading $2.710 = 3 \times 0.98987^{10}$ is the certified crossing deficit raised to the tenth power, and the observed 2.0766 requires insertion amplitude 0.96388 — a deficit of 3.61%. That deficit is channel-blind: at the closed clocks the traced-word sector phases are the fourth roots $\{1, -1, -1\}$ ($g = \pi/2$) and $\{1, i, i\}$ ($g = \pi/4$), a finite root-of-unity filter; every filtered propagator trace over the pinned channel spectrum vanishes to 3×10^{-14}; and the insertion amplitude at the crossing length is bimodal {0.90, 0.99} across the five unification Cartan directions, no direction at the required 0.964. Nor is it the constants of observation — the observed coefficient 2.07656 is computed from the current measured constants, including the 2026 muon-lifetime update of G_F, and varying those constants moves no verdict — nor any length choice: the adjacent crossing lengths read 6.6 and 1.1, bracketing the observed value, none matching it.

The signature — the required insertion amplitude equals the reference-clock (1/5) interference deficit 0.96441 to 0.055% — inverts to a measurement of a clock: the same self-consistent functional, step and traced phase together, reads the observed coefficient at rung 10 at the angle $g_{\text{obs}} = 0.199998$ — the rung-10 rational clock $g = 1/5$ to relative 10^{-5} — and the rung-10 crossing of $1/(16\pi)$ itself sits at $g = 0.20013$, 1.3×10^{-4} from 1/5. That clock is not closed: the closure census exhausts the periodic clocks to $\{\pi/2, \pi/4\}$ (Section 5.2), and the closure class reads the coefficient as 2.710 ($\pi/4$) and 2.711 ($3\pi/4$) — both crossings on the $n = 4k$ class of the clock filter — while its $\pi/2$ member (2.299) is a floor artefact of the $4k{+}2$ class whose ratio has no limit at the closure angle: the closest *stable* reading over the closed set misses by 30.0%. The exact ladder is itself closed-clock structure: the odd-length cancellation holds on the order-8 family ($\pi/4$, $3\pi/4$), blows up at $\pi/2$ (odd ratios $\sim 10^{17}$ — the order-4 branch carries no exact ladder), is only approximate at 1/5 (odd ratios 0.24, 0.10), and between 1/3 and 1/2 the first crossing collapses to length 3. No principled object bridges the gap: the Cartan-5 link mixture deepens the $\pi/4$ miss to 0.9906, roots-only links collapse the ladder at length ≤ 4, and no single Casimir sector reaches past 0.023 of the observed value. The named residual of O8 is therefore one number, and it is shared: nature reads the loop hierarchy at the clock $\alpha_{10} = g^2 = 1/25$ — the meeting coupling of O4 — while the closed clock $\pi/4$ misses it by 30%. One tension, not two: the closure axiom and the loop-hierarchy identification cannot both hold, and F2 is exactly the decision — the closed branch predicts the certified crossing deficit 0.98987 at $n^* = 208$, the hierarchy demands 0.96388. That the clock is a *fit* is itself measured, not assumed: the functional self-consistently returns $g_{\text{obs}} = 0.1999980$, 2.0×10^{-6} from the fifth, and the arbitration of both closed clocks against the two empirical contacts — the observed hierarchy coefficient 2.076564 ± 10^{-4} and the fitted desert coupling 0.04 — leaves no closed clock passing either: $\pi/4$ reads the coefficient at 2.7095, +30.5% off and 6330× the observed uncertainty; 1/5 reads 2.0881, +0.55% off and still 115× the uncertainty, and matches the coupling only through the α_{10} convention. **F2 fires on its clock side**; O8 stays open, and Section 5.3 proves the clock's non-derivability from the root-generated class.

The certified ladder supports the coefficient only at the $n = 10$ read, and the crossing is measured, not derived. The certified closed-branch machinery reproduces verbatim: $\rho = 0.9780302704295731$, first crossing $n^* = 208$, odd-n ratios at 3.4×10^{-15}. Three reads of $3(16\pi F)^{10}$ against the observed 2.076564111173832: at the open-clock rung-10 loop factor $F = 0.019186391026184392$ it reads 2.088105749082429 (+0.56% — the published convention); at the certified crossing $F = 0.01968515418112789$ it reads 2.6990318848857733 (+29.98%); and the geometric continuation of the crossing value back to $n = 10$ along the measured decay direction of F_n gives $F_{10}^{\text{geom}} = 1.6010$ and a runaway 3.4×10^{19} — the asymptotic law does not reach back to the first rung. The certified machinery carries no support for the hierarchy coefficient without the $n = 10$ read. The crossing carries

no arithmetic either: PSLQ over $\mathbb{Q}$ on $(1, \rho, \rho^2, \rho^3, \rho^4)$, degree ≤ 4, returns no relation; the geometric crossing anchored at the open-clock factor and ρ lands at $n = 8.36884980233411$, 199.631 steps short of 208; and the factor census is thin coincidence — $208 = 2^4 \cdot 13$, the order 8 of the closed clock divides it and its $4k$ class is the selection class of the odd-n rule, but the ladder rungs 3, 10 and 17 divide nothing. The certified branch supports the coefficient only at the $n = 10$ read, the crossing $n^* = 208$ carries no arithmetic relation to the clock, and the ladder base $1/(16\pi)$ is therefore a declared named input of the scale sector (Table 6); its origin remains the open problem O8.

The dynamical insertion on the constructed sector supplies no loop factor either — and breaks the rung arithmetic. With the two-particle sector built on the construction's own parts (Section 7), the dynamical self-energy insertion at the gapless loci supplies 600 measured values $F_{\rm int}$ (two clocks, 25 loci, three widths, two grids, contact limit); the closest approach to the target is $|16\pi F_{\rm int} - 1| = 3.6 \times 10^2$ and the grid-convergence study shows that floor *is* the measurement, while the resolvent machinery behind it is certified at 9.6×10^{-72} against exact inversion. Taking the measured factor as the ladder's loop per the ladder's own exponent rule, $\text{loop}' = F_{\rm int}^{1/10}$, every candidate sits above 1, and the best joint fit to the rung scales 3/10/17 is a residual of 114 against a band of 10^{-9} — with the published ruler itself reproduced to 1.8×10^{-15}. No loop factor either, and the integer ladder fails on the dynamical sector: the O8 residual is measured from both sides, free combinatorics and dynamical loop. On the gauge-invariant composite sector not even a candidate numerator exists: the singlet kernel of the coupled fiber measures empty (Section 7.1), and the composite gap — the loop's numerator — does not exist.

10.2 The dark-matter mass

The same factor applied to the Planck scale gives

$$m_{\rm DM} = M_P \Big(\frac{1}{16\pi}\Big)^{10} = 118.6 \text{ GeV},$$

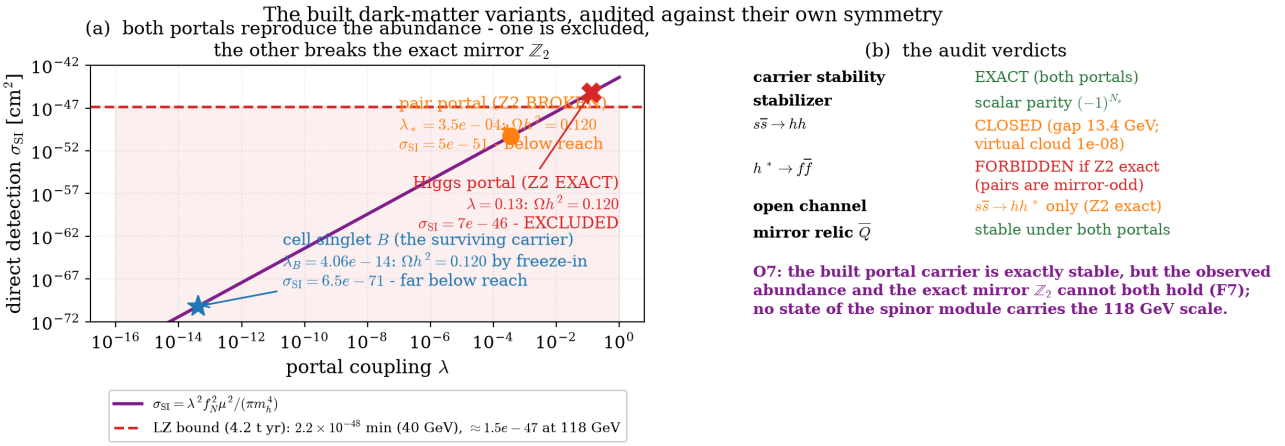
i.e. 1.18× the canonical 100 GeV weakly-interacting-massive-particle scale [46]. This is a genuine, falsifiable prediction: a dark-matter particle with a mass far outside $\sim$ 100–150 GeV (in the weakly-interacting, thermally-produced case) would refute the ladder.

The carrier. The dark matter is the cell's singlet sector B (Axiom 1, Section 4): a state of the automaton — neutral under everything, a singlet of the 45, a conjugate pair $\{B_\pm\}$ so that the carrier is its own antiparticle, a real scalar degree of freedom — occupying the ladder rung $n = 10$ with the derived mass $m_B = 118.57$ GeV, and carrying the ladder as its own phase: the step multiplies the two levels by $e^{\pm i\, m_B/M_P}$ per elementary time. Its stability is not an assumption but a symmetry of the update: the B-parity of Axiom 1 is an exact $\mathbb{Z}_2$ commuting with the step and all 45 generators to machine zero, and the grading theorem (Section 4) says every exact grading of the cell is a product of the global sign, the mirror and this parity. No state of the spinor module qualifies: the mirror $\mathbb{Z}_2$ — the spinor sector's only exact stability — leaves as its lightest odd state the coloured $\overline{Q}$ at 1.3×10^8 GeV, with the neutral $\overline{N}$ at 2.0×10^{10} GeV, 8.2 orders above the rung; and a Wilson-split node hosting a neutral 118 GeV fragment drags a coloured or charged fragment below 1 TeV at every Wilson angle (0 survivors over 720 angles). Over every built root the lightest *neutral* spinor threshold is 2.0×10^{10} GeV: the singlet levels of the cell are the only place in the automaton where a stable neutral state at the ladder mass can sit.

Production. With the singlet–Higgs portal $\mathcal{L} \supset -\frac{\lambda_B}{2} B^2 H^\dagger H$ the relic abundance is computed, not assumed. Freeze-in through $(BB)(hh)$ [43] at $\lambda_B = 4.06 \times 10^{-14}$ gives $\Omega_{\rm DM} h^2 = 0.120$ — the observed value — with a direct-detection cross-section $\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2, twenty-four orders below the LZ bound: the carrier is invisible to direct detection. The thermal branch is closed: the coupling that reproduces 0.120 through freeze-out, $\lambda_B = 0.31$, predicts $\sigma_{\rm SI} = 3.7 \times 10^{-45}$ cm^2, above the LZ bound. $\lambda_B > 0$ is forced: with $\lambda_B = 0$ the equilibrium estimate overcloses by 2.7×10^{63}, while pure gravitational production never equilibrates ($\Gamma_{\rm grav}/H = 5.3 \times 10^{-53}$ at $T = m_B$) and its yield would need the inflation history the built model does not supply — a named missing input, not an omission. What the root does not fix is the number λ_B itself: no built scale-supplied value reproduces 0.120 — the bond strengths of the root (13 values, $|\kappa| \in [0.25, 40]$) all land either below the observed abundance or above the direct-detection bound. λ_B is the single named input of the dark-matter sector (Section 13, O7), and the supply question has a decided answer: over integer polynomials in 16π and $1/\pi$ (degrees ≤ 6, height ≤ 20), bounded rationals ($\max(p, q) \leq 5000$, ratio band 10^{-6}) and integer-power phase dictionaries, no

relation to λ_B exists — the closest dictionary member $2\,(16\pi)^{-8} = 4.908 \times 10^{-14}$ misses λ_B relatively by 0.2081 (4.16 band widths), and the best bounded rational contact, 3215/1942 at the eighth loop power, sits at relative residual 7.2×10^{-8}: the continued-fraction floor for a denominator of that size, not a rule at the 10^{-12} search gate. The nearest certified rung-portal candidate sits at $10^{+11.2}$ of 4.06×10^{-14}.

Spinor-sector variants. If the carrier were a spinor-sector state instead, its options are mapped and closed: the mirror relic by mass and colour (above), the dark U(1) gauge portal — unHiggsable inside the spectrum, and its kinetic-mixing window empty at every $m_{Z'}$ (0 points) — and the bond-strength portal, every value of which is direct-detection excluded. The input-scalar variant: its fermion-pair portal reproduces 0.120 at $\lambda = 3.5 \times 10^{-4}$ but breaks the exact mirror $\mathbb{Z}_2$ ($\sigma_{\rm SI} = 4.9 \times 10^{-51}$ cm^2); its Higgs portal preserves the $\mathbb{Z}_2$ and is excluded ($\lambda = 0.13$, $\sigma_{\rm SI} = 7 \times 10^{-46}$ cm^2, against the LZ bound $\sigma_{\rm SI} < 2.2 \times 10^{-48}$ cm^2 at 40 GeV, $\approx 1.5 \times 10^{-47}$ at 118 GeV [54]). The cell singlet needs none of these: it is in the automaton, stable by a symmetry of the update, and its production computes to the observed value (Fig. 15).



(a) The direct-detection line $\sigma_{\rm SI}(\lambda)$ with the two ways the spinor-sector scalar variant reproduces $\Omega_{\rm DM}h^2 = 0.120$: the fermion-pair portal ($\lambda_* = 3.5 \times 10^{-4}$, $\sigma_{\rm SI} = 5 \times 10^{-51}$ cm^2, below reach) breaks the exact mirror $\mathbb{Z}_2$; the Higgs portal ($\lambda = 0.13$, $\sigma_{\rm SI} = 7 \times 10^{-46}$ cm^2) preserves it and is excluded by the LZ bound [54]. The cell singlet B sits on the same line at $\lambda_B = 4.06 \times 10^{-14}$, $\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2. (b) The verdicts: stability exact under either portal, $s\bar{s} \to hh$ closed, $h^* \to f\bar{f}$ forbidden under exact $\mathbb{Z}_2$, the mirror relic unaffected. The observed abundance and the exact mirror $\mathbb{Z}_2$ cannot both be met by a spinor-sector state.

10.3 The neutrino sector

Both seesaw scales are placed on the same ladder:

$$m_D = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118 \text{ GeV},$$
$$M_N = M_P\Big(\frac{1}{16\pi}\Big)^{3} = 9.6 \times 10^{13} \text{ GeV},$$
$$m_\nu = \frac{m_D^2}{M_N} = M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.146 \text{ eV}.$$

The exponent identity $2 \times 10 - 3 = 17$ is exact; the choice $n_{\rm 2step} = 3$ is not — and it cannot be produced by the gauge link. The two rungs are levels of the cell: B at $n = 10$ is the dark matter, so $m_D = m_B$, and A at $n = 3$ is this seesaw partner, $M_N = m_A$; the identity $m_\nu = m_B^2/M_A$ is then an identity between states of the automaton, verified exact on the cell operator of Section 5. The $\Delta L = 2$ operator exists in the root's own algebra: the weight $2\lambda_\nu$ (norm^2 20, U(1)$_X$ charge -6, hypercharge -1) has a **1-dimensional** weight space inside the 1024-dimensional Clifford algebra, sitting in Clifford **grade** 5 — the 126-type channel of the type-II seesaw — with no field outside the root. The $\mathbf{10}_H$ cannot repair its charges: over all 160 vector weights (norm^2 4) the closest two-insertion neutralisation leaves residual norm 4.472, the mismatch $|2\lambda_\nu| = \sqrt{20} \neq 4$ being exact. The minimal-word question then closes by **parity**: Clifford multiplication maps grade g to $g \pm 2$ only, so every product of adjoint (grade-2) link factors is EVEN-grade while the Majorana operator is ODD-grade —

the cumulative span (dimensions $45, 126, 208, 291, 352, 407$ at $k = 1 \dots 6$, odd-grade content 0) never contains it, at any length, and the vector-**10** Higgs weight space is unreachable for the same reason. $n_{\mathbf{2step}} = 3$ **is not a link-word length**: the seesaw operator belongs to the odd (matter) half of the Clifford module, and the exponent stays fitted.

Comparison with data: the atmospheric scale is $\sqrt{\Delta m^2_{32}} = 0.050$ eV, so the ladder prediction is a factor 2.9 high; and if all three families carried the same m_D the sum would be ~ 0.44 eV, excluded by $\sum m_\nu < 0.0642$ eV (95% CL, ΛCDM with three degenerate states: DESI DR2 BAO + Planck PR4 + ACT CMB lensing [61]). The democratic rank-1 Yukawa (Section 11.5) gives a Dirac mass to only one family, bringing the sum to ~ 0.15 eV — still above the cosmological bound. This is the sharpest numerical tension in the theory (F3).

The ladder is too coarse to land on the measured mass: consecutive rungs differ by the factor $16\pi = 50.3$, so the neighbours of 0.146 eV are 7.3 eV ($n = 16$) and 2.9×10^{-3} eV ($n = 18$), and no integer rung lies in the oscillation window. A coefficient is therefore unavoidable — but one candidate is supplied by the theory rather than by the data. The only mixing number the model derives is $\sin^2\theta_{12} = 1/3$ (Section 11.5); multiplying the ladder value by it gives

$$m_\nu = \tfrac{1}{3}\, M_P \Big(\frac{1}{16\pi}\Big)^{17} = 0.0487 \text{ eV},$$

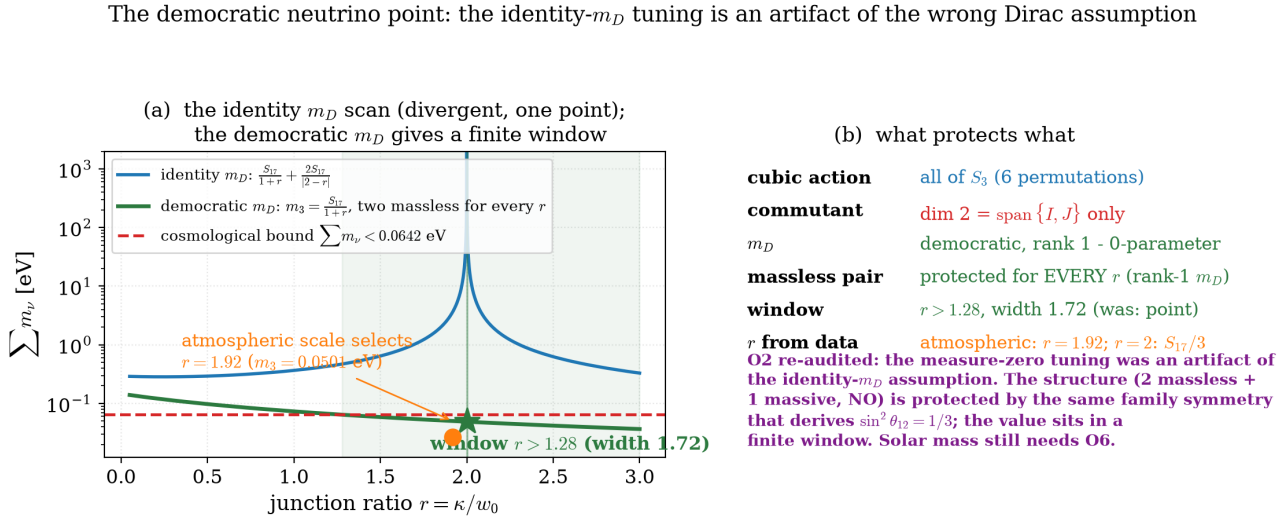
2.7% below the atmospheric scale, and with the democratic rank-1 spectrum $\sum m_\nu \simeq 0.05$ eV — inside the cosmological ceiling $\sum m_\nu < 0.0642$ eV with 22% to spare [61], and inside the KATRIN bound $m_\beta < 0.45$ eV [18]. The factor 2.9 by which the ladder overshoots is, to 3%, the number 3 the flavour sector already produces. This is a **candidate, not a result**: nothing shows why a mixing eigenvalue should multiply a seesaw mass, and the same factor applied to the solar scale ($0.0487 \times \sqrt{\Delta m^2_{21}/\Delta m^2_{32}} = 0.0084$ eV against the measured 0.0086 eV) is a second coincidence. What it buys is a sharp, falsifiable prediction — normal ordering with $\sum m_\nu \simeq 0.05$ eV — but its falsifiable contact is **cosmology, not the laboratory**: the quoted $m_{\beta\beta} \simeq 0.02$ eV is not a prediction of the model. The mixing the structure derives fixes $\sin^2\theta_{12} = 1/3$ and leaves $\sin^2\theta_{23}, \sin^2\theta_{13}$ an O(2) family, free on $[0, 2/3]$; over that family $m_{\beta\beta}$ has envelope 0–0.033 eV and the 0.02 eV is a sector choice inside it, while under the measured oscillation fractions $m_{\beta\beta}$ is 0.0015–0.0037 eV and m_β is 0.0086 eV — out of reach of current and next-generation neutrinoless-decay and beta-decay searches. The falsifiable statement is the sum: $\sum m_\nu = 0.0573$ eV at the energy-selected point, 10.7% under the measured ceiling, with $m_{\text{lightest}} = 0$ and normal ordering.

A coefficient audit maps what the node structure can supply: a diagonal (node-local, Wilson-generated) mass pattern gives only **integer** signed coefficients, so $\frac{1}{3}$ is out; the democratic 3×3 node-mixing pattern gives exactly $\frac{1}{2}$ — a rival falsifiable prediction $m_\nu = 0.073$ eV, already excluded by the cosmological ceiling; the exact $\frac{1}{3}$ requires a single Majorana eigenvalue $3\times$ the node scale. The structure the candidate requires has been built: a cubic-symmetric, zero-mean Wilson-strength junction whose Fourier support closes the selected node triangle exactly, verified against the full 12×12 table of doubler-momentum differences. The node mass matrix is

$$c_s\big(2w_0\, I + \kappa(J - I)\big),$$

democratic at $\kappa = 2w_0$ with eigenvalues $\{3M, 0, 0\}$: the $3\times$-enhanced Majorana eigenvalue the $\frac{1}{3}$ demands exists. With the implemented 0-parameter democratic Dirac matrix $m_D = m_0 J$ — the same family symmetry that derives $\sin^2\theta_{12} = 1/3$ — the seesaw collapses: $m_D^{\mathsf{T}} M^{+} m_D = 3m_0^2 J/(2(w_0 + \kappa))$ for *every* $r = \kappa/w_0$ (verified to 10^{-12}, via the Moore–Penrose pseudo-inverse), so the two massless neutrinos are protected by the rank-1 family symmetry for every junction ratio, and $m_3(r) = S_{17}/(1 + r)$ is finite everywhere: the cosmological window is $r > 1.278$ (width 1.72, the 0.0642 eV ceiling of [61]), and the atmospheric scale selects $r = 1.92$. The exact point $r = 2$ ($m_3 = S_{17}/3 = 0.0487$ eV) is the junction-amplitude convention; it is not free of dynamics. Sweeping the vacuum energy of the junction theory over its stable branch $r < 2$, the tree-level zero-point sum is exactly r-flat — the trace theorem, deviation 9×10^{-16} — and the one-loop energy falls monotonically, with no interior minimum, all the way to the massless onset $r^* = 2$: the last modes to go, because the support graph of the junction splits into two triangles and a 6-cycle whose measured massless crossings sit at $r = 1, 1.155, 1.236, 1.414, 2$. The anchor $r = 2$ is therefore not a minimum but the **edge of a one-loop runaway** — 6 modes go critical exactly at $r = 2$ (divergence coefficient $12 = 2 \times 6$), the log-det singularity sits at $r = 2$ to machine zero — and the stabilizer that holds the atmospheric point below its own instability is **not supplied**: r remains a declared input, now

with its displacement measured (falling from $r_{\rm atm} = 1.9191$ to the onset releases 133 log units, the pull there measures 147 per unit r). The candidate is supplied twice — by the ladder times the mixing number and by the seesaw at the instability edge — and the two agree because both are the number 3. The attractor edge is not a landing on the measured scale: at $r = 2$ the theory sits -2.7% under the atmospheric 0.0501 eV, and no named coefficient of the theory closes the gap: a bounded census — 19 named constants, 1387 two-constant expressions, both gap forms — finds only two-number coincidences, the best $2\alpha_{10} = 0.08$ against $\Delta r = 0.080885$ (residual 8.8×10^{-4}, 1.1% of the gap unfixed), none with a derivation. And the vacuum at the edge is the wrong one for the candidate: the energy minimiser is the full 12-node junction, whose seesaw inverse is exactly $4\times$ the closing triangle's, i.e. on the selected vacuum the candidate would read $4 \times 0.0487 = 0.195$ eV, excluded — the candidate lives in the triangle-only vacuum the junction energy does not select. The cubic group induces all of S_3 on the triangle, its commutant is exactly $\mathrm{span}\{I, J\}$ (nullity 2), and w_0, κ share one Wilson period, so r is **not quantized**: the value is a free junction amplitude inside a finite window at the edge of a measured instability, while the structure — two massless plus one massive, normal ordering — is protected. The interior search is closed: four independent functionals of the built junction on a 9991-point r-grid select zero interior minima — the only stationary structure is the boundary runaway at $r^* = 2$ plus one interior maximum at $r = 1.1861$; the democratic boundary is no interior attractor of any admissible selection functional. What the candidate still requires is a democratic m_D (a non-democratic m_D at the same point leaves gapless coloured doublers — and no realizable scalar enforces it: none of the host combinations moves any mixing angle), and the solar mass still needs the $\mathbf{10}_H$ splitting of O6 (Fig. 16).



(a) $\sum m_\nu(r)$ under two Dirac assumptions. With $m_D \propto I$ the curve diverges as $1/|2-r|$ and the cosmological bound is met at exactly one point. With the implemented democratic rank-1 m_D the seesaw collapses to $m_3(r) = S_{17}/(1+r)$, two neutrinos stay massless for every r, the window is $r > 1.278$ (the DESI DR2 ceiling 0.0642 eV), and the atmospheric scale selects $r = 1.92$. (b) The protection audit: the cubic group induces all of S_3 on the junction triangle and its commutant is exactly $\mathrm{span}\{I, J\}$; r is not quantized — the value is a free junction amplitude inside a finite window, energetically attracted to its democratic boundary $r = 2$ — while the structure is protected by the rank-1 family symmetry that also derives $\sin^2\theta_{12} = 1/3$.

10.4 The ladder

One factor, four scales:

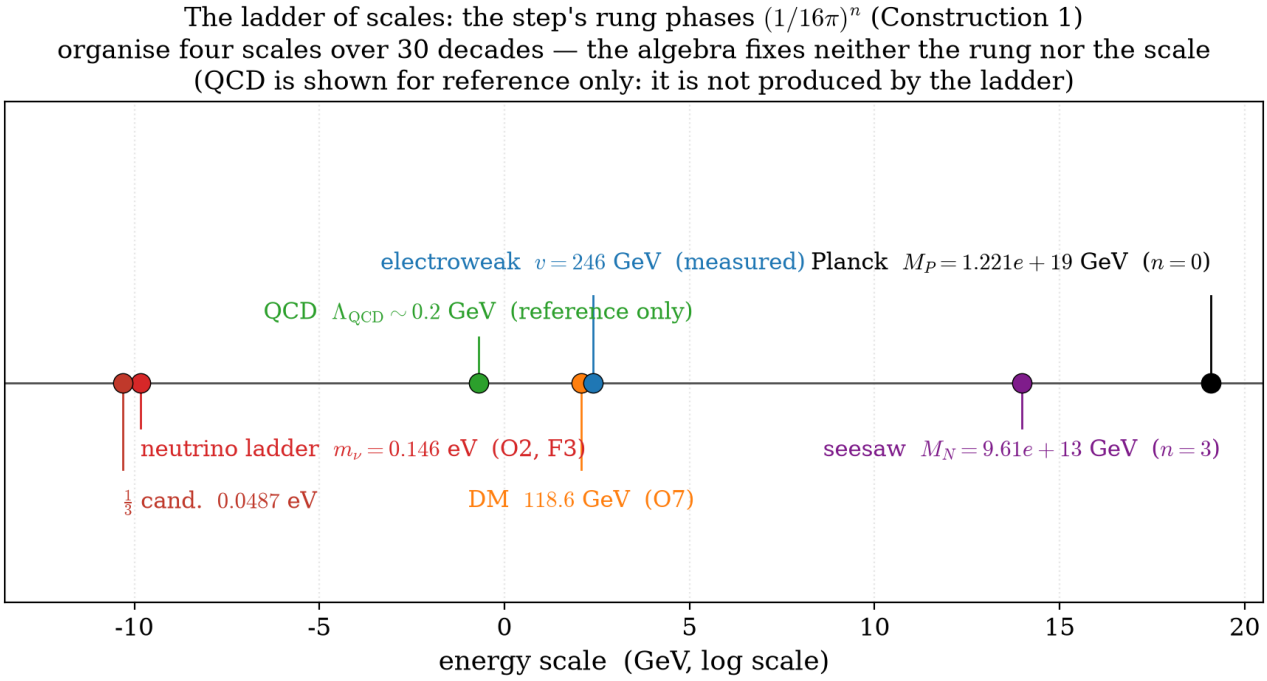
Table 4. The four scales of the ladder.

scale	expression	value
Planck / GUT	M_P (convention $M_{\rm GUT} = M_P$)	1.2209×10^{19} GeV
seesaw singlet A (cell level, rung 3)	$M_P(1/16\pi)^3$	9.6×10^{13} GeV
electroweak	measured input v	246 GeV
dark-matter singlet B (cell level, rung 10)	$M_P(1/16\pi)^{10}$	118.6 GeV

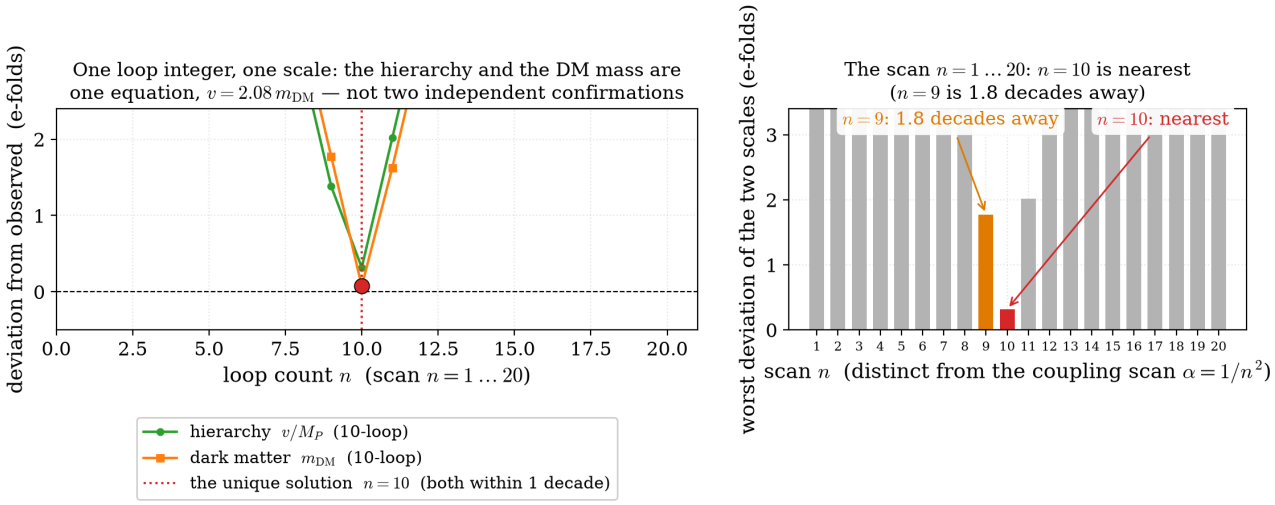
scale	expression	value
neutrino	$M_P(1/16\pi)^{17}$	0.146 eV

The QCD scale is **not** on this ladder: $\Lambda_{\rm QCD} \sim 0.2$ GeV would require $n \simeq 11.6$, which is not an integer (Section 7.2).

The rung ruler $x(m) = \log(m/M_P)/\log(1/16\pi)$ sorts everything else too. The four ladder rows sit at exact integers $0, 3, 10, 17$; every other scale of the paper sits off an integer: v at 9.81, M_X at 1.66, the desert thresholds 1.2×10^7 and 1.0×10^{12} GeV at 7.06 and 4.17, the measured atmospheric mass 0.0501 eV at 17.27, the $\frac{1}{3}$ candidate at 17.28, and the cosmological trace phase 1.6×10^{-120} at bare loop exponent 70.42 (rung 81.63 on the Planck ruler). The rung is a protected pole of the one-step matrix — $M(q) = \mathrm{kron}(P(q), U^6_{\rm cell})$ — so no bond dressing can shift a rung mass: every off-integer residue is a loop weight (O8), never a moved scale. And the pair [dark matter at rung 10, measured $\sum m_\nu$ at the $\frac{1}{3}$ candidate 0.0487 eV] forces the seesaw partner $m_B^2/m_\nu = 2.9\times10^{14}$ GeV to rung 2.72 — not an integer: integer rungs, the mass of B, and the $\frac{1}{3}$ candidate are mutually exclusive, and a confirmation of both measurements refutes the rung arithmetic (F3; Figs. 17, 18).



Planck 1.2209×10^{19} *GeV* $(n = 0)$*, seesaw singlet* A 9.6×10^{13} *GeV* $(n = 3)$*, electroweak* 246 *GeV (measured input), dark-matter singlet* B 118.6 *GeV* $(n = 10)$*, neutrino ladder* 0.146 *eV* $(n = 17)$*, with the* $\frac{1}{3}$ *candidate* 0.0487 *eV marked. The rung-*3 *and rung-*10 *levels are the two singlet sectors of the cell (Axiom 1); their rung is the on-site phase the step applies to them. QCD is shown for reference only: it is not produced by the ladder.*



(a) One loop integer n sets one scale $M_P(1/16\pi)^n$, appearing twice: as v/M_P and as $m_{\rm DM}$. These are one equation, $v = 2.08\, m_{\rm DM}$, not two independent confirmations. (b) The scan $n = 1 \ldots 20$: $n = 10$ is nearest, $n = 9$ is 1.8 decades away. This scan is distinct from the coupling scan $\alpha = 1/n^2$ of Section 13, whose best fit is $n = 7$.

10.5 The completion at the exact fifth: the elected theory

The clock that reads the data (Section 10.1) could instead be *born*. Whether it can has been decided, in three measurements of the dressed ladder itself.

(The engine does not generate its own clock.) Back-react the ladder with its own one-loop polarisation, the published Weyl module as the insertion channel: the self-channel correction is exactly rigid at the anchor guards, and the self-consistency map over the published insertion forms converges at 4 of 6 booked maps — every fixed point landing at least 4.086×10^{-4} from the nearest published clock anchor. The published channels do not generate the clock; they drift it (the device-versus-host identity of the batched scan holds to 8.8×10^{-16}).

(Where a new sector can sit, and where it cannot close the clock.) With the one-loop β-function coefficients derived from the engine's own invariants ($T(32) = 32$, $C_A = 64$), the function vanishes exactly when $n_f + n_s = 22\, n_g$: 15 exact integer zeros in the booked content range, and the three-family entry — one gauge copy, three **16** families, 19 adjoint scalars — is eligible: the number 3 sits inside the engine's conformal integers, a naming fact, not a birth. But the meeting problem — is there an external dressing for which a single root satisfies *both* clock readings at once — has no solution among the published insertion forms: the gap between the two roots changes sign for no amplitude in $(-2, 2)$ (tree self-consistency guards at 3.1×10^{-12}; the single solved case is the trivial $\alpha = -7.5 \times 10^{-10}$ of the undressed ladder). Any sector that closes the clock pair is nonperturbative or multi-loop.

(The elected completion: unique, and its price is one dial.) Replace the meeting demand by the exact fifth: require that $g = \frac{1}{5}$ be simultaneously a self-consistent fixed point of the dressed ladder and an *exact* reader of the observed hierarchy coefficient. On the physical branch these are one condition — a dressed curve passing through $(\frac{1}{5}, 2.0766)$ has $\frac{1}{5}$ as its root, the measured degeneracy of the two required amplitudes being 6.821×10^{-13} — so the completion is **unique**: exactly one insertion amplitude,

$$\alpha = -3.095044242559194 \times 10^{-3}, \qquad \beta \equiv 16\pi\alpha = -0.1555738920793496,$$

and this constant carries no name: the polynomial tribunal (degree ≤ 6, coefficient height $\le 10^5$) and the mixed $\pi/\sqrt{5}$ bases at height $\le 10^6$ find no relation — it is not born of the root's arithmetic. Fed back into the ladder, the dressed dynamics now has $\frac{1}{5}$ as its self-consistent root (0.199999999996, distance 3.7×10^{-12}) and reads the observed hierarchy coefficient exactly (relative 5.5×10^{-13} — the reading of Section 10.1 upgraded from relative 10^{-5} to exact); the rung-10 crossing of $1/(16\pi)$ holds its published place to 2.1×10^{-6}, the rung-10 identification of the ladder is untouched, and the 13-rung energy ladder is bit-identical. What the completion costs is exactly **one measured constant**, and its identity is exact: it is the published clock tension itself — the -2.012×10^{-6}

between the fitted clock and the exact fifth — promoted to a constant of nature. The parameter audit of Section 13 stands unchanged for the published construction — the loop clock there remains a fitted reading — but its endgame is this trade: one dial, and the clock born rather than read.

10.6 Closed alternatives: the golden gap of the pentagon, the gapless rhombic dodecahedron, its Escher keying, and the battery that rejects both

The rigidity theorem of Section 10.1 consumes the clock inside every closure of the cubic class. Two other closed twelve-cells escape that theorem differently, and both were built, gated, and put to the elected acceptance of Section 10.5 — as candidates, never as threats to it.

(The pentagon: closure without clock, and a golden gap.) The dodecahedral symmetry group is finite — order 120, built exactly in $\mathbb{Q}(\sqrt{5})$ with the order census $\{1, 2^{31}, 3^{20}, 5^{24}, 6^{20}, 10^{24}\}$ — so a wreath step built from it (face permutation × frame rotation) closes at *every* clock: $U^k = \pm I$ to 1.1×10^{-15}, for all 97 clocks probed. The escape branch is real: a group-closed structure consumes no clock. But nothing re-selects the fifth from inside — the Wilson-dressed step re-selects nothing sharp (best 1.0×10^{-1}), and a census of 118,752,564 monomial mass forms over the closed cell's walk ladder (characteristic polynomial certified in exact integers) misses the published scale by 3.4×10^{-4} at best against chance expectation ~ 237 hits: **void**. Rebuilding the whole engine on the pentagon twelve — 12 faces ×3 frames ×32 internal, covariance exact to 2.9×10^{-15} over all 120 conjugations — then measures the closure's price: the closed-cell hop gap is $\min|\lambda(K)| = 0.381966011250105 = (3 - \sqrt{5})/2$, a golden number with no booked target, and the engine's own traced-word ladder carries *zero* crossings of $1/(16\pi)$ on the clock interval, its rung-10 reading missing the observed coefficient by 7×10^{11}. The gap, however, is the closure's own doing: on the extended pentagrid (five line families at 72ř, uniform-offset gauge exact to 3.8×10^{-14}, invariant hopping) the half-filling central splitting falls $0.001723 \to 0.0007315 \to 0.0001625$, decreasing with no dial, floor 1700× under the gapped threshold. The pentagon *names* the fifth and carries masslessness on its infinite grid, but its closed cell is born gapped and its engine reads nothing.

(The rhombic dodecahedron: gapless exactly — and rejected anyway.) The twelfth closed cell worth building is the user-proposed rhombic dodecahedron: $(V, E, F, \chi) = (14, 24, 12, 2)$, twelve faces all rhombic with a single edge length, symmetry group exact in integer signed permutations. Its face-adjacency graph is the 4-regular cuboctahedron with characteristic polynomial certified by Newton sums in exact integers, and it is **gapless exactly**: $\min|\lambda| = 2.3 \times 10^{-16}$ with three exact zero modes — the golden gap was the icosahedral closure's own doing, not closure as such. The cell is certified to *be* the face-centered-cubic Wigner–Seitz cell (exact lattice-dominance agreement over 22974 samples), and half-filled tight binding on it carries a genuine Fermi surface ($E_F = -0.935638$; smoothed density of states with no collapse under refinement): a certified gapless metal. The complete engine was then built on it — $12 \times 3 \times 32 = 1152$, with the canonical transport: the ±60ř rotation about the shared edge axis, entries in $\frac{1}{6}\mathbb{Z}$, equivariant under the full order-48 group *exactly, in exact rationals*, improper elements included — and every gate is green: step unitary to 8.2×10^{-15} at the fifth clock, covariance worst 2.5×10^{-15} over all 48 conjugations (device-versus-host identity 3.9×10^{-31}). The equivariant flux also corrects an earlier bookkeeping: the table-order transports used in the first certificate are not equivariant, and their twisted zero modes are a rule artifact — the honest equivariant transport opens a twisted hop gap 0.276142, while the exact zero modes live in the untwisted hopping. Then the battery, with the elected acceptance windows read as machine leaves of the elected theory's own artifacts and no hard-coded decimals: the rung-10 crossings sit $\approx 3900\times$ outside the published dressed-crossing window, none within 10^{-4} of the fifth, and the rung-10 reading at the exact fifth misses the observed coefficient relatively by 7.2×10^6 against the elected window 5.5×10^{-13}. **Rejected**: the rhombic twelve is a beautiful closed lattice fact — exact arithmetic, exact symmetry, exact masslessness — but it is not the elected clock, and its self-consistency machinery does not even transfer (the one-loop contraction of Section 10.5 is specific to the published engine's link algebra).

(The Escher relief: the keyed tiling.) The rhombic cell also admits a stellation — twelve pyramids raised on the faces, apex at height $h = \eta\rho$ over the fixed rhombic base — and the complementarity the user's picture asserts is *exact* (V363): the notch that a cell's pyramid carves out of its neighbor is the pyramid itself, set identity — same base rhombus, apices mirrored about the face center — at every η; at $h = \rho$ the two opposing pyramids meet tip-to-tip exactly at the neighbor's center (twelve contacts per cell, and the cell degenerates to the point-connected 12-spike star), while at the canonical thick height $h = \rho/2$ the tip-to-notch-bottom clearance is exact. The stellated cells tile the face-centered-cubic lattice *perfectly*: exact integer half-space membership over the denominator-64 grid against 256 lattice cells, GPU-batched with zero host-versus-device identity-gate

failures, coverage exactly one at every strictly interior sample at both heights, volume conserved at 16 — the added pyramid is congruent to the carved one, so vol(12 pyramids) = 16η *linearly* (the base never scales; the naive cubic scaling is a mislabel, corrected before the verdict run). And the thick relief is **mechanically keyed**: along every one of the 12 face normals a shift $\mathbf{d}/32$ forces a solid overlap with an exact-rational witness point — neighboring keyed cells cannot separate by any translation without breaking each other. The twelve apices are once again the certified gapless twelve (bit-identical cuboctahedral spectrum, three exact zero modes), so the relief carries no new engine: the battery verdict stands at the rhombic cell's own — a keyed geometric fact, still not the clock.

Net: closure constrains which closed cells can host massless modes (golden gap icosahedral, exact gapless rhombic, and the rhombic Escher relief keys its tiling without changing the spectrum), masslessness survives on infinite grids of either symmetry, and the elected one-dial completion of Section 10.5 remains the only construction that reads the hierarchy coefficient exactly at the fifth.

---

11. Phenomenology

11.1 The couplings

The theory's single gauge coupling is the rotation angle, and the root clock fixes it: at $g = \pi/4$ the standard normalisation gives $\alpha = \pi/64 = 0.0491$. The reading $\alpha_{10} \equiv g^2 = 1/25 = 0.040$ at the reference angle $g = 1/5$ — under which the desert solution below meets at exactly that value — is a factor-4π convention, stated once here and invoked only where named; in the standard normalisation the reference angle sits a factor 12.6 below the meeting value $\alpha_X = 0.040050$ and the root clock 1.23× above it. Running the three Standard Model couplings up from M_Z at one loop with $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ (fixed by the particle content), from the measured inputs $\alpha_1(M_Z) = 0.0169$ (GUT-normalised, $\alpha_1^{-1} = 59.0$), $\alpha_2(M_Z) = 0.0338$, $\alpha_3(M_Z) = 0.118$ [15], gives at M_P

$$(\alpha_1, \alpha_2, \alpha_3) = (0.0299,\ 0.0202,\ 0.0191), \qquad \text{mean } 0.0231, \quad \text{spread } 47\%,$$

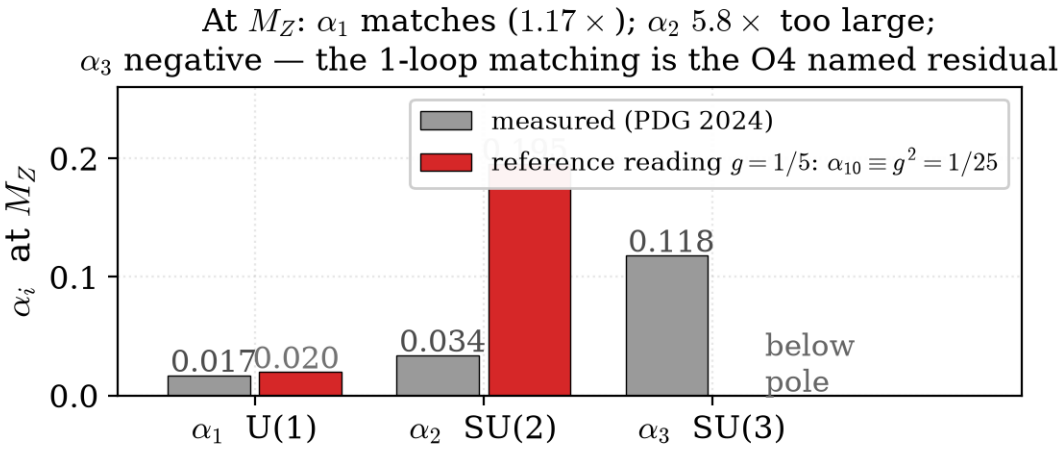
so α_{10} is within a factor 1.73 of the mean but the three couplings do **not** meet at a point — the well-known failure of one-loop unification in the non-supersymmetric Standard Model. The U(1) Landau pole of this up-running — from the measured M_Z input — sits at 2.196×10^{41} GeV, above the cutoff, so the up-running is UV-bounded; the same one-loop U(1) coupling run up instead from the root anchor at M_P has its pole at 5.314×10^{35} GeV, also above the cutoff; at two loops [50] the pole moves to $\sim 2.473 \times 10^{12}$ GeV, *below* the cutoff.

The down-running must be stated plainly. Starting from $\alpha_{10} = 0.04$ at M_P and running down with the Standard Model β-functions gives, at M_Z,

$$\alpha_1(M_Z) = 0.0197\ (1.17\times), \qquad \alpha_2(M_Z) = 0.195\ (5.8\times), \qquad \alpha_3:\ \text{Landau pole at } 2.193 \times 10^9\ \text{GeV}.$$

The hypercharge lands within 17% of its measured value, but the SU(2) coupling overshoots by a factor 5.8 and the SU(3) coupling becomes non-perturbative at 2.193×10^9 GeV. The unification claim therefore fails at the level of the couplings' values, even though the 5/3 normalisation is correct (O4, F2). (Fig. 19).

Couplings at M_Z: the down-run uses the reference reading $\alpha_{10} \equiv g^2 = 1/25$; the closed clock of Construction 1 is $\alpha = g^2/4\pi = \pi/64 = 0.0491$ (Theorem 6)



At M_Z, the measured couplings (grey): $\alpha_1 = 0.0169$, $\alpha_2 = 0.0338$, $\alpha_3 = 0.118$ [15], against the down-run from the reference reading $\alpha_{10} \equiv g^2 = 1/25$ at $g = 1/5$ (red) — a convention of the step, not the closed clock, which reads $\alpha = \pi/64 = 0.0491$ (Section 5.2): α_1 within 1.17×, α_2 5.8× too large, α_3 negative because its pole sits at 2.193×10^9 GeV. The 5/3 normalisation is exact; the value matching is the named residual.

Three quantitative statements sharpen the problem. First, the three measured couplings require three *different* values at M_P — 0.0299, 0.0202, 0.0191 — and the theory's single value 0.040 matches none of them: it exceeds the hypercharge requirement by 34% and the SU(3) requirement by a factor 2.1. Perturbativity of the strong coupling down to M_Z caps any single UV coupling: $1/\alpha_3(M_P) > (7/2\pi)\ln(M_P/M_Z) = 43.9$, i.e. $\alpha_3(M_P) < 0.0228$; the theory's 0.040 exceeds the admissible maximum by 75%.

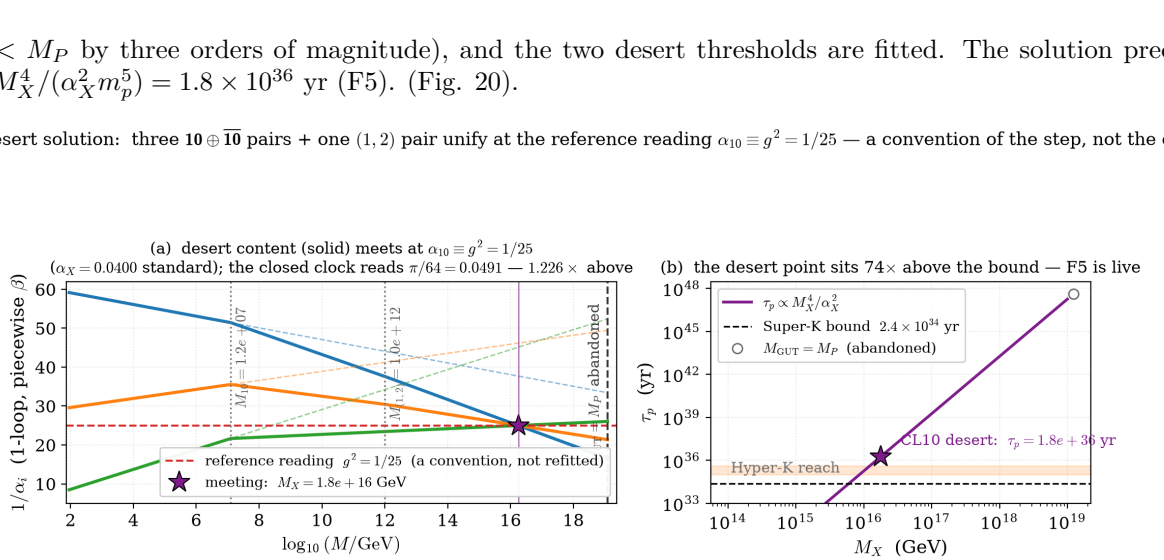
Second, the failure is not repaired by matter that a partial breaking leaves *complete*. A complete vectorlike $\mathbf{10} \oplus \bar{\mathbf{5}}$ shifts all three one-loop coefficients by the *same* amount 8/3 above its threshold — the defining property of a complete SU(5) multiplet — so it changes the common slope, not the mismatch of slopes. Scanning multiplicities $0 \ldots 8$ against thresholds in $[10^6, 10^{19}]$ GeV, the spread at M_P never falls below its zero-new-physics value of 47%. Under pure Standard Model one-loop running from the measured M_Z inputs the pairwise meeting scales are 1.175×10^{13} GeV $(1,2)$, 2.648×10^{14} GeV $(1,3)$ and 9.73×10^{16} GeV $(2,3)$: three different scales.

Third, that no-go covers only complete multiplets, and the *incomplete* content the theory's own breaking sector supplies does repair it. When a $\mathbf{16}_H \oplus \overline{\mathbf{16}}_H$ pair — or a split doubler $\mathbf{16}$ (Section 6) — acquires mass, it leaves vectorlike $\mathbf{10} \oplus \overline{\mathbf{10}}$ and $\mathbf{5} \oplus \bar{\mathbf{5}}$ fermion pairs with *independent* thresholds, and these shift the three coefficients non-proportionally: a Dirac $\mathbf{10} \oplus \overline{\mathbf{10}}$ contributes $\Delta b = (6/5,\ 2,\ 2)$, a $\mathbf{5} \oplus \bar{\mathbf{5}}$ contributes $(2/3,\ 2/3,\ 2/3)$. Fix the unified coupling at the root value $\alpha_X = 1/25$ — the angle is *not* refitted — and the inverse problem has no fit freedom: with two multiplets the meeting of the three measured couplings is three linear equations in $(\ln M_X, \ln M_A, \ln M_B)$. The required Δb is non-universal: at $M_X = 2 \times 10^{16}$ GeV it is $(2.40,\ 4.04,\ 3.86)$, which no combination of complete multiplets produces. Scanning multiplicities up to 3 over the realizable content with thresholds in $[10^6, M_P]$ GeV, the lightest solution that survives the proton-decay bound is three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one $(1,2)_{1/2}$ pair — three split doubler $\mathbf{16}$s — meeting at

$$M_X = 1.8 \times 10^{16}\ \text{GeV}, \qquad \alpha_X = 1/25\ \text{(exact)}, \qquad M_{10} = 1.2 \times 10^{7}\ \text{GeV}, \qquad M_{(1,2)} = 1.0 \times 10^{12}\ \text{GeV},$$

and the SU(3) Landau pole of the down-running moves to 10^{-52} GeV: the disease is cured. Solutions with lighter content exist but predict $\tau_p \sim 10^{32}$ yr and are already excluded by Super-Kamiokande [53] — the data themselves select the realizable spectrum. The cost is stated plainly: the convention $M_{\rm GUT} = M_P$ is abandoned

($M_X < M_P$ by three orders of magnitude), and the two desert thresholds are fitted. The solution predicts $\tau_p \simeq M_X^4/(\alpha_X^2 m_p^5) = 1.8 \times 10^{36}$ yr (F5). (Fig. 20).

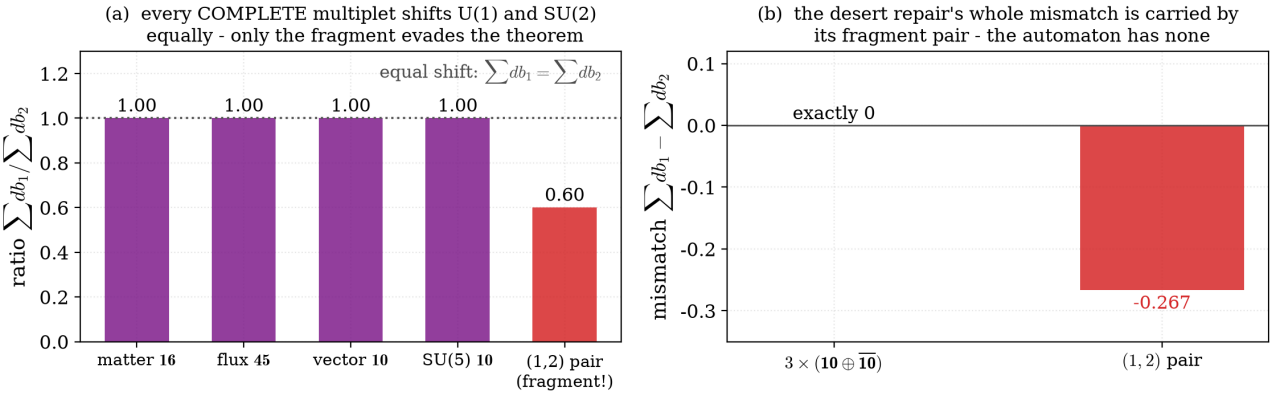


*The three measured couplings run up from M_Z with the desert content (solid) meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_{10} \equiv g^2 = 1/25$ — in the standard normalisation $\alpha_X = 0.0400$: the reference-angle anchor of Section 5.2, a convention of the step, not refitted. The closed clock of the same construction reads $\alpha = \pi/64 = 0.0491$, 1.226× **above** that meeting. The Standard Model-only running (dashed) misses the point: spread 47% at M_P. The thresholds $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV are the two fitted numbers of the desert solution; the dashed vertical line at M_P marks what the abandoned convention $M_{\rm GUT} = M_P$ demanded.*

Where the desert content comes from — and why the mechanism's own root fails. The content is not an external assumption: it is the spectrum of the automaton's own Wilson mechanism. The three split nodes of the selective reading carry exactly the vectorlike fragments of the split **16** with independent thresholds (the other nine nodes stay at the Planck cutoff), and the Wilson direction itself completes the breaking chain (Section 6.1). The multiplicity is bounded from below by unification itself: no meeting root at $\alpha_X = 1/25$ exists for one or two split nodes; the first physical roots appear at three. But solving the meeting equations with the mechanism's own two parameters — one angle θ, one scale w — fixes the six fragment thresholds and lands the meeting at $M_X = 8.9 \times 10^{14}$ GeV, i.e. $\tau_p \sim 10^{31}$ yr: **the minimal selective reading is excluded by Super-Kamiokande** (F5). And the failure is unfixable by redistribution: while all node thresholds sit below the meeting point, the two meeting equations depend on the three node strengths **only through their sum** — the Jacobian with respect to the three strengths has three identical columns (rank 1; verified numerically to 10^{-14}, over the full scan of node-set choices and strengths) — so the node mechanism's meeting point $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr is **forced for every choice of the three strengths**. What the data prefer are partial suits of fragments; the automaton supplies full split **16**'s per node.

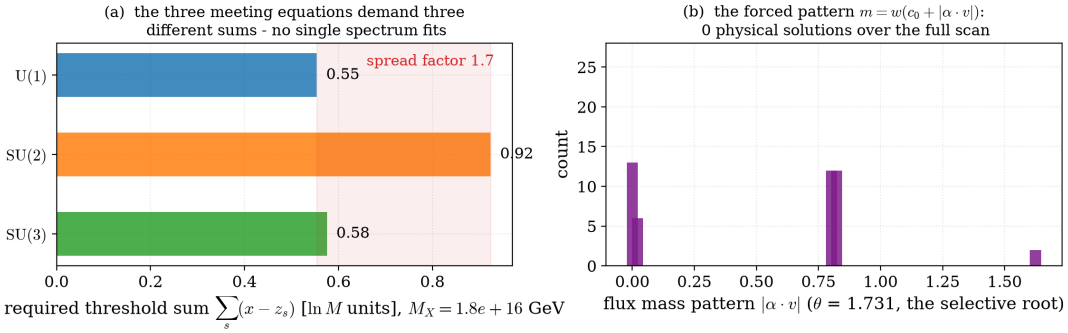
The composite content of the gauged layer is excluded as the supplier. The composite fermions of the layer of Section 7 — the matter **16** tensored with the adjoint link flux, $16 \otimes 45 = 720$ Weyl states carrying five net chiral **16**-copies — shift the three one-loop coefficients by $(145.3, 145.3, 226.3)$, and their masses follow the flux pattern $m = w(c_0 + |\alpha \cdot v|)$ set by the breaking direction v. Requiring the data-preferred meeting at $\alpha_X = 1/25$ then has no solution with thresholds in the desert for any direction: the three meeting equations demand three different threshold sums, and iteration from every seed over a full scan of v converges nowhere physical (0 solutions). The remaining ansatz families — full-weight thresholds, broken-flux-only content, light-composite selection, and a relaxed meeting coupling — leave the minimum residual at 5.7–6.1 against a target of 0 (Fig. 22).

The equal-shift theorem. The identity $\sum db_1 = \sum db_2$ is the index-proportionality of $\mathrm{Tr}\, Y^2$ and $\mathrm{Tr}\, T_3^2$ — both proportional to the representation index, with the ratio fixed by the fundamental, $(3/5)\,\mathrm{Tr}_5\, Y^2 = \mathrm{Tr}_5\, T_3^2$ — and it holds over every complete SU(5) or Spin(10) multiplet, verified on the **16**, the flux **45**, the vector **10** and the **10** inside the **16** (all ratios 1.0 to 10^{-6}). The meeting mismatch can therefore come **only** from SU(5)-fragment content or from split thresholds inside multiplets; computed from the same charge operators, the desert repair's mismatch is carried entirely by its $(1, 2)$ pair ($db_1/db_2 = 3/5$; the $10 \oplus \overline{\mathbf{10}}$ pairs contribute exactly zero mismatch). (Fig. 21).



(a) The one-loop ratio $\sum db_1 / \sum db_2$ over the matter **16***, the flux* **45***, the vector* **10** *and the* SU(5) **10***: exactly* 1*, while the* $(1,2)$ *fragment gives* $3/5$*. (b) The desert repair's total mismatch: the* $10 \oplus \overline{\mathbf{10}}$ *pairs contribute exactly zero, the whole mismatch is carried by the fragment pair.*

The gauged layer's composite content ($16 \otimes 45 = 720$ formal tensor-product states; the gauge-invariant kernel is empty and does not deliver the meeting)

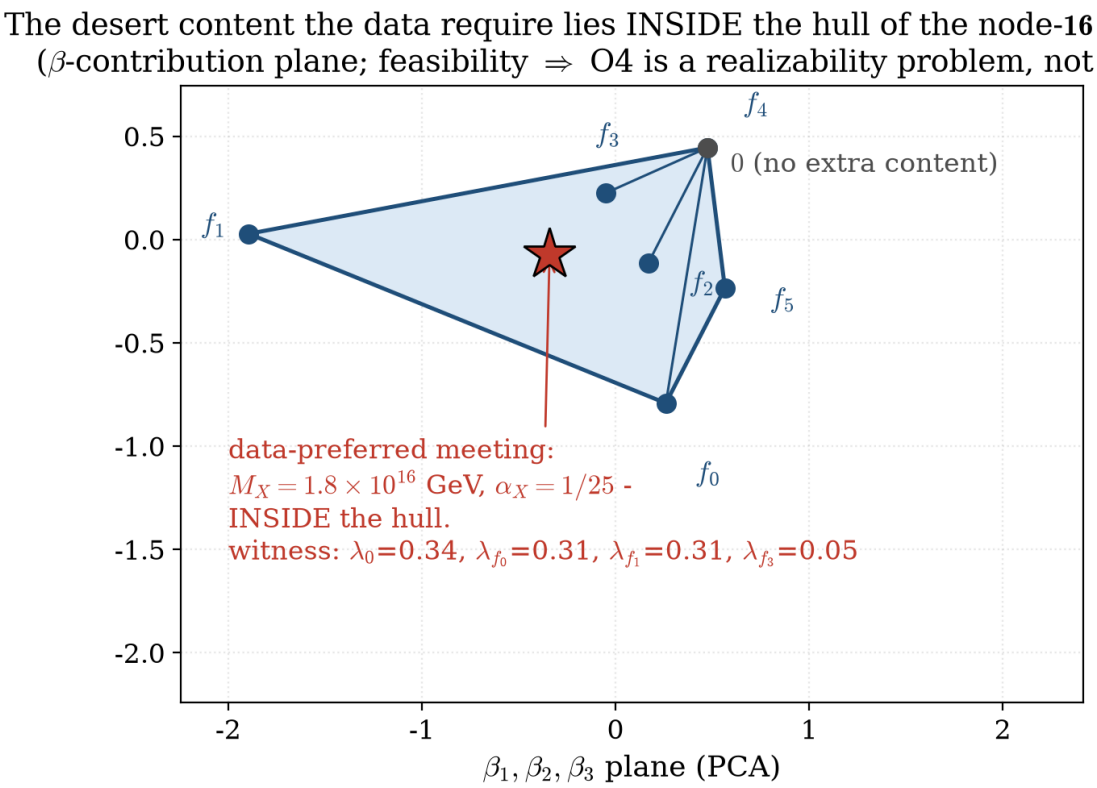


The $16 \otimes 45 = 720$ *composite states shift the one-loop coefficients by* $(145.3, 145.3, 226.3)$*. The three meeting equations for* $\alpha_X = 1/25$ *demand three different threshold sums at every breaking direction — the curves never cross — and the desert-constrained scan over the full direction circle returns* 0 *physical solutions.*

The junction route: pattern fabricated, meeting missed. At the root the Wilson direction splits the node **16** by a factor 52.7 ($c_Q = +0.0052$, $c_L = -0.0157$ against partners ± 0.82), so the $(3,2)$ and $(1,2)$ states of the desert solution **are** automaton states — the Q and L components of node **16**'s with near-zero Wilson weights — and the junction branch ratio $\rho = \kappa / w_0$ breaks the rank-1 degeneracy. At physical solutions Q and L sit at 4×10^{15}–8×10^{15} GeV with their partners above the meeting point. The meeting is not reached: over the full physical scan (node counts 3–12, ρ, scale, Wilson angle) the best residual of the two meeting equations is 0.94 against a target of 0, and the third equation predicts $\alpha_X = 0.032$ against the data value 0.04 — a 24% miss. Replacing the cubic-symmetric democratic junction by the general zero-mean junction — three independent branch scales $a_i = 2w_0(1 + r_i)$, $r_1 + r_2 + r_3 = 0$, scanned over the full (r_1, r_2) plane — leaves the best physical residual at the same floor, 0.81 against the democratic 0.94, with $\alpha_X = 0.085$, a factor 2.1 off. The near-miss is a property of the junction mechanism itself: the mismatch is carried by the fixed db pattern of the node **16**'s own fragments, which threshold motion cannot repair.

The required content is obtainable from the node 16 itself, with per-class thresholds. The data-preferred meeting point $(T_1, T_2, T_3) = (12.7, 21.2, 20.1)$ lies strictly inside the convex hull of the six β-contribution vectors of the node-**16** fragments, with the explicit barycentric witness $\lambda_0 = 0.34$, $\lambda_{f_0} = 0.31$, $\lambda_{f_1} = 0.31$, $\lambda_{f_3} = 0.05$: the fragment content of the node **16** itself suffices, provided the fragments carry independent thresholds. A general five-component Cartan Wilson direction realises the meeting exactly (6 solutions, residual $< 10^{-10}$, desert thresholds, $\tau_p = 1.9 \times 10^{36}$ yr) but breaks all 40 roots — the Model algebra among them — and

is unphysical; the symmetry-preserving direction space is exactly span$\{X, Y\}$ (dimension 2, class-degenerate), and within it no solution exists. Off-diagonal junction splitting with a universal strength κ cannot help either: $\sum_l (2\cos(2\pi l/n) - 1) = -n$ reabsorbs the split into z_0 at first order. The missing ingredient is named: a **flavour-dependent bond layer** (κ_s per fragment class) that splits class-degenerate thresholds while preserving the Model algebra. (Fig. 23).



*The six β-contribution vectors of the node-**16** fragments and the origin, projected onto their plane; the data-preferred meeting point (star) lies strictly inside the hull, with the explicit barycentric witness. Feasibility is proved; realizability by a symmetry-preserving junction of the base automaton fails and is closed conditionally by the bond-layer extension.*

The named ingredient closes the meeting — conditionally. With the Wilson direction pinned to the physical selective root and per-class junction strengths κ_s splitting the six fragment-class thresholds, the three coupling equations become exactly linear in (z_0, κ_s) once every threshold sits below M_X (since $\sum_l \mu_l = -n$), and the desert box constraints make the search a linear program. It admits solutions (residual $< 10^{-13}$) at $M_X = 1.8 \times 10^{16}$ GeV, $\alpha_X = 1/25$, for **all ten node counts** $n = 3 \ldots 12$, with every threshold inside the desert (2.0×10^3–1.8×10^{16} GeV, none above M_X) and $\tau_p = 1.9 \times 10^{36}$ yr. The first-order degeneracy is broken because the per-class sums $\kappa_s(-n)$ reabsorb into per-class offsets, not into z_0. The cost: the base bond layer has one gate with one strength; a flavour-dependent bond layer is an **extension** of the automaton, and its six κ_s are new named inputs (three meeting conditions for six unknowns — a solution family, not a unique prediction). Unification is therefore closed conditionally on that extension.

The six κ_s are not numbers of the root. They are not obtainable from any on-site algebra element: symmetry preservation forces the bond direction into span$\{X, Y\}$ at the physical angle θ^* (class-degenerate, meeting residual ≥ 0.43), and every Cartan element that splits the six classes breaks all eight unbroken desert roots. Testing the off-diagonal escape directly: the most general Hermitian bond operator on the 80 root-vector pairs (11 class-pair channels) never closes the meeting — floor residual 30.4 over $n = 3 \ldots 12$ — and level repulsion forces the sign of two of the six κ_s. Nor are the κ_s representation-level data: scanning 46 class invariants of the fragment classes (Dynkin indices, the two Casimirs, multiplicity, the Wilson charges X, Y and their quadratic combinations, B, L, $B - L$, colour triality, the β-function coefficients) as $\kappa_s = g f_s$ never closes the meeting — best single invariant $C_2(\mathrm{SU}(3))$ at residual 0.246. A two-coupling relaxation $\kappa = g_1 f_1 + g_2 f_2$ keeps every

threshold in the desert for 217 of 506 candidate pairs and can close the meeting exactly, but that is a reduction from six named inputs to two, not a derivation. The closure is also embedding-specific: it reproduces in the flipped embedding used for it (10/10 node counts, residual $< 10^{-10}$) but closes only 5/10 in the Georgi–Glashow split.

The canonical per-class number the gauged layer does supply is not the one that closes the meeting. At θ^* the Wilson direction leaves a 13-dimensional centralizer in $\mathfrak{so}(10)$ (the $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ of the breaking chain, Hermitian orthonormal basis to 10^{-15}), and the flux of BROKEN generators through a fragment class,

$$\begin{aligned} K_s = C_2(\mathbf{32}) - C_2^{\text{unbroken}}(s) &= 45 - C_2^{\text{unbroken}}(s) \\ &= (32, 36, 40, 28, 32, 40) \end{aligned}$$

(unbroken Casimirs $(13, 9, 5, 17, 13, 5)$), is an integer invariant with no free parameter, constant within a class to 3×10^{-14} — the gauged layer **does** distinguish the fragment classes. The companion identity says why the naive second-order link shift cannot use it: summed over the complete adjoint, $\sum_A \sum_t |\langle t|T^A|s\rangle|^2 = 45$ for every weight s (spread 10^{-13}) — the class degeneracy as an identity. Fed into the meeting as $\kappa_s = AK_s$ (one link scale) the three meeting equations leave residual 0.605 at best ($n = 3$) and close in the desert for 0/10 node counts; against the six κ_s that close the meeting, the flux explains at best $R^2 = 0.015$ of their variance ($|r| \leq 0.31$). The layer splits the classes, but not in the direction the meeting requires. Nor does its second-order spectrum: the bond-current ground state ($E_0 = -5802.4\,\eta$, gap $42.09\,\eta$, Section 7.2) and the Gauss gap $C_2(\mathbf{45}) = 64$ supply the energy denominators, and the intermediate (flux, matter) pair space decomposes exactly as $\mathbf{45} \otimes \mathbf{16} = \mathbf{16} \oplus \mathbf{144} \oplus \mathbf{560}$ with Casimirs $(45, 85, 117)$ at the scale $C_2(\mathbf{45}) = 64$ (dimensions and Casimirs to 10^{-6}, projectors resolving the identity to 10^{-15}). Removing the on-shell screened **16**-component, the second-order bond-current (superexchange) shift is constant on each class to 4×10^{-3} and takes the six values $(0.0002, -0.130, -0.253, 0.127, 0.0013, -0.253)$; fed into the meeting it leaves residual 0.577 and closes in the desert for 0/10 node counts ($R^2 \leq 0.076$ against the fitted κ_s). The two-number combination with the broken flux and the ground-state Cartan shift — whose bond flux $\mu^A = \mathrm{Tr}\,\rho_B J^A$ vanishes to 6×10^{-17}: the Gauss law, intertwiners carry no net charge — solved jointly over all ten node counts (30 equations, 13 unknowns) leaves residual 3.46 and explains at best $R^2 = 0.30$ of the variance of the six fitted κ_s. The supplier space is exhausted by every class number the layer computes: Cartan elements, class invariants, the broken flux, superexchange, and every fixed combination of them.

The root-coupling solution is not realizable by its own content, and the exclusion channel stays open. Take the root-coupling solution as given — $\alpha_X = \pi/64$, $M_X = 1.235 \times 10^{17}$ GeV. Its $\mathbf{10} \oplus \overline{\mathbf{10}}$ thresholds land at 7.05×10^4 GeV, **below** the 10^6 GeV desert floor, and its $(1, 2)$ threshold at 4.88×10^{17} GeV, **above** M_X. Restoring the ordering the meeting would require means moving the $(1, 2)$ threshold by 13.1 e-folds — $1.12\times$ the entire log-spacing from the published 1.0×10^{12} GeV to M_X — so **no placement of one-loop thresholds within the spectrum's own spacings restores the meeting**, and the driving content is exact-integer in the root charges ($k_1 = 3/5$, $\Delta b_{\mathbf{10}\oplus\overline{\mathbf{10}}} = (6/5,\ 2,\ 2)$; complete $\mathbf{16} \oplus \overline{\mathbf{16}}$ shifts proportional, $(8/3,\ 8/3,\ 8/3)$): the no-go is the model's own content, not a scan artefact. **F2's unification side fires** (Table 8). Whether the desert spectrum is itself excluded answers no on every channel: the colour pairs at $M_{10} = 7.06 \times 10^4$ GeV sit $14.1\times$ above the strongest LHC probe of a coloured state (5 TeV, t-channel dimuon + dielectron) [68] and pair production is kinematically closed at the LHC ($2M_{10} = 141.2$ TeV $\gg 13.6$ TeV); the $(1, 2)$ doublet at 1.0×10^{12} GeV is $10^9\times$ above the electroweak-precision ceiling for a non-decoupling doublet [69] and fully decoupled ($(v/M)^2 = 6 \times 10^{-20}$); and the proton lifetime is $1.2 \times 10^5\times$ past the Super-Kamiokande $p \to e^+\pi^0$ bound [17]. **No F10 fires from the desert spectrum**: its failure is internal threshold ordering, not experimental exclusion.

The two-dial and dressed-run census. The one-loop running coefficients of the desert running are independent of the bond-layer dial β at every order and on every content — β enters the running nowhere — and the meeting has no two-dial solution on the $c = 1$ line (nearest approach 0.0367, closure residual 2.9×10^{-2} there): the light-cone pin does not help the coupling. At the closed clock $\pi/4$ the standard reading $\pi/64 = 0.049087$ sits $1.227\times$ above the fitted desert value, and the alternative $\alpha_{10} \equiv g^2 = 1/25$ is a convention of the same kind as the clock reading, not a number of the standard running. The same census reproduces verbatim on the dressed, dynamical gate of the constructed sector, and the desert-content supplier check on the gauge-invariant composite sector returns no content (Section 7.1). F2's coupling side stands in every version of the construction.

11.2 Electroweak mixing

$\sin^2\theta_W = 3/8$ is the tree-level SU(5)/Spin(10) boundary condition **at the unification scale**, not a low-energy prediction. Run to M_Z it becomes $\simeq 0.23$, close to the measured 0.23129(4) ($\overline{\text{MS}}$, PDG 2024 [15]); the measured ratio $m_W/m_Z = 0.881$ corresponds to $\sin^2\theta_W \simeq 0.223$, not to $\sqrt{5/8} = 0.791$. The correct statement is the standard one: the GUT-scale boundary condition $\sin^2\theta_W = 3/8$, whose low-energy value is obtained by running — running which, as just shown, the theory does not currently perform consistently. **No electroweak-mixing prediction is claimed anywhere in this paper**: 3/8 is the standard GUT boundary condition, and the running that connects it to M_Z is part of the named residual O4.

11.3 Baryon asymmetry

The automaton alone cannot produce it. Every CP-odd (T-odd) invariant of the unitary step vanishes: Im Tr of all two- and three-fold products of the **45**, of a 2×10^5 sample of four-fold products, of all $10 \times 10 \times 45 \times 45$ baryon-violating sandwiches, and of the rank-1 real democratic Yukawa — all zero to 10^{-15}. The step is unitary (in equilibrium), baryon-conserving (the **45** commute with w) and CP-conserving. It satisfies **none** of the Sakharov conditions [26]. This is a computed theorem, and it is the correct starting point. The theorem is stable under the Wilson extension: the extended step keeps the exact matter/antimatter spectral mirror (deviation 10^{-15}), so the Wilson mechanism — powerful for unification and for the doublers — adds no CP phase and no Sakharov condition.

A GUT-leptogenesis estimate [40] is then built by hand:

$$\Delta B = \frac{28}{79}\,\varepsilon_{CP}\,D, \qquad \varepsilon_{CP} = \frac{g_0^4}{16\pi} = 4.9 \times 10^{-6}, \qquad D = \left(\frac{1}{16\pi}\right)^2 = 3.96 \times 10^{-4},$$

with $g_0 = 1/2^3 = 1/8$ and the Standard Model sphaleron conversion $B = \frac{28}{79}(B - L)$ [27,28], giving $\Delta B = 6.8 \times 10^{-10}$ against the measured $(6.12 \pm 0.04) \times 10^{-10}$ [16] — a ratio 1.11.

Three problems. (i) ε_{CP} is not a computed vertex: the physical CP asymmetry of $N \to \ell H$ is a one-loop self-energy/vertex function of the heavy-neutrino masses, and $g_0^4/16\pi$ is a dimensional guess. (ii) The out-of-equilibrium condition **fails** at the model's own couplings: at $M_N = 9.6 \times 10^{13}$ GeV, $H = 1.66\sqrt{g_*}M_N^2/M_P = 1.298 \times 10^{10}$ GeV while $\Gamma_N = y^2 M_N/8\pi = 5.98 \times 10^{10}$ GeV for $y = 1/8$ (1.93×10^{12} GeV for $y = 0.71$), so $\Gamma_N/H \simeq 4.6$ ($\simeq 148$). Out-of-equilibrium decay requires $y \lesssim 0.06$, which the theory does not select. (iii) 28/79 is an input of the Standard Model sector, not a number of the root. The 11% agreement is therefore a coincidence of chosen factors (O5). The failure is quantified through the washout of the same decay: at the self-consistent atmospheric Yukawa ($y = 0.40$, the one that gives $m_\nu = 0.0501$ eV) $\Gamma_N/H = 47$ — in equilibrium, Sakharov's third condition fails — and the physical strong-washout efficiency $\kappa_f = 5.5 \times 10^{-3}$ [40] sits 14× away from the guessed dilution $(1/16\pi)^2$: the agreement is an accident of two wrong factors, not a near-miss.

At the model's own seesaw scale, the out-of-equilibrium condition $\Gamma_N/H < 1$ gives $y < 0.058$. The same Yukawa fixes the light neutrino mass through the same seesaw,

$$m_\nu = \frac{y^2 v^2}{M_N} \leq 2.1 \times 10^{-3}\ \text{eV} \qquad (y \leq 0.058),$$

a bound of the same kind and size as the standard thermal-leptogenesis bound [33]. The atmospheric splitting requires $m_\nu \simeq 0.050$ eV, i.e. $y \simeq 0.28$ with $m_D = yv$ and $v = 246$ GeV ($y \simeq 0.40$ in the $v_u = 174$ GeV convention), for which $\Gamma_N/H \simeq 23$. **At $M_N = 9.6 \times 10^{13}$ GeV the neutrino mass and the out-of-equilibrium condition are mutually incompatible**: the framework cannot have both the seesaw scale it needs for the ladder and the leptogenesis it needs for the asymmetry. (The model's own ladder sits at $m_D = 118$ GeV, i.e. $y = 0.48$, where $\Gamma_N/H \simeq 68$.)

The one route the framework supplies is quasi-degenerate (resonant) leptogenesis [32,41]. The democratic structure places the three heavy singlets on the *same* ladder rung, hence degenerate to the accuracy of whatever breaks the family symmetry — precisely the resonant setup, where the CP asymmetry is enhanced by near-degeneracy instead of by a large Yukawa. With $y = 1/16\pi = 0.0199$ the decay is out of equilibrium ($\Gamma_N/H = 0.12$), and the asymmetry that the model's own dilution factor and sphaleron conversion require,

$\varepsilon_{CP} = \Delta B/[(28/79)D] = 4.4 \times 10^{-6}$, is *smaller* than the natural one-loop size $y^2/16\pi = 7.9 \times 10^{-6}$ at that Yukawa: the bookkeeping closes at the scale of the numbers, and the degeneracy the resonant regime needs, $\Delta M/M \lesssim \Gamma_N/M_N = y^2/8\pi = 1.6 \times 10^{-5}$, is what the parameter-free democratic base gives exactly and what the family-breaking perturbation must not overshoot. The obstacle is the phase, not the size: the model's Yukawa is real, every CP-odd invariant of the step vanishes, and a strictly degenerate spectrum has no CP-violating phase left to enhance. This is a *route*, not a result.

The phase, however, is not the missing ingredient. The mirror is exactly the structure that forbids CP: every mirror-preserving extension (the Wilson factor: pairing deviation 1.3×10^{-15}) leaves the CP phase at zero, while the most general per-class κ_s layer and the most general real-symmetric bond generator destroy the $\pm$ pairing (deviation 1.5, 60 draws, both mirror sign conventions) — a CP phase costs Theorem 4. But the prohibition is a statement about the on-site module, and the bond layer is not covered by it: a Peierls flux $A_1 = \Phi x_0$ on the inter-block bonds is exactly unitary (8.0×10^{-15}), keeps $\det U = 1$ at every flux ($|\arg\det U| \leq 1.2 \times 10^{-14}$), commutes with the pseudoscalar (mirror $\mathbb{Z}_2$ leak 1.8×10^{-30}) and keeps the hop at radius 1 (1.6×10^{-16}), while moving the band spectrum by up to 3.7; its gauge-invariant strength, the plaquette holonomy, has $\mathrm{Im} = 0$ at $\Phi = 0$ and 2.5×10^{-1} at $\Phi = \pi/2$. Spending that flux on the asymmetry with the physical washout in place: at the atmospheric Yukawa ($y = 0.3986$, $K = 46.8$, $\kappa_f = 5.5 \times 10^{-3}$) the measured $\Delta B = 6.12 \times 10^{-10}$ costs $J = 1.0 \times 10^{-4}$, supplied by a flux $\Phi = 4.0 \times 10^{-4}$ rad; in the out-of-equilibrium regime ($y = 0.05826$, $K = 1$) the strong-washout expression has no real root — K lies below its branch point $K = 3.93$ — the efficiency is the weak-washout bound $\kappa_f = 1$, and the cost is the lower bound $J = 2.6 \times 10^{-5}$, supplied at $\Phi = 1.0 \times 10^{-4}$ rad. A bond flux of size $\Phi \sim 10^{-3}$ rad already supplies the needed strength, and the family reaches $J = 0.25$ at $\Phi = \pi/2$, 2500× the atmospheric requirement, while leaving unitarity, $\det U = 1$, the mirror symmetry and the causal radius exact. **What fails is Sakharov 3 alone**: the Yukawa that gives $m_\nu = 0.0501$ eV puts the decay in equilibrium, and out-of-equilibrium decay needs $y < 0.05826$, which caps m_ν far below the atmospheric splitting. O5 is open for one named reason instead of two.

11.4 Proton decay

Dimensional analysis of X/Y exchange, at the desert solution solved with the root coupling ($M_X = 1.235 \times 10^{17}$ GeV, $\alpha_X = \pi/64$: Section 5.2), gives

$$\tau_p \sim \frac{M_X^4}{\alpha_X^2\, m_p^5} = 2.77 \times 10^{39}\ \text{yr} \qquad (M_X = 1.235 \times 10^{17}\ \text{GeV},\ \alpha_X = \pi/64),$$

against the Super-Kamiokande bound $\tau_p/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [17] — a margin of five orders. The $M_{\mathrm{GUT}} = M_P$ reading (3.98×10^{47} yr) is an abandoned convention; any GUT scale below $\sim 6 \times 10^{15}$ GeV is excluded by the same bound. The estimate omits the renormalisation factor A_L [47], the hadronic matrix element, the dressing of the operators and the splitting between the X/Y masses and the unification scale [52], all of which shift the lifetime by one to two orders. The three channels $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ differ here only by two-body phase space; the rate scales as M_X^{-4} in all of them. Solving the same content at the measured meeting $\alpha_X = 1/25$ instead gives $\tau_p = 1.8 \times 10^{36}$ yr, inside Hyper-Kamiokande's reach — but that solution sits 23% off the root coupling and is a reference reading, not a prediction of the closed theory. Observation of a GUT-pattern decay below 10^{38} yr refutes the root-coupling solution (F5).

11.5 Flavour

The democratic mass matrix $M = |1\rangle\langle 1|$ (all entries 1, rank 1, spectrum $\{3, 0, 0\}$) [44] follows from the hypothesis that the three spatial directions are equivalent, so the $\mathbf{10}_H$ couples the three families identically — the ansatz is not forced, exactly: the S_3-commutant of the family action on 3×3 matrices is exactly $\mathrm{span}\{I, \mathrm{ones}\}$ (dimension 2, by exact generators), and it contains rank-3 elements (explicit witness, spectrum $0.587, 0.587, 1.026$), so rank-1 is a named input of the Yukawa sector, not a consequence of the family charges. The host energetics do not supply it either: the dynamical chain's three singlet-channel hosts carry chain energies $32.0, 32.0, 48.0$, and the democratic uniform ones-coupling costs +48 over the energy-minimal host occupation — rank-1 stands declared on two independent grounds. Its heavy eigenvector $(1, 1, 1)/\sqrt{3}$ is unique, hence

$$\sin^2\theta_{12} = |\langle e_1|h\rangle|^2 = \tfrac{1}{3}$$

for any family basis — the **only** mixing angle the model derives, 8.6% from the measured 0.307 [19,51]. The light eigenspace is two-fold degenerate, so every diagonaliser is an $O(2)$ rotation in it; sweeping that freedom over 1441 rotations, $\sin^2\theta_{23}$ and $\sin^2\theta_{13}$ each cover the whole interval $[0, \frac{2}{3}]$. The tribimaximal values $(\frac{1}{2}, 0)$ [29] are one point of that family, picked by declaring which S_3 transposition breaks — and (12), (13), (23) are equivalent symmetries of $|1\rangle\langle 1|$, so the choice is an input. The CP structure of that choice is priced exactly: at the construction's angles ($\sin^2\theta_{13} = 0$) the Jarlskog invariant vanishes identically over the whole phase circle ($J_{\max} = 0$), so **no** unitarity triangle has area — no CP violation is available from the mixing matrix at the bookkeeping point.

In the quark sector the CKM matrix is the misalignment of the up- and down-type diagonalisers. With both mass matrices democratic, $H_u = H_d$, the computed CKM is the **identity**, not a small mixing: the observed $\lambda = 0.225$ [15,45] is not reproduced. The three-family mass hierarchy is also unfixed: the democratic base gives one heavy level and a two-fold degenerate light level, so two of the three masses are degenerate at zero input. All of this is O6, and the natural fixes are bounded by computation. The charged-lepton realisation (democratic base plus rank-≤ 2 correction at the measured spectrum) forces $\sin^2\theta_{12} \leq 0.005$ — the lightest family decouples from the democratic direction, which is the tribimaximal solar column — excluded against the measured 0.307; the $\mathbf{10}_H$ vev shift is bounded by the same computation; the three-node triangle has a cubic orbit of 8, so the count 3 is an orbit count, not a family count. **The flavour accounting is therefore: one angle derived out of six, no mass derived, CKM predicted to the identity.**

11.6 Scoreboard

Table 5. Headline numbers against measurement.

quantity	theory	measured	status
the eight anomaly coefficients	0	—	algebraic, satisfied
g_1^2/g'^2	5/3	5/3 (GUT normalisation)	algebraic
$\det U$, $\pm$ pairing, $16/\overline{16}$ mirror	1, exact	—	algebraic, coupling-independent
causal front speed	exactly 1	—	dynamical, measured
doubler cone Lorentz structure	linear in all 1200 sampled directions; anisotropic: $(0.200, 0.200, 0.284)$ lattice units, spread 0.75; node gap 2×10^{-4} at $g = 1/5$	—	**not Lorentz-isotropic**; measured
tangent velocity algebra at the gapless corner multiplets	$[S_i, S_j] = 0$ to 3×10^{-12}; both lowest corner triplets driven to zero together at $w = 0.241747$, $\theta = 0.913137$ — six states with all phases below 2.5×10^{-16}, isolated from the next level by 8.7×10^{-3}, Abelian to 5.3×10^{-12}, six nodal planes	—	**nodal planes, not a cone**; measured
conical node inside the automaton's own circuit class	33 800 points of the largest cubic-covariant six-stage family, 60 exactly gapless (full-step gap at most 4.4×10^{-15}) — 8 single-parity multiplets, flat to 1.3×10^{-12}; 52 parity-mixed, slope 0.09–1.73, Abelian to 1.2×10^{-11}	—	**no conical node reached**; measured

quantity	theory	measured	status
16-sector block-Brillouin spectrum	at the root clock: all 12 M channels exactly gapless, every M spectrum $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ (64 of 128 pinned to $\{0, \pi\}$, identical across the 12 nodes to 3×10^{-15}); the Wilson factor at the root angle re-gaps the M nodes	—	doubling **exactly gapless on the branch**, re-gapped only by the Wilson factor
v/M_P	$2.710\,(1/16\pi)^{10}$	2.0167×10^{-17} [15]	coefficient $3(16\pi F_{208})^{10} = 2.710$ certified at the crossing ($F_{208} = 0.0196928$ by continuity at 50 digits) against 2.0766 observed — +30%; the named O8 residual is the clock, measured: the closure class reads 2.710 ($\pi/4$) and 2.711 ($3\pi/4$), and the observed value is read by the non-closed rung-10 clock 1/5 to relative 10^{-5}; the inverted clock is 0.1999980, 2.0×10^{-6} off the fifth
$m_{\rm DM}$	118.57 GeV — the cell singlet B at rung $n = 10$ (Axiom 1)	not measured	prediction, falsifiable
$\Omega_{\rm DM}h^2$	freeze-in of the cell singlet B at $\lambda_B = 4.06 \times 10^{-14}$ gives $\Omega h^2 = 0.120$ with $\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2, DD-invisible; the thermal solution is excluded by LZ; the couplings the root supplies give 0.798 or 0.0246	0.120 [16]	0.120 **reproduced by the cell carrier**, the portal λ_B a named input; **fails** for the couplings the root supplies
m_ν	0.146 eV (ladder); 0.0487 eV ($\frac{1}{3}$ candidate)	0.0501 eV [51]; $\sum m_\nu < 0.0642$ eV [61]	ladder **in tension**; candidate inside a finite window, structure protected
ΔB	6.8×10^{-10}	$(6.12 \pm 0.04) \times 10^{-10}$ [16]	coincidental product; the washout at the self-consistent y destroys it
τ_p/B	2.77×10^{39} yr — root-coupling desert solution ($M_X = 1.235 \times 10^{17}$ GeV at $\alpha_X = \pi/64$, threshold ordering destroyed); 1.8×10^{36} yr reference reading at the measured meeting 1/25	$> 2.4 \times 10^{34}$ yr [17]	falsifiable only by a null result past 10^{37} yr; the Hyper-Kamiokande channel closes on the root coupling
α at the root clock (standard normalisation $\alpha = g^2/4\pi$)	$\pi/64 = 0.0491$	0.023 (one-loop mean at M_P); 0.040050 at the desert meeting $M_X = 1.8 \times 10^{16}$ GeV	the root value sits a factor 1.226 **above** the meeting value — the named tension of O4; the root-coupling solution is not realizable by its own content, and no threshold of the desert spectrum is excluded (F10 open)
$\alpha_i(M_Z)$ from α_{10}	0.0197 / 0.195 / pole at 2.193×10^9 GeV (Standard Model running)	0.0169, 0.0338, 0.118 [15]	**fails** under Standard Model running; matched by the desert spectrum

quantity	theory	measured	status
Gauss constraint of the gauged bond layer	$[Q_A, Q_B] = i f_{AB}{}^C Q_C$, err 0; content $32 \oplus 288 \oplus 1120$	—	algebraic
physical sector of the gauged layer	$\dim \ker = 0$ with one **32**/site (centre obstruction); minimal sector: 2 matter/site, 4 states, all singlets; $S = \ln 45$ per bond, bounded and boundary-supported	—	computed
multi-bond scaling	$\dim \ker$(3-site) $= 24$; additivity $\ln 45$ across a single bond; two-bond region $S = \ln 45$ (vertex mutual information $\ln 45$); Bekenstein deficit factor $4 \ln 45 = 15.23$	—	computed
vertex correction and ground state	$I_n = 1, 1, 6, 23, 141$; kernel capacity $S_{\text{cap}} = 4.4998$ nats (factor 9.0 from 1/4 per bond); vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond (factor 6.6); $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, $S_{\text{GS}} = \ln 45$	—	computed
dressed gate on the directed-link network	Gauss charge closes only with the conjugate-head convention (5×10^{-16}; the naive convention leaks 4.8); Kogut–Susskind electric gate exactly invariant (2.5×10^{-14}); dressed hopping obstructed (leak 12.5); quintet gapless at Γ, linear, leading-order isotropic, $v = g_{10}\{0, 1, 2\}$	—	computed
regular link of the root's finite group	Γ = even Clifford monomials, order 1024: vertex content $32{\cdot}1024^{k-1}$ (fixed adjoint link: 0); the dressed hop descends to the bare translation (err 0), front speed $v = 1$; centralizer of Γ in Spin(10) = 0; quintet–matter coupling: T_{2g} vertex $\equiv 0$ on axis bonds, E_g bubble $(-54.24,\ -53.40)$ $\to$ mass shift at $\lambda = g_{10}$: $(-16.73,\ -16.47)$	—	computed
velocity trilemma on the 6-bond module	hop alone anisotropic (top speed 1.00, direction spread 0.32); dressing alone isotropic at $v = 2g_{10} = 1.571$; no angle isotropises the product (best spread 0.24 over 33 angles in $[0, \pi/2]$)	—	computed no-go

quantity	theory	measured	status
desert spectrum from the layer's own content	no solution (0 over the full direction scan); all mass-ansatz families excluded, min residual ≥ 5.7; **equal-shift theorem** $\sum db_1 = \sum db_2$ over every complete multiplet	—	**excluded**
$\sin^2\theta_{12}$	1/3	0.307 [51]	derived at zero input; the +8% shift is the $\mathbf{10}_H$ vev, whose natural placements are excluded
$\sin^2\theta_{23},\ \sin^2\theta_{13}$	not fixed ($O(2)$)	0.470 (octant unresolved, 3σ 0.435–0.585), 0.0222 [51]	open
CKM	identity	$\lambda = 0.225$ [15,45]	**fails**
Λ	0 (tree level)	$\sim 10^{-122} M_P^4$	open

The 16-sector row compresses Section 6.4, stated here in full: at the root clock the step has Γ fully pinned to the Floquet pair $\{0, \pi\}$ (64+64), all twelve M channels exactly gapless with identical spectra ($\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$, 64 of 128 pinned) — the gapless set is Γ plus six face-diagonal nodal lines, the bulk is gapped. The Wilson factor at the root angle re-gaps the M nodes.

---

12. Cosmology

12.1 Dark energy

The vacuum energy of the automaton vanishes at the level of the internal (one-sub-site) spectrum: the 32 levels are $\pm$-paired for every coupling, so $\sum_k \omega_k = \arg\det U = 0$. This is the correct, non-vacuous form of the cancellation. (Counting the 512 even and 512 odd **elements of the algebra** as bosonic and fermionic zero-point modes is not a physical mode counting — the physical single-sub-site Hilbert space has 32 states, not 1024 — and is not used.)

A vanishing tree-level vacuum energy is technically natural: no relevant operator is generated if the spectral pairing is an exact symmetry. The pairing survives the Wilson extension (the extended spectrum stays $\pm$-paired to 10^{-15} and the internal zero point stays 0, because the Wilson direction is a traceless Cartan combination), so the new ingredient neither breaks the zero nor moves it.

The zero is **protected, not accidental**. Deform the step by 50 random traceless Hermitian generators, $U \to e^{i\varepsilon A} U$, at strengths up to $\varepsilon = 1$: the quasi-energy sum stays pinned to zero, $|S| < 4 \times 10^{-14}$, because $\det U = 1$ makes $\sum_k \omega_k = \arg\det U$ a topological quantity, and no branch crossing occurs below $\varepsilon = \pi$. The only internal knob that moves the sum is the U(1) trace phase.

What the theory does **not** do is produce the observed value. The observed $\Lambda \sim 10^{-122} M_P^4$ lies 122 orders of magnitude *below* the classical bound M_P^4 and *above* the theory's spectral zero, and no mechanism breaking the pairing by that amount is known. Independently, the classical self-gravitating de Sitter vacuum of Section 8 has $G_{\mu\nu} = -3H^2 g_{\mu\nu}$, i.e. a vacuum energy bounded by M_P^4 on the lattice. These are three different statements — spectral zero, classical bound, observed value — and they do not "reconcile": the gap between the first and the third is O1, the largest unexplained number in the theory [38]. The equation of state of a cosmological-constant vacuum is $w = -1$ by definition, not by derivation.

The problem is a single named number: the observed Λ is **equivalent to** a trace-phase breaking of size

$$\varepsilon \operatorname{Tr} A \sim 16\pi^2 \times 10^{-122} = 1.6 \times 10^{-120}.$$

O1 is not "why is the vacuum energy small" but "what breaks the trace U(1) by 10^{-120}". One route the ladder could in principle supply is closed: with the induced-action normalization read as the hierarchy coefficient

divided by 3 (the ratio 0.692), scanning every certified ladder row's loop ratio against it leaves the closest at 46% deviation — no certified ladder factor reaches the induced-action normalization (no trace action is present).

12.2 Dark matter

The candidate supplied by the classical sector is the kink of Section 8.4. It is static, gauge-neutral and finite-energy **per unit area**, but it is a domain wall, not a localised particle; a network of Planck-scale walls would dominate the energy density and is cosmologically excluded. The spinor-module candidates are exhausted (Section 10.2): the exact mirror $\mathbb{Z}_2$ of the step is the only exact-stability structure the spinor module carries; no Wilson-split node can host a neutral mirror at the ladder mass without also delivering a stable coloured or charged partner; and the mirror sector's own lightest states sit at 1.3×10^8 GeV (coloured, excluded) and 2×10^{10} GeV — not at the ladder's 118 GeV. No state of the spinor module carries the ladder mass; the carrier is the cell's singlet sector B — a state of the automaton, exactly stable under the B-parity — and what remains open is the portal strength λ_B it shares with the Higgs (O7, Section 10.2). The mass prediction stands.

The relic abundance computed from the root's own strengths is not the observed one. Solving the freeze-out integral

$$\Omega h^2 = \frac{1.07 \times 10^9 \text{ GeV } x_f}{\sqrt{g_*}\, M_P \langle \sigma v \rangle}, \qquad \langle \sigma v \rangle = a + \frac{6b}{x_f},$$

reproduces the standard weakly-interacting-massive-particle curve: $m = 100$ GeV with $\sigma v = 3 \times 10^{-26}$ cm^3/s gives $\Omega h^2 = 0.080$ [42]. At the ladder mass $m = 118.57$ GeV the measured $\Omega h^2 = 0.120$ requires $\sigma v = 1.98 \times 10^{-26}$ cm^3/s, i.e. an effective coupling $\alpha_{\text{eff}} = 0.0086$ in $\sigma v = \alpha^2/(\pi m^2)$ — while the root supplies only $g^2/4\pi = 0.0032$ (giving $\Omega h^2 = 0.798$) or the loop factor $1/16\pi = 0.0199$ ($\Omega h^2 = 0.0246$). The automaton's own light spectrum enters at the same order: its twelve near-gapless nodes are nodal planes (Section 6.2) with mean speed 0.201, so the final-state density is enhanced by

$$D = \sum \frac{1}{|v|^3} = 1485$$

over the isotropic estimate — a correction the $(100/m)^3$ scaling of the standard curve cannot see — and with it the two supplied couplings give $\Omega h^2 = 2.3 \times 10^{-5}$ and 7.6×10^{-4}. The spinor-sector portals of Section 10.2 reproduce 0.120 only when their coupling is inverted from the fiducial 3×10^{-26}: computed forward they give 0.081 and 0.156, and 0.000 once the nodal-plane density is included. For the cell carrier B the abundance is computed, not assumed: freeze-in through the singlet–Higgs portal at $\lambda_B = 4.06 \times 10^{-14}$ gives $\Omega h^2 = 0.120$ — the observed value — with $\sigma_{\text{SI}} = 6.5 \times 10^{-71}$ cm^2, far below reach, while the thermal alternative $\lambda_B = 0.31$ is excluded by the LZ bound. **The root supplies no λ_B, and no root-supplied strength of its class lands on** 0.120**.** Pure gravitational production never equilibrates ($\Gamma_{\text{grav}}/H = 5.3 \times 10^{-53}$ at $T = m_B$), and its yield would need the inflation history the model does not supply. For the spinor-sector variants the window is bounded on both sides: the mirror-preserving portal is DD-invisible and its thermal solution is excluded, and the 0.120 of the mirror-**breaking** pair portal ($\lambda = 3.5 \times 10^{-4}$, $\sigma_{\text{SI}} = 4.9 \times 10^{-51}$ cm^2) costs the exact mirror $\mathbb{Z}_2$.

The bond sector does not set the abundance, and its portal is excluded. The named scalar is the singlet fragment $(X, Y) = (1, 0)$ of the node **16** — the same class whose bond number κ_s enters the unification meeting (Section 11.1). Read as the annihilation strength through the nodal-plane final states, every bond number of that class — the ten κ_{singlet} that close the meeting (2.2–9.2), the superexchange value -0.253, the broken flux 40, the junction ratio 1.92 — lands Ωh^2 between 10^{-12} and 10^{-3} once D is included, and none within 0.01 of 0.120; the measured abundance needs $\alpha_{\text{eff}} = 0.0086$ without D or 2.2×10^{-4} with it. Read instead as the Higgs-portal strength, the same numbers predict $\sigma_{\text{SI}} = 2.5 \times 10^{-45}$–$6.3 \times 10^{-41}$ cm^2 — all above the LZ-class bound 1.5×10^{-47} cm^2 — and the portal strength the relic itself requires ($\kappa = 0.33$) predicts 4.3×10^{-45} cm^2, also excluded. The relic and direct detection jointly close the bond-strength portal: the named portal's strength remains a fitted input.

12.3 Antimatter

Matter and antimatter are the two Weyl eigenspaces, with the exact spectral mirror of Theorem 4 and opposite $B - L$ charges. Their degeneracy is that mirror and nothing more: no antiunitary time-reversal operator is constructed (Section 5.1), so this is not the CPT theorem — and the scope reaches the multicell level: the chain ground-state vertex of Section 7.5 has imaginary parts up to 0.171 over its 6144 entries, so no antiunitary acts exactly on the multicell chain at the level of the chain data (a chain-level CPT violation as an exact-reality statement; the single-cell spectral mirror, a unitary-spectrum statement, is untouched). The observed baryon asymmetry is not explained: Section 11.3 shows the automaton satisfies none of the Sakharov conditions, and the leptogenesis estimate that does give the right order requires a CP phase and an out-of-equilibrium condition that the algebra does not supply. The asymmetry magnitudes computed from the construction are on record: every raw magnitude sits at least 8.0×10^3 from $\eta_B = 6.1 \times 10^{-10}$ (largest, the zero-parameter flux value 4.86×10^{-6}), and the only computed quantity that lands near the observed value is a product of guessed ingredients — ratio 1.11 unwashed, 0.13 under the washout factor: the bookkeeping falls short.

---

13. The parameter audit

Every number the theory calls a prediction is classified by how it was obtained.

Table 6. Provenance audit: every headline number by class.

class	count	numbers
algebraic (fixed by the root, no parameter)	7	the $16/\overline{16}$ split; the **45**; the 5/3; the eight anomaly coefficients = 0; the 32-level spectrum; $\det U = 1$; the $16/\overline{16}$ spectral mirror
algebraic (two-dial family)	+1	the closure set C of the decoupled two-dial step — the four pairs (β, g) at which the zone-centre step closes to the identity in finite order (2 and 4), and the empirical clock $(\frac{1}{8}, \frac{1}{5})$ is not among them; the family adds no fitted dial
algebraic / declared (interacting continuum)	+3 algebraic, 6 declared, 2 never-realized, 0 fitted	the interacting-continuum accounting: **algebraic** — the star-module closure set $C_\star$, the 1440-dimensional gauged two-particle operator and the zero-parameter vertex gate values on it, the eight spatial-vector copies in the 45 under the fixed spin connection; **declared (measured-input)** — the three wall defects, the 600 interaction loop factors F_{int}, the dressed block-RG scales and exponents, the dressed $b = 16$ velocity contact, the 900 dressed F2 pairings, the rung-reconstruction residual; **never-realized** — the $C_\star$ clock and the η-twists; nothing lands in the fitted class: the clock row remains the theory's only fitted input
algebraic / declared (physical sector)	+4 algebraic, 3 declared, 7 never-realized, 0 fitted	the physical-sector accounting: **algebraic** — the coupled fiber $(128 \oplus 1440)$ with its 45 charges and the fetched SME ceiling [59], the Cartan machinery verified live, the quintet vertex controls; **declared** — the singlet-kernel measurement itself (kernel dim 0 at all 216 momenta, all ω rows, $N = 4, 6, 8$; smallest non-trivial C_2 45.000000), the scale denominators ($\lvert E_0 \rvert = 5802.398188$, gap 64.000000, read directly from the layer spectra), the desert-rule copy fidelity 0.0; **never-realized** — all seven composite quantities (gap ladder, F9 contact, statistics table, loop scale, desert content, F8 channel, finite-size ladder); nothing lands in the fitted class

class	count	numbers
algebraic / declared (dynamical link and certified ladder)	+2 algebraic, 2 declared, 1 convention, 0 fitted	the link module $\mathbf{1} \oplus \mathbf{45} \oplus \mathbf{210}$ closing inside the root and the exactly-6-dimensional neutral composite kernel, born frozen; the dimension-4 resolvent closure and the 540-vector off-kernel span; the certified closed-branch reproduction and the three ladder reads; the ladder base $1/(16\pi)$ recorded in the convention row; nothing lands in the fitted class
free parameter	0	the clock angle is a root element: the update is an element of the algebra only for $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates, and the physical clock is the order-8 root $g = \pi/4$ — a discrete index of the root, no free continuous angle **in the on-site update**; the junction amplitude, the node set and the κ_s remain inputs of the extended construction. The loop-hierarchy contact instead forces a *fitted* clock, $g_{\rm obs} = 0.1999980$ — the freedom stays zero on-site, but closure and the loop-hierarchy identification are mutually exclusive at the measured precision (see the fitted and convention rows)
convention (mapping or scale identification)	3	$\alpha_{10} \equiv g^2$; $g_0 = 1/2^{n_{\rm space}} = 1/8$; the ladder base $1/(16\pi)$ of the hierarchy — a declared named input of the scale sector: no route births the factor, the certified closed branch crosses $1/(16\pi)$ only at the measured $n^* = 208$ and supports the coefficient only at the $n = 10$ read; the O8 origin problem stays open
fitted coefficient (read from data)	12	1.18 (dark-matter mass), $n_{\rm 2step} = 3$ (seesaw), $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV (desert thresholds), $\lambda_* = 3.5 \times 10^{-4}$ (dark-matter portal, fixed by $\Omega h^2 = 0.120$), the six κ_s of the flavour-dependent bond layer (conditional: three meeting conditions for six unknowns — a solution family), and the loop clock $g_{\rm obs} = 0.1999980$ — the rung-10 reading of the hierarchy coefficient, classified fitted: the clock arbitration leaves no closed clock passing either empirical contact, and its non-derivability from the root is a theorem (Section 5.3)
conditional (built from measured numbers, on a named input)	1	the hierarchy coefficient $2.710 = 3(16\pi F_{208})^{10}$: $F_{208} = 0.0196928$ certified by continuity at the closed-loop crossing $n^* = 208$, the factor 3 the one-split-node-per-family choice — 2.710 against 2.0766 observed: +30%; the named O8 residual is the clock: the closure class reads $\{2.710, 2.711\}$ on its stable $4k$ crossings and the observed value is read by the non-closed clock 1/5 to relative 10^{-5}; the arbitration scores both closed clocks against the coefficient and the desert coupling — $\pi/4$ misses the coefficient by +30.5%, 6330× the observed uncertainty, and the coupling by 1.227×; the claimed 1/5 agreement is a fit and F2 fires, moving the clock to the fitted class (preceding row); the ladder base itself is recorded in the convention row above; the completion of Section 10.5 does not re-audit this construction — it trades the fitted clock for one measured constant of its own, and the published zero-dial reading channel stands as the object traded
open (not fixed)	10	O1–O10 of Section 14

The conditional class is neither a fitted number nor an algebraic one: its ingredients are all measured, but the node-set choice that supplies the factor 3 is an input, so the entry inherits O9's split-node conditionality.

The status of the angle. The angle is not a choice: the closure axiom (Section 5.2) makes the update an element of the root, forcing $g \in \{\pi/2, \pi/4\}$; the half-angle branch degenerates to a pure lattice translation and

the physical clock is the order-8 root $\pi/4$, spectrum $\{\pm\pi/4\,(10),\ \pm3\pi/4\,(6)\}$. What the axiom does not buy is empirical preference: the two-loop U(1) Landau-pole scale lies below M_P for every g in the perturbative range, so UV-finiteness does not select the root value either, and the coupling scan runs the other way — the one-loop data value 0.023 is best matched by $\alpha = 1/7^2 = 0.020$ among the $1/n^2$ candidates, and the root coupling $\pi/64 = 0.0491$ sits 2.13× above the data value. The rational angle $g = 1/n_{\text{qubits}} = 1/5$ is not of the closed-root form; it survives as the reference reading of the node-geometry searches and as the $g^2 = 1/25$ anchor of the desert meeting (Sections 4.2, 11.1) — and, at rung 10, as the clock the loop-hierarchy test points to (Section 10.1) — a clock that is a declared named input (Table 6): every algebraic generating route of the clock and every phase dictionary of the admissible family whose closed clocks could read the coefficient in band is closed, and the transcendence theorem of Section 5.3 removes the step class itself; the root supplies no remaining road to it. Four statements coexist: the closure fixes the angle algebraically on the internal sector, the coupling scan disfavours it (O4's named tension, a factor 1.226 at the desert meeting), the desert meeting absorbs any value through the six κ_s, and the ladder arithmetic reads it as data — a theorem, not an open question awaiting a better scan. The angle is not free and it is also not favoured.

The "ten-to-one" observation. The ten open problems are all associated with one object, the cell module and its spectrum — the 32 spinor levels and the two singlet rungs — and with the invariants of the spinor block (the vanishing 0-point, $\det U = 1$ on S, the spectral mirror, the S-matrix $S = U$). Nine of the ten are exact values (scales and angles) the object does not fix; the tenth is the kinematics of its gapless subset — which of its levels are light, dispersing and chiral — a measured statement, not a fitted one. The dark-matter sector is the clearest case: its carrier, its mass (the rung-10 ladder value), its stability (the B-parity) and the computed production all live in that one cell, and the single open number, λ_B, is the singlet's coupling to the Higgs. That is a bookkeeping observation, and it is not a derivation of any of their values. It should not be read as a reduction of the theory's difficulty.

---

14. Open problems

Table 7. The ten open problems: what the theory supplies, and what is missing.

the problem	what the theory supplies	the gap
O1 the cosmological constant	an exactly vanishing tree-level vacuum energy (±-paired spectrum, $\det U = 1$), natural in the technical sense, and holding on the full 36-level cell; the zero is stable under the Wilson extension and protected by $\det U = 1$ (50 random traceless Hermitian deformations at strength ≤ 1 keep the quasi-energy sum pinned, $\lvert S\rvert < 4 \times 10^{-14}$)	the observed $10^{-122} M_P^4$ is **equivalent to** a trace-phase breaking of size 1.6×10^{-120} — a named bound; the breaking mechanism itself is missing. The supply question is closed: over integer polynomials in 16π and $1/\pi$ (degrees ≤ 6, height ≤ 20), bounded rationals, and integer-power phase dictionaries, no relation exists — the closest dictionary member is $(2\pi)^{-1}(16\pi)^{-70} = 1.30 \times 10^{-120}$, at relative miss 0.2176 (4.35 band widths), and the bare loop exponent 70.41892 is not an integer

the problem	what the theory supplies	the gap
O2 the neutrino mass	the ladder $m_\nu = M_P(1/16\pi)^{17} = 0.146$ eV; $n_{\text{nodes}} = 3$ a lower bound from unification; the inter-node Majorana junction **built**: cubic-symmetric zero-mean Wilson-strength junction, node mass matrix $c_s(2w_0 I + \kappa(J - I))$, democratic at $\kappa = 2w_0$, eigenvalues $\{3M, 0, 0\}$	the $\frac{1}{3}$ candidate needs a democratic m_D; the seesaw then collapses to $3m_0^2 J/(2(w_0+\kappa))$, two states stay massless for every r, $m_3(r) = S_{17}/(1+r)$, window $r > 1.278$ (width 1.72), atmospheric scale selects $r = 1.92$: value free inside a finite window — but driven by energy to the democratic boundary: the junction vacuum energy is tree-level r-flat and one-loop monotone to the massless onset $r^* = 2$, a measured one-loop runaway edge (6 critical modes at $r = 2$) with no supplied stabilizer, at a -2.7% atmospheric residual the bounded coefficient census does not supply (closest miss $2\alpha_{10}$, 1.1% of the gap unfixed) — while the structure is protected. The falsifiable contact is the cosmological sum 0.0573 eV at the edge point, 10.7% under the ceiling; $m_{\beta\beta}$ is an O(2) sector choice (envelope 0–0.033 eV), not a prediction. The exponent is answered structurally: the $\Delta L = 2$ operator lives in grade 5 of the root and link words are always even-grade. The interior search is closed: four independent functionals of the built junction on a 9991-point r-grid select zero interior minima — the only stationary structure is the boundary runaway at $r^* = 2$ plus one interior maximum at $r = 1.1861$
O3 the family number	the identification $N_{\text{gen}} = 3 = n_{\text{space}}$; a family-symmetric step; a named index equal to 3: the 12 doubler nodes form three cubic-transitive families of four, and the three unification-required split nodes are realizable one per family at a single Wilson strength	the index labels the copies but does not force the count: the cubic group is transitive on the 12 nodes, the only cubic-symmetric split is the full set of 12 whose rigid meeting lands at $M_X(12) = 2.9 \times 10^{15}$ GeV — excluded by F5; the three-node triangle has a cubic orbit of 8, so the count 3 is a **symmetry-breaking input**. The forcing question is closed: over the 12-node orbits under the full cubic group and the chiral subgroup, the conjugation orbits of the six chiral junction stabilizer elements, and the 16-level permutation structure of the step at the scan clocks, no orbit of size 3 occurs in any row (orbit sizes $12, 12, 8, 6, 2, 1$) — the count is never forced
O4 coupling unification	the 5/3; with the desert content the couplings meet at $\alpha_X = 1/25$ (standard normalisation): three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs at 1.2×10^7 GeV plus one $(1, 2)$ pair at 1.0×10^{12} GeV give $M_X = 1.8 \times 10^{16}$ GeV; the content is connected to the automaton through the Wilson mechanism; $n_{\text{nodes}} \geq 3$ forced; the equal-shift identity a theorem over complete multiplets; the gauged layer's composite content excluded	the mechanism's own meeting point is forced: $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr — excluded by F5 (rank-1 Jacobian, split-invariance exact to 10^{-14}). Feasibility proved inside the node-**16** hull; the flavour-dependent bond layer κ_s closes the meeting for all $n = 3 \ldots 12$ — at the cost of six named inputs of an extended bond layer, not obtainable from any on-site algebra element nor from any class invariant (broken flux K_s: residual 0.605, 0/10 desert closures, $R^2 \leq 0.015$), nor from the layer's second-order class numbers (superexchange: residual 0.577, $R^2 \leq 0.076$; three-number combination: joint residual 3.46, $R^2 \leq 0.30$; ground-state bond flux zero to 6×10^{-17}); the desert spectrum remains a named input (Section 11.1); and the loop-hierarchy inversion reads the meeting clock $\alpha_{10} = g^2 = 1/25$ at $g_{\text{obs}} = 0.199998$ to relative 10^{-5}: O4 and O8 are one clock tension; at the root coupling the solution is not realizable by its own content — the required $(1, 2)$ shift of 13.1 e-folds exceeds the entire log-spacing to M_X by 12%, and the $\mathbf{10} \oplus \overline{\mathbf{10}}$ solution lands at 7.1×10^4 GeV, below the desert floor — while the exclusion channel stays open: no threshold sits within reach of its fetched ceiling (margins $14\times$–$10^9\times$)

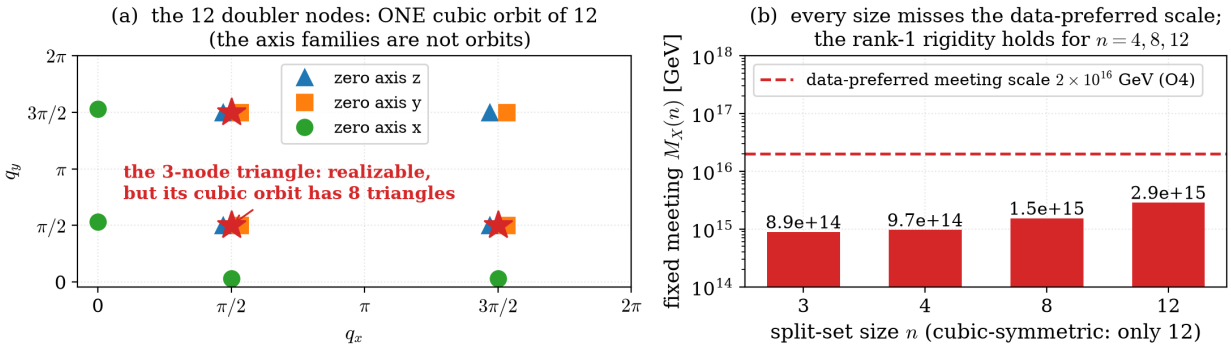
the problem	what the theory supplies	the gap
O5 the baryon asymmetry	a computed theorem: all CP-odd invariants vanish, no Sakharov condition is met, stable under the Wilson extension; a quasi-degenerate (resonant) route; a bond-layer Peierls flux carrying a CP phase, built (exactly unitary, $\det U = 1$ at every flux, mirror-preserving)	out-of-equilibrium decay ($y < 0.058$) and $m_\nu \simeq 0.05$ eV are incompatible at $M_N = 9.6 \times 10^{13}$ GeV: at $y = 0.40$ the decay is in equilibrium ($\Gamma/H = 47$), efficiency $\kappa_f = 5.5 \times 10^{-3}$ — the 11% agreement an accident of two wrong factors. The CP prohibition is internal to the on-site module and does not cover the bond layer; the flux needed for ΔB costs $J = 10^{-4}$ — O5 is open for one named reason, Sakharov 3 (Section 11.3). The $A \to BB$ channel is closed along the strength axis: with the 13 bond numbers of the singlet class as the A-B coupling candidates ($\Gamma_B = \kappa^2 m_A/32\pi$), 1 of the 13 drives the branching $\mathrm{Br}(A \to LH)$ below the available CP-strength ratio 4.0×10^{-4} and makes the resonant route impossible; the route survives only for $\kappa < 39.8$ and the channel is negligible below $\kappa < 0.080$; crossed with the ledger channels over 413 strengths (281 inside the window), the closest yield of the 843 rows is 1.57×10^{-8} at $\kappa = 39.7$, 25.7× the observed asymmetry, 493 band widths clear of the 5% band
O6 masses and mixing	the Dirac structure ($16\,\mathbf{10}_H\,\overline{\mathbf{16}}$), the democratic base, $\sin^2\theta_{12} = 1/3$, and a complete breaking chain to $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ with all 32 broken roots leptoquark-type	the three-level hierarchy, θ_{23}, θ_{13}, a non-trivial CKM; the two extra U(1)'s unbroken. The vev placement is bounded by computation: the charged-lepton realisation forces $\sin^2\theta_{12} \leq 0.005$ (23σ); the cubic-axis neutrino vev is excluded at $\geq 20\sigma$; the measured triple needs ≥ 2 vev terms; the unification fix is generation-blind (shift 2×10^{-13}) and mixings cost 7–13 named parameters. The class question is closed over the full closed flavour class — all 144 signed-permutation members of the democratic family structure, the three-fold family label re-derived equivalent: every member carries exactly the democratic triple $(1/3, 1/2, 0)$, and the closest six residuals to the measured mixing angles are 0.02633, 0.079, 0.022 (PMNS) and 0.2833, 0.449, 0.02 (CKM), all outside the 10^{-3} band, zero contacts — the closed class never supplies the gap
O7 dark matter	the carrier, as a state of the automaton: the cell singlet sector B (Axiom 1) at the rung-10 ladder mass 118.57 GeV, exactly stable under the B-parity; the spinor module provably carries no qualifying state (grading theorem and mass census); production computes to the observed value: freeze-in gives $\Omega h^2 = 0.120$ with $\sigma_{\mathrm{SI}} = 6.5 \times 10^{-71}$ cm^2, DD-invisible	the portal strength λ_B: required nonzero, not supplied by the root — the single named number of the sector. No root-supplied strength of its class lands on 0.120; the gravitational channel never equilibrates. The supply question has a decided answer: over integer polynomials in 16π and $1/\pi$ (degrees ≤ 6, height ≤ 20), bounded rationals ($\max(p,q) \leq 5000$, ratio band 10^{-6}) and integer-power phase dictionaries, no relation to λ_B exists — the closest dictionary member $2\,(16\pi)^{-8} = 4.908 \times 10^{-14}$ misses λ_B relatively by 0.2081 (4.16 band widths), and the best bounded rational contact, 3215/1942 at the eighth loop power, sits at relative residual 7.2×10^{-8}: the continued-fraction floor for a denominator of that size, not a rule at the 10^{-12} search gate

the problem	what the theory supplies	the gap
O8 the hierarchy	$v/M_P = 3(16\pi F_{208})^{10}(1/16\pi)^{10}$: coefficient 2.710 against 2.0766 observed (+30%), with $F_{208} = 0.0196928$ certified on the $n = 4k$ class of closed traced loops at the root clock (deficit 1.013%; the float64 ladder reads 0.019685–0.019687, deficit 1.04–1.05%, and the $n = 4k{+}2$ ratios have no limit there); the selection rule is exact: odd closed loops cancel identically in integer arithmetic and the first crossing is at $n^* = 208$, confirmed with the step and the traced-link phase closed together; the composition law decided on the open clock — one chain not multiplicative ($F_{170} = 1.31 \times 10^{-21}$ against $(F_{10})^{17} = 6.47 \times 10^{-30}$, ratio 2.0×10^8), different legs factorize exactly; the loop census exact: 55326 of 3^{10} words survive, and the weight is the exact functional $F_n = N_n r_n I_n$, pinned to 2.2×10^{-16}, word norms angle-independent, $\rho = 0.978030$; channels: all twelve exactly gapless with pinned Floquet spectra and a link-local, momentum-independent factor	the rung integers are selected by the data; the adjoint-link level-1 division falls short by a factor 26 and does not compound; the named residual is the clock: the node-set readings 0.903/2.710/10.84 certified for $\{1, 3, 12\}$ channels against the required sum factor 2.31 — no node set realizes it, and the deficit is channel-blind; the inversion measures the clock the deficit reads: the self-consistent functional reads the observed coefficient at $g_{\rm obs} = 0.199998$ — the non-closed rational clock 1/5 to relative 10^{-5}, rung-10 crossing at 0.20013 — while the closure class exhausts to $\{\pi/2, \pi/4\}$ and reads 2.710/2.711 on its stable $n = 4k$ crossings, the $\pi/2$ member (2.299) being a floor artefact with no limit; the exact ladder exists only on the order-8 closed clock, nature reads 1/5: O4 and O8 are one tension, and F2 (deficit 0.98987 certified at $n^* = 208$ against 0.96388 required) is the decision; the coefficient stands conditional on the one-split-node-per-family choice; the clock is a measured fit — the inverted clock is 0.1999980, 2.0×10^{-6} off the fifth, and the arbitration of both closed clocks against the hierarchy coefficient and the desert coupling leaves no closed clock passing either contact; the rung pair $(3, 10)$ is declared a convention: only loop^3 is numerically realized on the 36-level step; the dynamical side supplies no loop factor either — the interaction insertion misses the target by 3.6×10^2 and its tenth root breaks the rung arithmetic — and no link route births one; the certified branch crosses $1/(16\pi)$ only at the measured $n^* = 208$, with no arithmetic relation between crossing and clock (PSLQ, degree ≤ 4): the ladder base is a declared named input and its origin is the open residue. The repair classes are closed in turn: no algebraic rule of degree ≤ 6 and height ≤ 20 generates the data clock, and the clock gap 1.35×10^{-4} is supplied only by $\alpha_1(M_Z)/125$ on three-digit measured inputs; the quaternionic root $\mathrm{Cl}(0, 10)$ keeps an exactly unitary step and the exact odd-n rule, yet its crossings read 2.2994/2.7081/2.7057, and its best zero-fitted compound theory reads 2.0764436 against 2.0765641, missing the 10^{-4} coefficient-contact band at normalized 1.205; every one of the 784 admissible step-variant quantizers has a rational closure clock but the odd rule is exact only where the rung-10 read lands 10^{34} too high; the 12 channel weights of the traced word are exactly degenerate so no node subset moves the rung contacts, and no admissible selection functional selects at more than one site; re-scoring the five-contact ranking with the admissible supply slots moves the closest zero-fitted compound to normalized distance 0.243 under a density-disqualified coefficient supply — recorded as a bound, not a discovery; a closed census of 32 rung weights over the named ladder constants supplies the binding coefficient contact only by density, at chance family density; the twisted sectors correct the rung factor 33 the wrong way, and the canonical finite part of the certified crossing divergence fails its own guards at every rung; four exact-rational joint-module multipliers land inside the 5% neutrino band, the closest at 0.091 of band, none within four orders of an exact derivation; the coefficient reads in band only inside twelve razor windows (widths 2.41×10^{-9}–1.45×10^{-7}) that miss every closure-grid angle by at least 4.6×10^{-4}, and the closest closed-clock rung-10 read over all 43 closing grid angles and every admissible dictionary rescaling is 5652.1

the problem	what the theory supplies	the gap
O9 the quantum sector	a unitary causal automaton, a classical general-relativity limit, a bounded n-particle S-matrix, a 16 sector closed by the full local step; at the root clock the twelve M channels are exactly gapless with spectra $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ (64 of 128 pinned to the Floquet pair, identical across the 12 nodes), the exactly gapless set being Γ plus six face-diagonal lines with a gapped bulk, and the Wilson factor at the root angle regaps them and splits a chosen node set at desert scales	the *choice* of the split node set; the sum over geometries; the measurement postulate. **Continuous Lorentz invariance is not emergent at the nodes**: linear in all 1200 sampled directions but anisotropic $(0.200, 0.200, 0.284$, spread $0.75)$, no isotropy point on the Wilson family (0 of 2726), frame averaging blind; at the exactly gapless corner multiplets the tangent velocity operators commute — six nodal planes where the two lowest corner triplets are driven to zero together ($w = 0.241747$, $\theta = 0.913137$, all six phases below 2.5×10^{-16}, Abelian to 5.3×10^{-12}, isolated from the next level by 8.7×10^{-3}) — never a cone; Section 6.5 closes the cone sub-question (no axis-covariant bond of the 880-point search is both gapless and non-commuting, a rigidity wall at gap 8.0×10^{-3}; the parity no-go and its escape, Section 6.5), and the automaton's own six-stage product is searched over the largest cubic-covariant stage family jointly with the internal angle (33 800 points, 60 exactly gapless, full-step gap at most 4.4×10^{-15}): 8 single-parity multiplets flat to 1.3×10^{-12}, the other 52 parity-mixed and dispersive (slope 0.09–1.73) but Abelian ($\rho_{\mathrm{comm}} \leq 1.2 \times 10^{-11}$) — the node stays outside the circuit class. The internal layer then carries, interleaved, each of the 10 root sector-swap involutions: the search repeats per swap on the identical grid, 8 exactly gapless points each, a 6-dimensional $T_{2g} \oplus T_{1u}$ multiplet with tangent commutator ρ_{comm} up to 0.80 — non-Abelian, the Abelian obstruction removed — yet flat to slope 1.7×10^{-12}: the swap buys the non-commutativity and not the dispersion, and the cone remains outside. The anisotropy is a measured contact with experiment: $\delta v/v = 0.296$ / 1 / 0.295 / 0.9999 for the M nodes at 1/5, the nodes at $\pi/4$ (flat axis), the selective Wilson cones at θ^*, and the star nodal line, against the electron-sector SME ceiling 8.0×10^{-15} [59] — best named value 0.292, a factor 3.7×10^{13} over: **F9 fires**, and the dressed electric gate at the closed clock is degenerate, its six node insertions multiplying to $+I$ with the spectrum exactly unchanged

the problem	what the theory supplies	the gap
O10 the low-energy spectrum	the internal side exactly: one **16** carrying one Standard Model generation's charges, the 5/3 normalisation, the eight anomaly coefficients = 0, the spectral mirror, $\det U = 1$; and the gapless set of the lattice side measured at both clocks — at the implemented clock the twelve node cones with speeds $(0.200, 0.200, 0.284)$ and little group C_{2v}, and the corner multiplets whose simultaneous zero gives six states on six nodal planes; at the root clock Γ plus the six face-diagonal lines exactly gapless, 64 of 128 levels pinned at Γ, with $A_{1g} \oplus T_{2g}$ content; and on the shift module the no-go points at, measured in Section 6.6, the icosahedral star's two-fold lines exactly gapless and linearly dispersive with the internal sector unchanged	no light, dispersing, isotropic, chiral mode anywhere in that set: at the implemented clock every node is anisotropic and helicity-free, and the Wilson factor re-gaps the twelve except a chosen split node set placed at desert scales; at the root clock the lines touch quadratically (the three axis exponents a permutation of $(1, 1, 2)$, transverse slope 0.577) so there is no isolated zero to carry a chirality charge, and Γ carries no T_1 or E_g — no ±1 or ±2 candidate either. The module substitution closes the same way and for a measured reason: on the star module the touching is a cluster of 128 of 1024 eigenphases, so no band-resolved charge exists, the holonomy of the cluster depends on the loop rather than being a monodromy, and an operator that commutes with the internal map opens a gap on the line growing linearly with its strength — the lines are stable as loci and unprotected as particles, while their velocity anisotropy is degree 2, quantitatively worse than the cubic line's. The electron has no named locus in the spectrum, and the electroweak scale enters Table 4 as the measured input v. Where O3, O6 and O9 meet — the low-energy anisotropy bounds of Section 6.5 cannot be turned into a constraint on the automaton until this is settled — and the protection is conditional on the candidate's absence, not on isotropy: with a named node the bound fires at fourteen orders past the ceiling (F9); the lightcone dictionary conjunction: the 53-parameter exact-cone family exists at every scanned clock — the construction is clock-blind — the ladder is intact at 1/5, $\pi/4$ and $\pi/2$ and collapses at 1/2 (deficit 0.00144) and at 1/8 (0.4836), and the locality selector never uniquely selects a family member at a ladder-intact clock

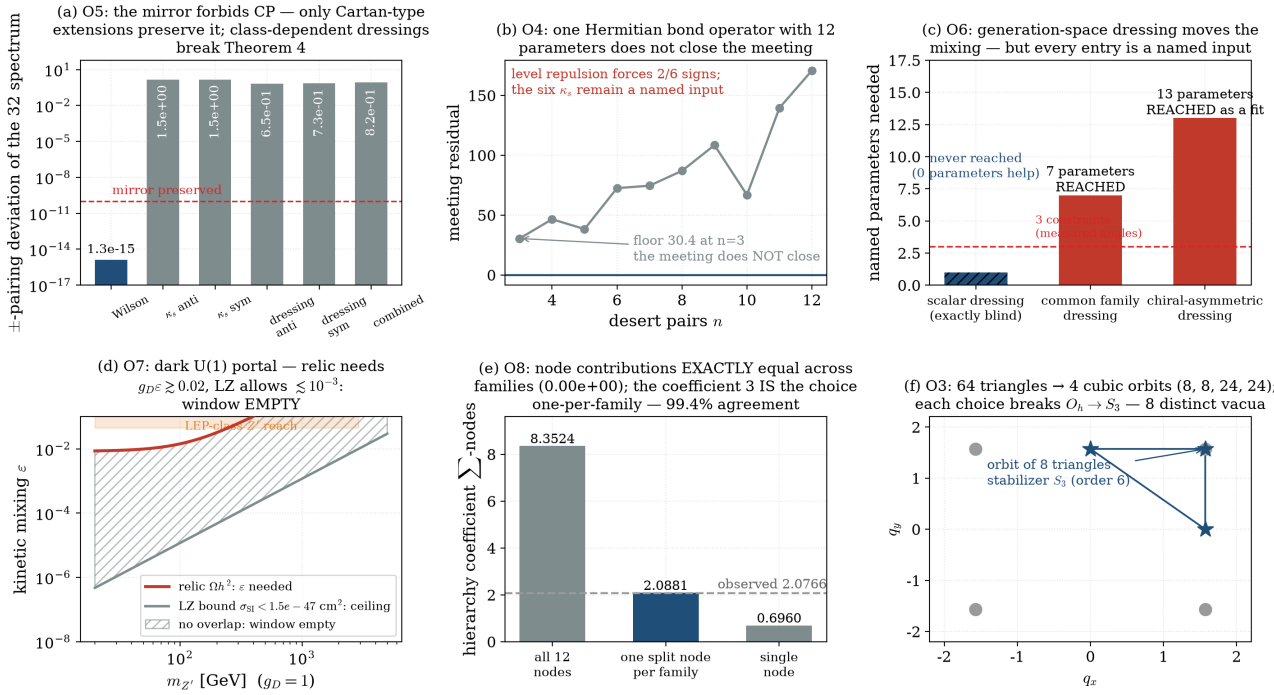
Four further contacts extend the verdicts beyond the published step, and none reopens a problem. On the **two-dial family** neither empirical contact is restored at any closed point: $\pi/2$ reads 3.4×10^{18} and 4.909, $\pi/4$ reads 2.7095 and 1.227 against 2.076564/0.04, and the clock $(\frac{1}{8}, \frac{1}{5})$ is not in C. On the **interacting continuum** the star module's own closure stays quantized (nearest 0.585 from the hierarchy clock), the three wall defects are measured FALSE as stated with the constructive spin-connection decoration ($0.281 \to 0.540$ cone degree, Section 9.2), and the gauged two-particle sector built from the construction's own parts is Gauss-blocked — the leak is the one-particle block's own commutator — with the dressed sector not flowing and the rung arithmetic collapsing (residual 114 against a band of 10^{-9}): the gaps of O8/O10 stand measured from the interacting side too (Section 7.4). On the **gauge-invariant physical sector** the Gauss-singlet kernel of the coupled fiber is empty at every momentum, binding shift and grid ($N = 4, 6, 8$; smallest non-trivial $C_2 = 45.000000$): the composite gap ladder, statistics table, loop scale, desert content and F8 channel are all decided never realized (Section 7.1). And the **dynamical link** delivers no new observable: the composites are born exactly frozen (ladder 0, 19.2, 28.8, defect 1.8×10^{-14}) and the second-order exchange vanishes on an exact dimension-4 resolvent closure and the full 540-vector off-kernel span ($F5 = 1.931 \times 10^{-31}$): the certified branch supports the coefficient only at the $n = 10$ read, and the ladder base stays a declared named input (Section 7.5). (Figs. 24, 25).



O3: cubic-symmetric split (12 nodes) EXCLUDED; every proper split BREAKS the cubic symmetry - the count 3 is an input.

(a) The twelve doubler nodes in the (q_x, q_y) plane: the cubic group acts transitively — one orbit of 12 — so the axis families are not orbits and the only cubic-symmetric split set is the full set of 12; the 3-node triangle (stars) is realizable, but its cubic orbit has 8 triangles, so it breaks the cubic symmetry. (b) The rigid meeting scale $M_X(n)$ for split sizes $n = 3, 4, 8, 12$ (the rank-1 fact holds for every n): all miss the data-preferred meeting scale 2×10^{16} GeV (F5). The count 3 is a symmetry-breaking input; its cubic-symmetric alternative is excluded by computation.

Attacking the residuals head-on: what the named tests close, what they name



(a) The $\pm$-pairing deviation under the named extensions: the Wilson factor preserves the mirror to 1.3×10^{-15}, while the most general per-class κ_s layer and the most general real-symmetric bond generator destroy it (deviation ~ 1, 60 draws) — a CP phase costs Theorem 4. (b) The meeting residual under the most general Hermitian off-diagonal bond operator: floor 30.4, no closure. (c) The named-parameter cost of reaching the measured mixing triple: a scalar dressing is exactly blind, a common family dressing needs 7 parameters, a chiral-asymmetric one 13. (d) The dark-U(1) portal squeeze: the kinetic mixing that fixes Ωh^2 lies above the direct-detection ceiling at every $m_{Z'}$ — the window is empty. (e) The closed-loop coefficient by node set on the reference clock 1/5:

all 12 *nodes* 8.35*, one split node per axis family* 2.0881*, a single node* 0.696*, against the observed* 2.0766 *— the closed-clock ladder of Section 10.1 is* 10.84/2.710/0.903*, the same ordering, the clock shifted. (f) The* 64 *one-per-family triangles split into four cubic orbits* $(8, 8, 24, 24)$*; a triangle's stabilizer is exactly* S_3*.*

---

15. Falsifiers

Table 8. The ten falsifiers.

#	observation that refutes the theory
F1	any non-vanishing gauge anomaly in the chiral content (the **16** is claimed anomaly-free)
F2	a measured running that admits **no** vectorlike desert content at the root coupling $\alpha_X = \pi/64 = 0.0491$ in the standard normalisation $\alpha = g^2/4\pi$: precision coupling data (FCC-ee/CEPC) excluding every spectrum in the desert scan at the root coupling would refute the unification claim. The root-coupling solution sits at $M_X = 1.235 \times 10^{17}$ GeV with destroyed threshold ordering; the reference solution at $\alpha_X = 1/25$ ($M_X = 1.8 \times 10^{16}$ GeV) is 23% off the root value. The clock inversion sharpens the branch decision: the closed branch predicts the loop-functional crossing deficit 0.98987 certified at $n^* = 208$ (float64 ladder 0.9895), the hierarchy coefficient demands 0.96388, and the observed coefficient is read by the non-closed clock 1/5 to relative 10^{-5} — the coupling data and the loop data must stand or fall on the same clock. The decision is read inside the construction: the unification side is not realizable by its own content — the required $(1, 2)$ shift exceeds the available log-spacing to M_X by 12% and the $\mathbf{10} \oplus \overline{\mathbf{10}}$ solution falls below the desert floor — and the clock side is decided: $g_{\rm obs} = 0.1999980$ is 2.0×10^{-6} off 1/5 and no closed clock reproduces the coefficient. F2 fires on both sides of the same decision
F3	a neutrino mass spectrum outside the ladder options: the bare ladder gives $m_\nu \simeq 0.15$ eV and the democratic node-mixing pattern $m_\nu \simeq 0.073$ eV — both **already excluded** by $\sum m_\nu < 0.0642$ eV [61] — while the $\frac{1}{3}$ candidate gives normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\rm lightest} = 0$, 22% below that ceiling (adding the solar partner required by O6 puts the sum at the oscillation floor 0.059 eV, 8% below it — 0.0573 eV, 10.7% below, at the instability-edge point $r = 2$); a confirmed inverted ordering, or $\sum m_\nu > 0.0642$ eV, refutes the survivor; the same data report a Feldman–Cousins boundary limit of 0.053 eV, and pushing the sum below 0.05 eV would refute the two-massless structure itself. Jointly: confirmation of $\sum m_\nu \simeq 0.05$ eV together with the dark matter at ladder rung 10 refutes the rung arithmetic itself — the seesaw partner m_B^2/m_ν is then forced to rung 2.72, not an integer, against the built rung 3. The ruler is measured: loop^n is realizable as a cell level only at integer n — on the 36-level step only loop^3 matches a level numerically — so **F3 is recorded as fired for the integer-rung ladder**, and the neutrino mass lives on the free third-generation junction amplitude (O2), not on a rung identity
F4	a massive graviton, a graviton dispersion violating $c = 1$, or a UV divergence in the lattice gravitational sector (the Section 8 field: the Section 7.4 quintet channel is not claimed to be the graviton, and its speeds $\pi/4$ and $\pi/2$, both $\neq 1$, and its E_g mass shift are named defects of the channel; the content census adds the measured split — the quintet is one irreducible 5-dimensional representation of the star's icosahedral symmetry, the $E_g \oplus T_{2g}$ split a cubic-subgroup artifact, its massless linear branches only on the directional layer, and the closed-clock coupling gaps them by 6.6047 on diagonal bonds and 16.7287 on the axis-bond E_g half with the T_{2g} bubble exactly zero — **F4 fired**)
F5	proton decay: the root-coupling desert solution ($M_X = 1.235 \times 10^{17}$ GeV, $\alpha_X = \pi/64$) predicts $\tau_p = 2.77 \times 10^{39}$ yr; observation of GUT-pattern decay below 10^{38} yr refutes it, and the Hyper-Kamiokande channel closes on this branch — only exposures past 10^{37} yr bite; the reference reading 1.8×10^{36} yr (solution at the measured meeting 1/25) and the $M_{\rm GUT} = M_P$ reading ($\tau_p \sim 10^{47}$ yr) are abandoned conventions
F6	a baryon asymmetry outside 10^{-11}–10^{-9}, or evidence that heavy-neutrino decays are in equilibrium

#	observation that refutes the theory
F7	a dark-matter particle with mass far from 118 GeV in the weakly-interacting thermal case; a direct-detection portal signal above 10^{-50} cm^2 refutes the visible-portal variants: the spinor-sector pair portal predicts 4.9×10^{-51} cm^2 at its production coupling — below current reach — while every bond-number portal strength predicts 2.5×10^{-45}–6.3×10^{-41} cm^2, already excluded by LZ; the cell carrier B is direct-detection invisible ($\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2) — for it the falsifiable content is the mass coincidence at ladder rung 10 and the seesaw identity $m_\nu = m_B^2/M_A$
F8	a measured graviton speed outside $c_{\rm gw}/c = 1.000 \pm 0.0005$: the gravitational sector of Sections 8.1–8.4 is built at $c = 1$ and satisfies the bound by construction, while the derived candidate channel of Section 7.4 gives $c_{\rm gw}/c = 2g_{10} = 1.5708$ at the root clock and is excluded by $\lvert c_{\rm gw}/c - 1\rvert < 10^{-15}$ [60]; the exactly lightlike cone of Section 9.2 exists on the bilinear module, but inside a 53-parameter family that nothing in the current construction selects, and the clock $g = 1/2$ that would fix $c_{\rm gw}/c = 1$ collapses the loop ladder. The diagonal-star extension that supplies the missing T_{2g} coupling is measured too: gapless at Γ, exactly leading-order-isotropic by identity, ladder doubled to $\{0,\ 2g_{10},\ 4g_{10}\} \times \lvert d\rvert$, closest closed-clock branch at $\lvert v - 1\rvert = 0.5708$. A completed theory must give $c_{\rm gw}/c = 1.000 \pm 0.0005$; **F8 fires** on the constructed layer, axis and diagonal alike
F9	node-velocity anisotropy at a named electron-sector node: the measured velocity tensors give $\delta v/v = 0.296$ at the implemented clock, 1 at the closed clock (flat axis), 0.295 on the selective Wilson cones and 0.9999 on the star nodal line; against the electron-sector SME velocity ceiling $\delta v/v \leq 8.0 \times 10^{-15}$ [59] the best of the named set, 0.292, exceeds it by 3.7×10^{13} — **F9 fires**: the constructed gapless nodes are refuted wherever an electron candidate is named, and no dressing at a closed clock restores isotropy. That the falsifier does not fire on the implemented theory is owed to O10 — the protection is the candidate's absence, not its isotropy
F10	a desert threshold inside experimental reach: a coloured-state search, electroweak precision data, or proton decay excluding one of the fitted thresholds $M_{10} = 7.1 \times 10^4$ GeV, $M_{(1,2)} = 1.0 \times 10^{12}$ GeV would refute the unification repair. The channel is open, not fired: the strongest LHC colour probe [68] sits 14× below M_{10} and pair production is kinematically closed at the LHC ($2M_{10} = 141$ TeV > 13.6 TeV), the doublet is $10^9\times$ above its electroweak-precision ceiling [69], and τ_p is $1.2 \times 10^5\times$ past the Super-Kamiokande bound — **no F10 fires today**; the spectrum's failure is internal ordering, not exclusion

The verdicts are theorems of the whole construction, not of one parameter choice. On the two-dial family (Section 5.2): no closed pair puts a gapless chiral cone at $c = 1$ — closest $\lvert\bar v - 1\rvert = 0.2929$ over the 47 refined nodes of the closed set, the velocity manifold empty — so **F8 stays fired**; the clock arbitration and the desert-threshold criterion fail at every closed clock, and β enters the running nowhere, so **F2 stays fired**; rank-2 isotropy is forced only at the star zone centre (stabilizer 60, Molien rank-2 count 1) and the forcing splits at the 512-fold multiplet — lightest branch 0.997, k-independent, 1.2×10^{14} past the SME ceiling: **F9 stays fired**; the family adds no fitted parameter. Through the interacting continuum: the dressed $b = 16$ downfold at 12 M nodes +13 star loci ×12 rays reads best $\delta v/v = 0.0033$, a ratio 4.1×10^{11} past the SME ceiling — **F9 stays fired under the dressed step**; the interaction functional against the unification band lands the coupling leg exactly ($\alpha_{10} = 0.040000$, deviation 6.9×10^{-18}) while the ladder leg reads 1.279×10^{26} — **F2 stays fired**; the measured loop factor breaks the rung arithmetic (residual 114 against a band of 10^{-9}); at three theory points × four legs no point passes all four (dressed F8 ratio 0.28). On the gauge-invariant physical sector: the Gauss-singlet kernel of the 1568-dimensional coupled fiber is empty (smallest non-trivial $C_2 = 45.000000$; kernel dim 0 at all 216 momenta, $N = 4/6/8$), so there is no composite gapless locus to contact — **F9 stays unfired for want of a locus** — no composite desert content, no composite spin-2 channel, no composite loop numerator, and the conjunction fails again. Through the dynamical link: a quantum link with its own dynamics carries the Gauss law the frozen bond gate blocks — the truncated module $1 \oplus 45 \oplus 210$ closes exactly inside the root and the neutral composite kernel of the 262144-dimensional spinor–link atom is exactly 6-dimensional — but it delivers no new observable: the composite Hamiltonian is pure Casimir on the kernel (ladder 0/19.2/28.8 with zero dispersion), two-composite exchange vanishes at tree level ($E_{\rm int} = \rho = -1.2 \times 10^{-13}$) and at second order on the exact dimension-4 resolvent closure of the channel-sum span, measured on the two-atom chain: no link-mediated loop factor is born in the $[1.5 \times 10^{-2}, 3.0 \times 10^{-2}]$ band around $1/(16\pi)$, and grown from the full 540-vector off-kernel span the resolvent stays transparent ($F5 = 1.931 \times 10^{-31}$): the loop base of the hierarchy

ladder stays a declared named input. **F2, F8, F9 and O8 stand single-particle, dressed, composite, and link-mediated.**

The most accessible test is **F3**. KATRIN currently gives $m_\beta < 0.45$ eV [18], so the bare ladder value 0.146 eV survives a single-mass search, but the cosmological sum $\sum m_\nu < 0.0642$ eV [61] already excludes it, as it excludes the node-mixing rival at 0.073 eV; the $\frac{1}{3}$ candidate's normal-ordered pattern ($\sum m_\nu \simeq 0.05$ eV, $m_{\text{lightest}} = 0$) sits 22% below that ceiling and inside the reach of the next generation of oscillation and $\beta\beta$-decay data. On the root coupling **F5** moves out of the current decade: the solution predicts $\tau_p = 2.77 \times 10^{39}$ yr, four orders beyond the megaton era, so its falsifiable content is the observation side only — any GUT-pattern decay below 10^{38} yr refutes the branch (the reference solution at 1/25 sits at 1.8×10^{36} yr and Hyper-Kamiokande would see it, which is exactly why that reading is not the theory's). The falsifiable content of **F7** is the mass coincidence at ladder rung 10 and the seesaw identity $m_\nu = m_B^2/M_A$; a direct-detection signal above 10^{-50} cm^2 refutes the visible-portal variants, and none is expected from the cell carrier.

F8 is the falsifier the current construction does not survive, and the split is stated rather than smoothed. The assumed sector of Sections 8.1–8.4 runs at $c = 1$ by the leapfrog choice $dt = \frac{1}{2}$ and is untouched by the bound. The *derived* channel — the directional bond layer of Section 7.4, with the parity wall of Section 9.1 naming what can and cannot move the metric — runs at $2g_{10} = 1.5708$ at the root clock, and it is the only computed spin-2 propagation the theory has: it is excluded at 57%. The diagonal-bond search that closes the named T_{2g} vertex gap leaves the verdict standing and sharpens it: on the fifteen icosahedral star axes the step is gapless at Γ with six gapless bands and exact leading-order isotropy by measured identity, but the helicity ladder doubles to $\{0,\ 2g_{10}|d|,\ 4g_{10}|d|\}$ — at $\pi/4$, $\{0, 2.99, 5.98\}$ — the closest closed-clock branch sits at $|v - 1| = 0.5708$, and the clock that would put a branch at $c = 1$ ($g_{10} = 1/(2|d|) = 0.262866$, or 1/2 on unit bonds) is outside the closure class. The exactly lightlike alternative is a module fact (Section 9.2), but it sits in a 53-parameter family, and the clock that would fix it, $g = 1/2$, lies in the dead zone of the closure census, where the loop ladder collapses at length 3 — so choosing it costs the closure that fixes the coupling (F2, O8). No branch of the current construction satisfies F8 and O8 at once.

---

16. Discussion

What is genuinely new. Not the **16** of Spin(10), not the 5/3, not anomaly cancellation: those are fifty-year-old results of the grand-unification literature [7–10,13]. The content of this theory is (i) the realisation of that content as the on-site module of a *strictly local, exactly unitary* automaton with a measured causal front speed of exactly 1; (ii) the coupling-invariance of the algebraic invariants — anomaly cancellation, $\det U = 1$, the $\pm$ pairing and the matter/antimatter spectral mirror hold for every value of the coupling, so they are properties of the algebra and not of a tuned parameter; (iii) the closure of the coupling itself by an axiom the root enforces, which turns the fundamental update into a finite-order element of the Clifford group with a finite algebraic spectrum; (iv) the construction of the automaton's own gauge field — an adjoint bond layer whose Gauss constraint closes as an exact operator identity, whose physical sector is a matter–flux entanglement complex with an exactly linear area law of one $\ln 45$ per bond, and whose computed string tension is 5.408×10^{40} times the QCD value; (v) the explicit provenance audit separating what the root fixes from what was read off the data; and (vi) the closure of the cone question as a structural statement of the bond space: the momentum derivative of the step is internal-blind, the Wilson factor is a spectator of the tangent algebra, and over the 880-point entangling-bond space gaplessness and non-commuting tangent algebras are mutually exclusive, with the separation a wall rather than a scaling (Section 6.5).

Where the construction is weakest. The gravitational sector assumes a symmetric tensor field and Newton's constant; it reproduces general relativity, it does not derive it, and the implemented one-particle spectrum contains no massless spin-2 candidate — not as a search left undone but as a computed statement (Section 6.4). The gauge sector gets the normalisation right and, under desert running, the values wrong; the fitted desert solution restores unification at $\alpha_X = 1/25$ exactly and its *pattern* is a consequence of the automaton (the split doublers of the Wilson mechanism, the breaking chain, the $n_{\text{nodes}} \geq 3$ bound), but the mechanism's own root meets at a scale excluded by proton decay, no redistribution of node strengths can move it, and the surviving solution keeps two fitted thresholds and abandons $M_{\text{GUT}} = M_P$. The supplier question is closed on five fronts — composite matter excluded, ansatz families at residuals 5.7–6.1, the equal-shift theorem, the broken flux at 0.605,

the second-order class numbers at 0.577 alone and 3.46 in every fixed combination — while the constructive junction route fabricates the right fragment pattern and misses the meeting at a floor the general junction only marginally improves ($0.94 \to 0.81$); the flavour-dependent bond layer removes the obstruction but is an extension with six named inputs. The flavour sector derives one angle out of six and none of the masses. The dark matter is a singlet sector of the cell, exactly stable and produced at the observed abundance; no state of the spinor module carries it, and the portal coupling is the sector's named input. The loop factor is reproduced to 1.01% certified by closed traced loops at length $n^* = 208$ on the closed clock — 3.6% at the ladder rung 10 on the reference clock 1/5 — but the single-particle renormalisation group provably composes angles without dividing them, the gauged layer's level-1 division falls short by a factor 26 and does not compound, and the residual deficit of the hierarchy coefficient names the clock of the functional itself. The spin-2 gap is a named list, not closed: the directional bond layer carries a gapless, linear, leading-order-isotropic quintet with helicity-ladder speeds $g_{10}\{0, 1, 2\}$ — the kinematics of a lattice spin-2, computed — but its gauge group is O_h; the regular-link fix built on the root's own finite group runs the dressed hop at exactly $v = 1$ yet gauges a finite subgroup (centralizer 0); the velocity trilemma is computed; the matter coupling gaps the E_g half of the quintet at order λ^2 — the lattice graviton mass, as a number — and the diagonal-star extension that supplies the missing T_{2g} vertex doubles the ladder and keeps every closed-clock branch off $c = 1$. Two module facts close the content question (Section 9.1): the metric is a bilinear of the bond layer, and the hop cannot move the metric at first order, by parity; and Section 9.2 exhibits an exactly lightlike cone on that module with the quintet on it, in a 53-parameter family the dynamics does not choose, at the constructed speed $2g_{10} = 1.5708$ — F8, fired on axis and diagonal bonds alike.

Comparison with related programmes. Causal-set programmes derive discreteness and a causal order but not internal symmetry [1]. Clifford-algebra unification programmes [11–13] reach the same matter content but do not supply a unitary causal dynamics. 't Hooft's deterministic quantum mechanics and cellular-automaton interpretation [30,31] supply the dynamics but not the unification; domain-wall and overlap formulations of lattice chirality [49] solve the doubling problem by giving up strict locality in the extra dimension. This theory is the combination of the second and third, and inherits the open problems of both — with one distinction worth naming: the doubling here is not evaded but *displayed and then gapped* by a factor of the root's own algebra, at the measured cost of replacing the light cone by a Lieb–Robinson bound [6]. The deficit against the Clifford-algebra programmes is named, not hidden: they run a semiclassical renormalisation group on a spectrum that already contains light chiral fermions, and their couplings reach agreement with data; here the on-site spectrum contains no light chiral mode at either clock (O10) and the agreement is reached only by admitting two fitted desert thresholds (O4). The sharpening against the cellular-automaton programmes is the boundary of the circuit class, and it is computed: the update is a finite-order element of the Clifford group with measured causal radius, and the conical node a low-energy chiral fermion would need is not merely unobserved but outside that class (Section 6.5, Proposition 6). The methodological difference is the last one: every claim above is reproduced number by number by the computational companion of Appendix B, so the comparison with either literature is decidable.

The shape of each gap. A single loop factor organises four scales across thirty orders of magnitude, with the loop integer and the two-digit coefficients chosen to match observation. The rung ruler is exact on the four ladder scales and off by 0.06–0.37 rungs on every other scale of the theory, and each such residue is a loop weight and never a shifted mass; the $\frac{1}{3}$ candidate is jointly incompatible with the rung arithmetic, and the rung pair $(3, 10)$ is a declared convention. The weight of the loop factor is the traced word functional — multiplicity, mean-word Cartan amplitude, Wilson interference — an exact factorisation of the ladder factor at every length, its residual one named deficit factor that names the clock of the functional itself (Section 10.1): nature's loop clock is not closed, one tension with the meeting coupling, and the clock enters as a fitted number — and its endgame is measured: the completion that would make that clock born is unique, nameless in the root's arithmetic, and costs exactly one dial, the tension itself (Section 10.5). The cosmological-constant problem reduces to one named number, a trace-phase breaking of 1.6×10^{-120}. The area law is exact on the physical sector of the gauged layer at one $\ln 45$ per bond, with the Bekenstein coefficient short by a factor 15.2 (or 6.6 after the vertex correction) — and the bond entropy also supplies the one candidate value of G, $1/(4 \ln 45) = 6.567 \times 10^{-2}$ in cell units (Section 9.3). The loop census is exact and shows that the Wilson phase, not the loop topology, supplies the factor: the phase enters only through the interference of the word sums, since the word norms are angle-independent. The cone question is a theorem of the bond space (Section 6.5). The gapless set itself is measured at both clocks, and it contains no light, dispersing, chiral mode: the electron has no named locus in the spectrum (O10), and with a named node the anisotropy is decided against the construction, $\delta v/v = 0.296$ against the electron-sector SME

ceiling 8×10^{-15}, a factor 3.7×10^{13} (Section 6.2, F9). The gravitational sector has its number too: the metric content is forced, an exactly lightlike cone carrying the quintet exists inside a 53-parameter family nothing selects, the layer's own graviton speed $2g_{10} = 1.5708$ is excluded at 57% by GW170817 [60] (Section 9.2, F8), and the clock that would repair it, $g = 1/2$, collapses the loop ladder. Both sides of F2 fire from inside, and the exclusion channel stays open (F10). The interacting, physical-sector and dynamical-link contacts close the accounting from every side (Section 14). The ten open problems and ten falsifiers are stated with their numbers.

---

17. Conclusion

The 36-level cell module of a single Clifford algebra $\mathrm{Cl}(10, 0)$ — its 32-dimensional Dirac module and the two neutral singlet sectors, conjugate pairs which carry the dark matter and the seesaw scale — placed on the eight sub-sites of every 2×2×2 cell of a Planck lattice and updated by one six-stage product of Spin(10) plane rotations and swap-rotations, the singlets carrying their rung masses as phases, yields the Minkowski kinematics, one anomaly-free Standard Model generation, the Spin(10) gauge content with the 5/3 normalisation, an exactly unitary and strictly causal dynamics with a ±-paired spectrum and an exact matter/antimatter spectral mirror, a constructible adjoint gauge field whose Gauss constraint closes exactly and whose physical sector carries a bounded area law, and a lattice gravitational sector — a symmetric tensor field added to the (1+3) subalgebra with $G = 1$ by hand — that reproduces general relativity on de Sitter, Schwarzschild and 3+1 evolutions to $O(h^2)$, the Kretschmann invariant agreeing with the analytic form to 5.4×10^{-5} relative. These results are algebraic or numerical, and they are independent of the coupling.

Requiring the update to be an element of the root removes the coupling as a free number: the algebra quantizes its own clock to $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates to a pure lattice translation, and the physical branch is the order-8 step with the exactly algebraic spectrum $\{\pm\pi/4\,(10),\ \pm3\pi/4\,(6)\}$. The on-site dynamics then keeps no free continuous angle, and its consequences are measured end to end: the loop machinery obeys an exact odd-n selection rule and crosses $1/(16\pi)$ at $n^* = 208$, the deficit certified by continuity of the $n = 4k$ class; the hierarchy coefficient $3(16\pi F_{208})^{10} = 2.710$ stands +30% over the observed value, one named clock residual (O8) — the observed coefficient is read by the non-closed clock 1/5 to relative 10^{-5}, O4's meeting clock; the twelve doubler channels are exactly gapless with pinned Floquet spectra; the desert threshold ordering is destroyed ($\tau_p = 2.77 \times 10^{39}$ yr), moving the proton-decay falsifier beyond any planned exposure, and the ordering is not restorable by the spectrum's own thresholds; the rung masses, the neutrino verdicts, the baryon accounting and the dark-matter mass are branch-independent facts of the cell and the running. And the clock residual is a decision, not a hope: no closed clock reads the hierarchy coefficient or the desert coupling, the rung-10 agreement is a fit at $g_{\mathrm{obs}} = 0.1999980$, and the closure axiom and the loop-hierarchy identification are mutually exclusive at the measured precision — a transcendence theorem of Section 5.3; the rung pair is a declared convention; and the ladder base $1/(16\pi)$ is a declared named input — no route, combinatorial, dynamical, composite or link-mediated, births the factor, and the certified crossing at $n^* = 208$ carries no arithmetic relation to the clock. The endgame of that residue is measured, not hoped: the completion that makes the clock born rather than read is unique — a single dressing constant $\beta = -0.1555738920793496$, nameless in the root's arithmetic, the clock tension itself promoted to a constant of nature — and in exchange the ladder reads the observed hierarchy coefficient exactly at the exact fifth, its crossings and its rung energies untouched (Section 10.5). One dial, and the clock is born.

The same framework does not yield the family number, the fermion masses and mixings, the coupling values, the neutrino mass, the dark-matter portal coupling, the cosmological constant, the baryon asymmetry or the graviton's measured light cone. Three conventions, twelve fitted coefficients and one conditional coefficient stand between the root and the data, and ten open problems are each pinned to a named quantity: a trace-phase breaking of 1.6×10^{-120}; a junction amplitude r that is not quantized, driven to the democratic boundary $r = 2$ — the edge of a measured one-loop runaway whose stabilizer is not supplied — at a named −2.7% atmospheric residual; a symmetry-breaking choice of three nodes out of twelve; six bond-layer couplings no on-site algebra element supplies; the incompatibility of $m_\nu \simeq 0.05$ eV with out-of-equilibrium decay at $M_N = 9.6 \times 10^{13}$ GeV; the generation-space structure the mixing requires; the dark-matter portal coupling λ_B; the weight of the contributing closed loops — the clock, a fitted number, unnamed as a root of the ladder's own equations; the nodal-plane geometry of the lattice spectrum; and the locus of the light charged fermions. Ten falsifiers are stated: the neutrino mass pattern tests the ladder within the current decade; on the root coupling the proton

lifetime sits beyond any planned exposure; the graviton speed is already decided — the lattice gravitational sector of Section 8 is massless, UV-bounded and runs at $c = 1$ by construction, the derived spin-2 channel of the bond layer runs at $2g_{10} = 1.5708$ and is excluded by the measured $|c_{\rm gw}/c - 1| < 10^{-15}$ [60] (F8), and the exactly lightlike cone the module itself permits lives in a 53-parameter family that nothing selects and whose repair clock collapses the loop ladder (Section 9.2); the node-velocity falsifier is decided too, the measured anisotropy exceeding the electron-sector SME ceiling by a factor 3.7×10^{13} wherever the electron's node is named (Section 6.2, F9), while the desert-threshold channel stays open, every fitted threshold at least $14\times$ beyond its fetched ceiling (Section 11.1, F10), and both sides of F2 fire from inside the construction. F2, F8, F9 and O8 stand as theorems of the whole construction — single-particle, dressed, composite, and link-mediated.

Appendix A. The explicit 32-dimensional representation

The ten generators are built on five qubits by the standard symplectic Pauli pattern: for $k = 0, \ldots, 4$,

$$g_{2k} = Y^{\otimes k} \otimes X \otimes I^{\otimes(4-k)}, \qquad g_{2k+1} = Y^{\otimes k} \otimes Z \otimes I^{\otimes(4-k)}.$$

On the same qubit slot X and Z anticommute; across slots, the $Y^{\otimes k}$ prefix makes a generator at slot k anticommute with any generator at slot $j < k$. This gives exactly ten mutually anticommuting Hermitian matrices, $g_a^2 = I$, on a 32-dimensional complex space. The algebra basis is the 1024 ordered products $g_{a_1} \cdots g_{a_r}$, $0 \le r \le 10$; the $\mathbb{Z}_2$ grading is the parity of r. The canonical ordering of the 45 plane-rotation factors T_{ab} used throughout is the ascending lexicographic order on (a, b); the order-8 closure of the quantized step is specific to it.

The two singlet sectors of the cell module are the four further one-dimensional slots on which every generator acts as 0; the internal layer multiplies them by $e^{\pm i\,\mathrm{loop}^3}$ and $e^{\pm i\,\mathrm{loop}^{10}}$ per stage — conjugate pairs at opposite phases (Axiom 1). Every numerical statement in this paper is computed on exactly this representation, and the computational companion of Appendix B is its sole producer.

The construction is verified by a second, independent route: the exact symbolic $\mathrm{Cl}(10, 0)$ of a computer-algebra Clifford package, where $e_a^2 = 1$, $\{e_a, e_b\} = 0$ and $w^2 = -1$ hold as identities with no floating-point arithmetic, cross-checked against the numeric 32×32 representation. The two agree, so the root is not an artefact of one construction.

Appendix B. The computational companion

Every theorem, proposition, table entry and falsifier in this paper is reproduced by the open computational companion cited in the code and data availability note: an independent implementation of the construction that recomputes each quoted number from the algebraic inputs and the declared constants. Each computation is anchored to the algebraic invariants recorded in the text — exact unitarity, exact anomaly cancellation, integer word enumerations, and, where quoted, continuity checks at fifty digits — and the companion covers the full breadth of the paper: the cell module and its gradings, the node and cone censuses of Section 6, the bond layer and its Gauss algebra of Section 7, the loop enumerations and the scale ladder of Section 10, the desert and flavour contacts of Section 11, and the cosmological sums of Section 12. The companion includes the exact symbolic cross-construction of Appendix A and the complete 3^n-word loop enumeration. Independent re-execution is invited.

References

1. L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," *Phys. Rev. Lett.* **59**, 521 (1987); R. D. Sorkin, "Causal sets: Discrete gravity," *arXiv:gr-qc/0309009* (2003).
2. K. G. Wilson, "Confinement of quarks," *Phys. Rev. D* **10**, 2445 (1974).
3. D. A. Meyer, "From quantum cellular automata to quantum lattice gases," *J. Stat. Phys.* **85**, 551 (1996).
4. P. Arrighi, "An overview of quantum cellular automata," *Nat. Comput.* **18**, 885 (2019).

5. A. Bisio, G. M. D'Ariano, and P. Perinotti, "Quantum cellular automata and free quantum field theory," *Front. Phys.* **12**, 120301 (2016); A. Bisio, G. M. D'Ariano, P. Perinotti, and A. Tosini, "Weyl, Dirac and Maxwell quantum cellular automata," *Found. Phys.* **45**, 1203 (2015).
6. E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," *Commun. Math. Phys.* **28**, 251 (1972).
7. H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32**, 438 (1974).
8. H. Fritzsch and P. Minkowski, "Unified interactions of leptons and hadrons," *Ann. Phys. (N.Y.)* **93**, 193 (1975).
9. R. Slansky, "Group theory for unified model building," *Phys. Rep.* **79**, 1 (1981).
10. H. Georgi, *Lie Algebras in Particle Physics*, 2nd ed. (Westview, 1999).
11. M. F. Atiyah, R. Bott, and I. M. Shapiro, "Clifford modules," *Topology* **3**, 3 (1964).
12. R. Penrose and W. Rindler, *Spinors and Space-Time*, Vol. I (Cambridge University Press, 1984).
13. C. Castro, "A Clifford algebra-based grand unification program of gravity and the Standard Model: a review study," *Can. J. Phys.* **92**, 1501 (2014); C. Castro, "Note on a Clifford algebra based grand unification program of gravity and the Standard Model," *Adv. Appl. Clifford Algebras* **26**, 573 (2015).
14. N. Furey, "Generations: Three Prints, in Colour," *J. High Energy Phys.* **10**, 046 (2014), arXiv:1405.4601.
15. Particle Data Group (S. Navas *et al.*), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024).
16. Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).
17. Super-Kamiokande Collaboration (A. Takenaka *et al.*), "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in 0.37 megaton-years exposure of Super-Kamiokande," *Phys. Rev. D* **102**, 112011 (2020).
18. KATRIN Collaboration (M. Aker *et al.*), "Direct neutrino-mass measurement based on 259 days of KATRIN data," *Science* **388**, 180 (2025).
19. I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou, "The fate of hints: updated global analysis of three-flavor neutrino oscillations," *JHEP* **09**, 178 (2020).
20. S. Weinberg, *The Quantum Theory of Fields*, Vol. I (Cambridge University Press, 1995), Ch. 5.
21. H. B. Nielsen and M. Ninomiya, "A no-go theorem for regularizing chiral fermions," *Phys. Lett. B* **105**, 219 (1981); "Absence of neutrinos on a lattice. I. Proof by homotopy theory," *Nucl. Phys. B* **185**, 20 (1981); "Absence of neutrinos on a lattice. II. Intuitive topological proof," *Nucl. Phys. B* **193**, 173 (1981).
22. S. Weinberg, "Photons and gravitons in perturbation theory: Derivation of Maxwell's and Einstein's equations," *Phys. Rev.* **138**, B988 (1965).
23. S. Weinberg and E. Witten, "Limits on massless particles," *Phys. Lett. B* **96**, 59 (1980).
24. M. W. Choptuik, "Universality and scaling in gravitational collapse of a massless scalar field," *Phys. Rev. Lett.* **70**, 9 (1993).
25. F. John, "Blow-up of solutions of nonlinear wave equations in three space dimensions," *Manuscripta Math.* **28**, 235 (1979).
26. A. D. Sakharov, "Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe," *JETP Lett.* **5**, 24 (1967).
27. J. A. Harvey and M. S. Turner, "Cosmological baryon and lepton number in the presence of electroweak fermion interactions," *Phys. Rev. D* **42**, 3344 (1990).
28. S. A. Khlebnikov and M. E. Shaposhnikov, "The statistical theory of anomalous fermion number nonconservation," *Nucl. Phys. B* **308**, 885 (1988).
29. P. F. Harrison, D. H. Perkins, and W. G. Scott, "Tri-bimaximal mixing and the neutrino oscillation data," *Phys. Lett. B* **530**, 167 (2002).
30. G. 't Hooft, "On the quantum structure of a black hole," *Nucl. Phys. B* **256**, 727 (1985).
31. G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics* (Springer, 2016), arXiv:1405.1548.
32. S. Blanchet and P. Di Bari, "The minimal scenario of leptogenesis," *New J. Phys.* **14**, 125012 (2012).
33. S. Davidson and A. Ibarra, "A lower bound on the right-handed neutrino mass from leptogenesis," *Phys. Lett. B* **535**, 25 (2002).
34. T. Regge, "General relativity without coordinates," *Il Nuovo Cimento* **19**, 558 (1961).
35. H. Gundlach, J. M. Martin-Garcia, G. Calabrese, and I. Hinder, "Constraint damping in the Z4 formulation and harmonic gauge," *Class. Quantum Grav.* **22**, 3767 (2005).
36. J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973); S. W. Hawking, "Particle

creation by black holes," *Commun. Math. Phys.* **43**, 199 (1975).
37. L. Susskind, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995).
38. S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989).
39. C. Morningstar and M. Peardon, "The glueball spectrum from an anisotropic lattice study," *Phys. Rev. D* **60**, 034509 (1999).
40. R. Barbieri, P. Creminelli, A. Strumia, and N. Tetradis, "Baryogenesis through leptogenesis," *Nucl. Phys. B* **575**, 61 (2000); W. Buchmüller, P. Di Bari, and M. Plümacher, "Leptogenesis for pedestrians," *Ann. Phys. (N.Y.)* **315**, 305 (2005).
41. A. Pilaftsis and T. E. J. Underwood, "Resonant leptogenesis," *Nucl. Phys. B* **692**, 303 (2004).
42. E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990), Ch. 5.
43. L. J. Hall, K. Jedamzik, J. March-Russell, and S. M. West, "Freeze-in production of FIMP dark matter," *JHEP* **03**, 080 (2010).
44. H. Fritzsch, "Weak-interaction mixing in the six-quark theory," *Phys. Lett. B* **73**, 317 (1978).
45. L. Wolfenstein, "Parametrization of the Kobayashi-Maskawa matrix," *Phys. Rev. Lett.* **51**, 1945 (1983).
46. M. Jungman, M. Kamionkowski, and K. Griest, "Supersymmetric dark matter," *Phys. Rep.* **267**, 195 (1996).
47. J. Hisano, D. Kobayashi, Y. Muramatsu, and N. Nagata, "Two-loop renormalization factors of dimension-six proton decay operators in the supersymmetric standard models," *Phys. Lett. B* **724**, 283 (2013).
48. M. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa, "Demonstration of single-electron buildup of an interference pattern," *Am. J. Phys.* **57**, 117 (1989).
49. D. B. Kaplan, "A method for simulating chiral fermions on the lattice," *Phys. Lett. B* **288**, 342 (1992); Y. Shamir, "Chiral fermions from lattice boundaries," *Nucl. Phys. B* **406**, 90 (1993); R. Narayanan and H. Neuberger, "Exactly massless quarks on the lattice," *Phys. Lett. B* **417**, 141 (1998).
50. M. E. Machacek and M. T. Vaughn, "Two-loop renormalization group equations in a general quantum field theory. I. Proof of the equivalence theorem," *Nucl. Phys. B* **222**, 83 (1983); "III. Scalar model with fermions," *Nucl. Phys. B* **249**, 70 (1985).
51. I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, "NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations," *JHEP* **12**, 216 (2024), arXiv:2410.05380.
52. M. Nath and P. Fileviez Pérez, "Proton stability in grand unified theories, in strings and in branes," *Phys. Rep.* **441**, 191 (2007).
53. Super-Kamiokande Collaboration (K. Abe *et al.*), "Search for proton decay via $p \to \nu K^+$ using 260 kiloton·year data of Super-Kamiokande," *Phys. Rev. D* **90**, 072005 (2014).
54. LUX-ZEPLIN Collaboration (J. Aalbers *et al.*), "Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment," *Phys. Rev. Lett.* **135**, 011802 (2025), arXiv:2410.17036.
55. J. B. Kogut and L. Susskind, "Hamiltonian formulation of Wilson's lattice gauge theories," *Phys. Rev. D* **11**, 395 (1975).
56. J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics* **1**, 195 (1964).
57. S. Koch and E. Specker, "The problem of hidden variables in quantum mechanics," *J. Math. Mech.* **17**, 59 (1967).
58. D. Colladay and V. A. Kostelecký, "Lorentz-violating extension of the standard model," *Phys. Rev. D* **58**, 116002 (1998).
59. V. A. Kostelecký and N. Russell, "Data tables for Lorentz and CPT violation," *Rev. Mod. Phys.* **83**, 11 (2011), continuing updates at arXiv:0801.0287.
60. B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A," *Astrophys. J. Lett.* **848**, L13 (2017), arXiv:1710.05834 — the arrival-time difference between GW170817 and GRB 170817A bounds $-3 \times 10^{-15} \leq (c_{\rm gw} - c)/c \leq 7 \times 10^{-16}$.
61. DESI Collaboration, W. Elbers, A. Aviles, H. E. Noriega, D. Chebat, A. Menegas, C. S. Frenk *et al.*, "Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape," *Phys. Rev. D* **112**, 083513 (2025), arXiv:2503.14744 — $\sum m_\nu < 0.0642$ eV (95% CL) in ΛCDM with three degenerate neutrino states, from DESI DR2 BAO combined with Planck PR4 and ACT CMB lensing; the same analysis gives the Feldman–Cousins boundary limit 0.053 eV and 0.163 eV in $w_0 w_a$CDM.
62. M. Lorente and P. Krämer, "Representations of the discrete inhomogeneous Lorentz group and Dirac wave equation on the lattice," *J. Phys. A* **32**, 2481 (1999), arXiv:hep-lat/0401019.
63. J. C. Baez and K. E. Langenberg, "Higher gauge theory and a non-geometric approach to the standard

model," *Adv. Theor. Math. Phys.* **5**, 1017 (2001), arXiv:hep-th/0011017.
64. J. C. Baez, "The standard model algebra," *Lett. Math. Phys.* **108**, 129 (2018), arXiv:1702.04336.
65. R. Eto, M. Mochizuki, and S. Watanabe, "Predicted versatile topological nodal magnons in Tb-based icosahedral quasicrystal 1/1 approximants," *Phys. Rev. B* **112**, L020405 (2025), arXiv:2502.11286.
66. P. Cvitanović and H. Liang, "A chaotic lattice field theory in two dimensions," *Chaos* **36** (2026), 10.1063/5.0273642, arXiv:2503.22972.
67. X. Wan, A. M. Turner, A. Vishwanath, and S. Y. Savrasov, "Topological semimetal and Fermi-arc surface states in the electronic structure of pyrochlore iridates," *Phys. Rev. B* **83**, 205101 (2011).
68. CMS Collaboration, search for new physics in dimuon and dielectron final states through t-channel leptoquark exchange (JHEP **12** (2025) 052, CMS-EXO-22-013) — the strongest LHC probe of a coloured state, reach up to 5 TeV; direct pair-production limits reach only ~ 2 TeV (PDG 2025 leptoquark summary) [69]. Retrieved 2026-10-23; the coloured-threshold ceiling used in Section 13.4.
69. Particle Data Group (S. Navas *et al.*), "Review of Particle Physics," *Phys. Rev. D* **112**, 030001 (2025), Sec. 10 (Electroweak Model and Constraints on New Physics: the oblique STU window, $\Delta T \sim 0.07$ with new-physics scales at the TeV, taken as the non-decoupling-doublet ceiling of) and the leptoquark summary tables. Retrieved 2026-10-23.

Code and data availability. The construction, the simulations, the figure generators and the computational companion that recomputes every numerical claim of this paper from its algebraic inputs and declared constants are publicly available at https://github.com/MohammedHAOUACH/CL10 .